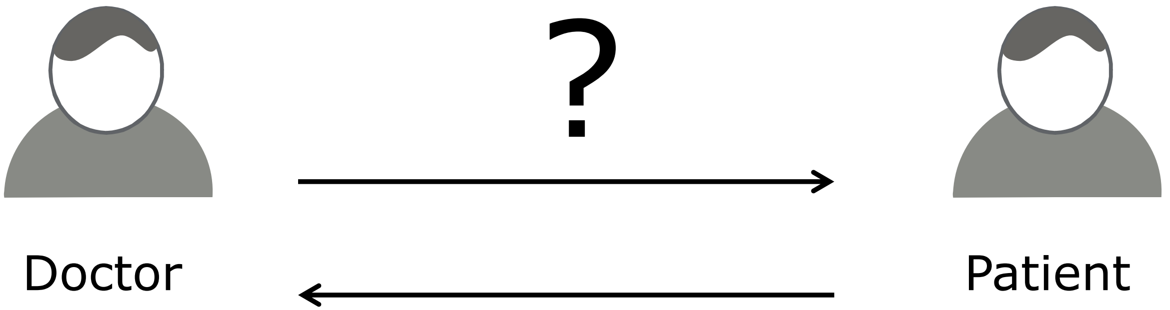
UDS Overview

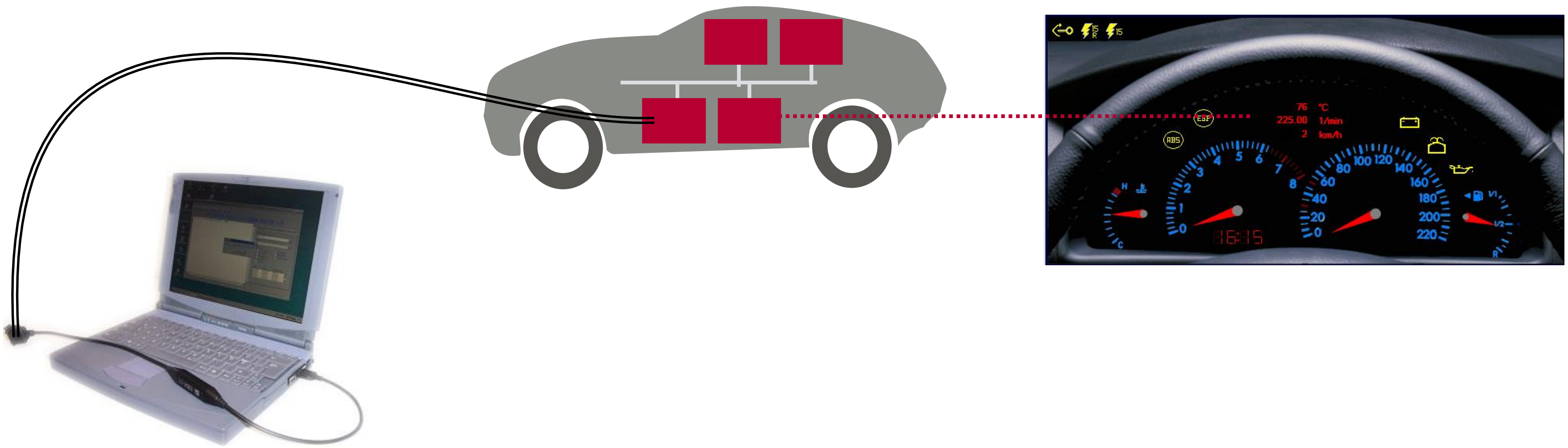
Diagnostics



Diagnostics is to

- determine, check and classify symptoms of a system
- with the purpose to get an overall picture

Diagnostic Goal



General Goal of Diagnostics:

- Continuous supervision of the vehicle and precise failure recognition in order to have a one-stop repair/maintenance at minimum cost and time.

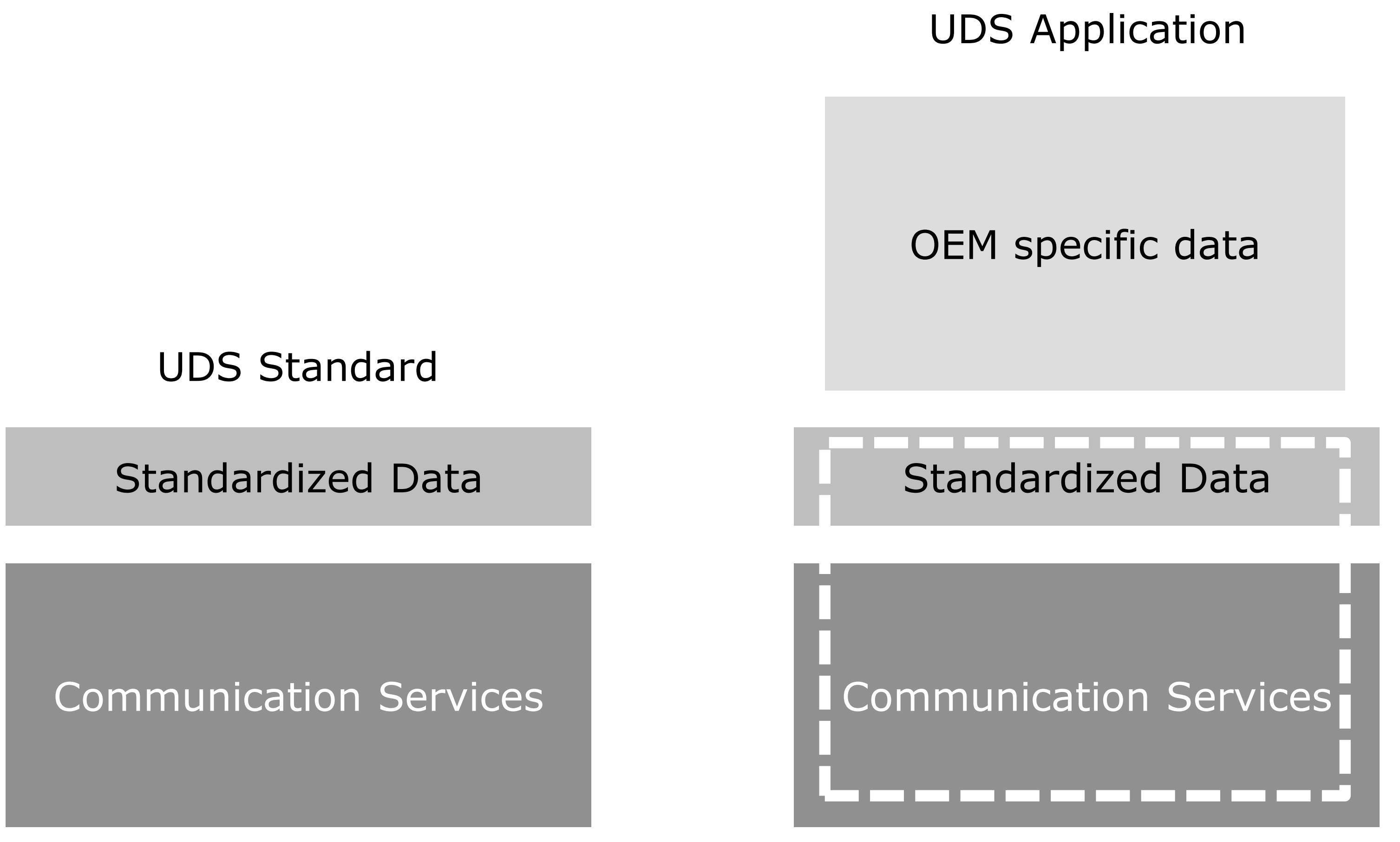
Agenda

- **Diagnostic Overview**
- Diagnostic Protocol UDS
  - Services Overview
  - Diagnostic Session Control
  - Security Access
  - Read/Write Data By Identifier
  - Fault Memory
  - Communication Method in Diagnostics
  - Review/Summary
- Vector Diagnostic Solution
  - Solution
  - Tool-chain
  - Summary

Vehicle Network Diagnostic

- A “diagnostic protocol” is necessary for communication between an ECU and a diagnostic tester.
- Diagnostic protocols: ISO 14230/ISO 15031/ISO 15765/ ISO 14229 …
- Diagnostic protocol defines diagnostic request/response rules, ECU handling behavior for request, and content of request/response message.

UDS Term



- Unified Diagnostic Services
- ISO 14229-1

UDS Range

- **ISO 14229/2013**: Unified diagnostic services (UDS)

  -1 Specification and requirements

  -2 Session layer services

  -3 Unified diagnostic services on CAN implementation (UDSonCAN)

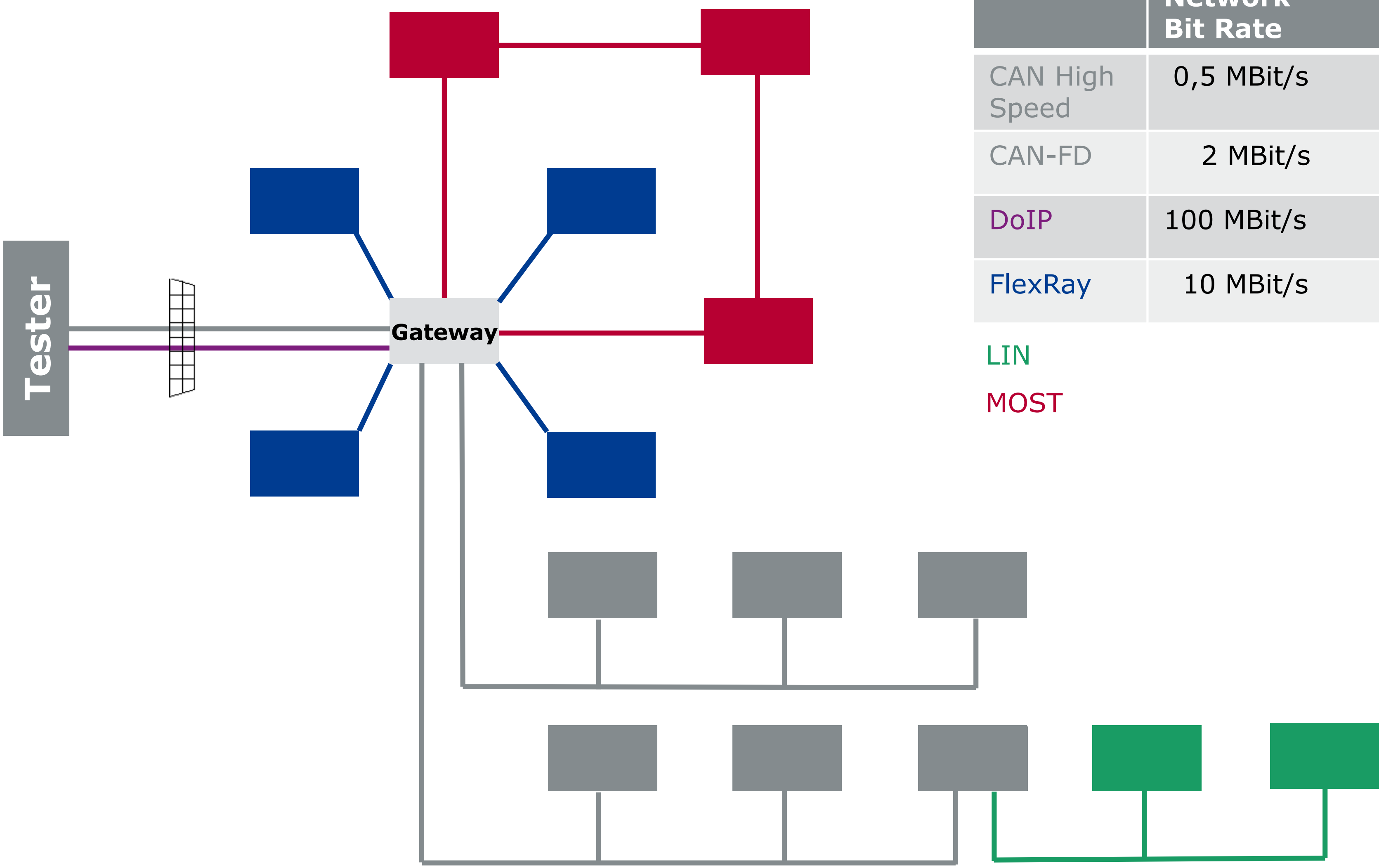
  -4 Unified diagnostic services on FlexRay implementation (UDSonFR)

  -5 Unified diagnostic services on Internet Protocol implementation (UDSonIP)

  -6 UDS on K-Line implementation (UDSonK-Line)

  -7 UDS on local interconnect network (UDSonLIN)

Network Example



Network	Typical Network Bit Rate
CAN High Speed	0,5 MBit/s
CAN-FD	2 MBit/s
DoIP	100 MBit/s
FlexRay	10 MBit/s

LIN

MOST

Diagnose Application with UDS

5 - 7	Diagnostics with UDS
3 - 4	Transport layer
1 - 2	Physical network

UDS in OSI Model

Layer					
7	ISO 14229-1: UDS Services				
7	14229-3	14229-4	14229-5	14229-6	14229-7
6	Data				
5	ISO 14229-2: Timings				
4, 3	**Diagnos-tics on CAN** 15765-2	**CoFR** 10681-2 **AUTOSAR TP**	**Diagnos-tics over IP** 13400-2		**Diagnos-tics on LIN** 17987-2
2, 1	**CAN** 11898	**FlexRay** 17458	**Ethernet** 13400-3 IEEE 802.3	**K-Line** ISO 14230	**LIN** 17987

VECTOR

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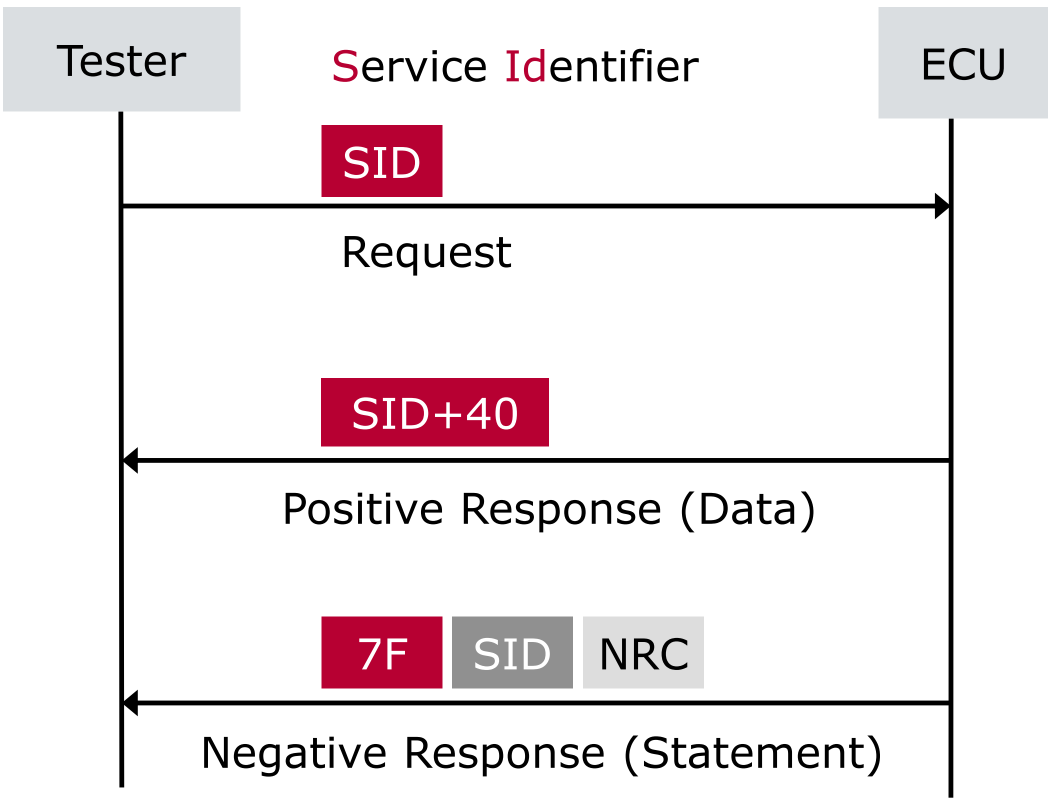
Services Request / Response Behavior



Principle:

Directed communication

Services can be diagnostic communication management requests, data requests, fault code requests, input/output control, security access, reprogramming requests, etc.

Protocol service values

Service identifier (hex value)	Service type	Where defined
00 – 0F	OBD service requests	ISO 15031-5
10 – 3E	ISO 14229-1 service requests	ISO 14229-1
3F	Not applicable	Reserved by document
40 – 4F	OBD service responses	ISO 15031-5
50 – 7E	ISO 14229-1 positive service responses	ISO 14229-1
7F	Negative response service identifier	ISO 14229-1
80	Not applicable	Reserved by ISO 14229-1
81 – 82	Not applicable	Reserved by ISO 14230
83 – 88	ISO 14229-1 service requests	ISO 14229-1
89 – 9F	Service requests	Reserved for future expansion as needed
A0 – B9	Service requests	Defined by vehicle manufacturer
BA – BE	Service requests	Defined by system supplier
BF	Not applicable	Reserved by document
C0	Not applicable	Reserved by ISO 14229-1
C1 – C2	Not applicable	Reserved by ISO 14230
C3 – C8	ISO 14229-1 positive service responses	ISO 14229-1
C9 – DF	Positive service responses	Reserved for future expansion as needed
E0 – F9	Positive service responses	Defined by vehicle manufacturer
FA – FE	Positive service responses	Defined by system supplier
FF	Not applicable	Reserved by document

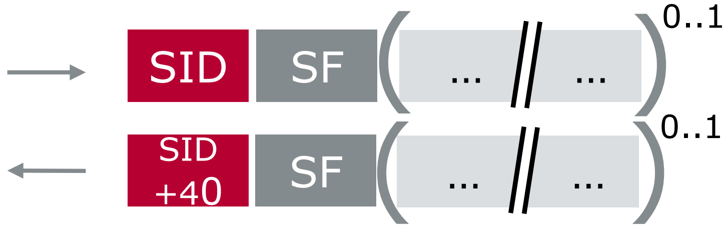
Service Identifier

($10) DiagnosticSessionControl
($11) EcuReset
($14) ClearDiagnosticInformation
($19) ReadDTCInformation
($22) ReadDataByIdentifier
($23) ReadMemoryByAddress
($24) ReadScalingDataByIdentifier
($27) SecurityAccess
($28) CommunicationControl
($2A) ReadDataByPeriodicIdentifier
($2C) DynamicallyDefineDataIdentifier
($2E) WriteDataByIdentifier
($2F) InputOutputControlByIdentifier
($31) RoutineControl
($34) RequestDownload
($35) RequestUpload
($36) TransferData
($37) RequestTransferExit
($38) RequestFileTransfer
($3D) WriteMemoryByAddress
($3E) TesterPresent
($83) AccessTimingParameter
($84) SecuredDataTransmission
($85) ControlDTCSetting
($86) ResponseOnEvent
($87) LinkControl

Total : 26

Subfunctions and Identifier

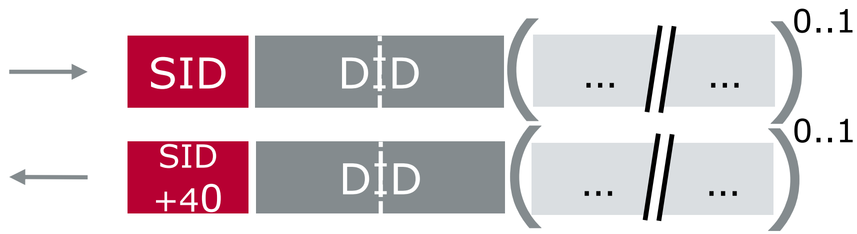
- Several UDS services support a subfunction (SF).



PosResponse repeats SF value of request

- Several additional UDS services support an identifier (DID)

to get access to data via a logical number (DID) which is used for the UDS communication.



PosResponse repeats DID value of request

There are 2 services which support both features:

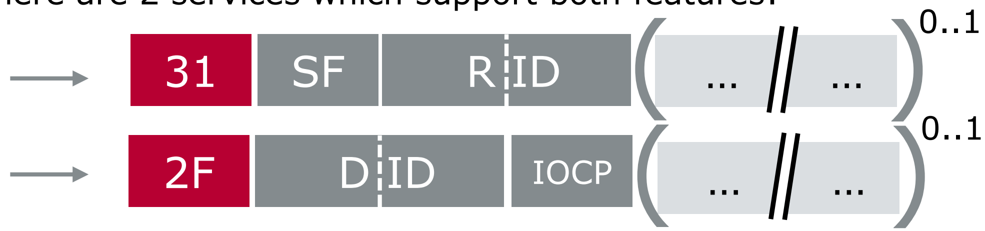
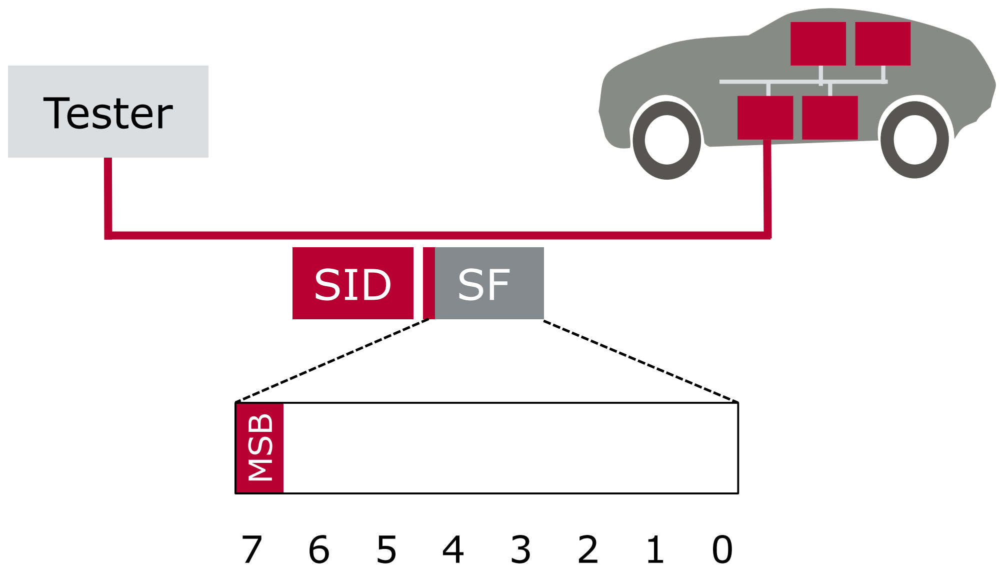


Abb.	Description
SF	Subfunction
DID	Data Identifier
RID	Routine Identifier
IOCP	Input Output Control Parameter (similar as sub function but no sub function according to definition)

Suppress Positive Response Message Indication Bit (SPRMIB)



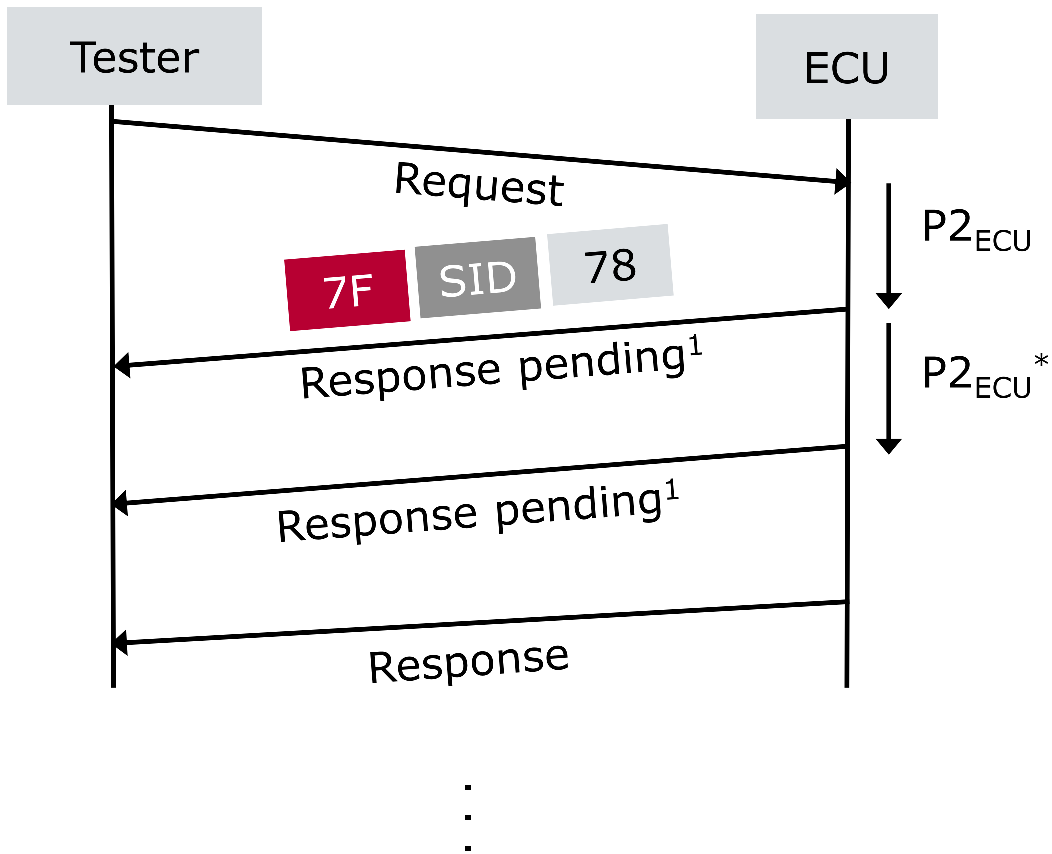
Value of Bit 7 of SF (SPRMIB)	Beschreibung
0	Positive Answer – not suppressed
1	Positive answer – suppressed

Abb.	Description
SPRMIB	Suppress Positive Response Message Indication Bit

For Rejecting a Request

NRC	Description
10	generalReject
11	serviceNotSupported
12	subFunctionNotSupported
13	incorrectMessageLengthOrInvalidFormat
22	conditionsNotCorrect
31	requestOutOfRange
7E	subFuntionNotSupportedInActiveSession
7F	serviceNotSupportedInActiveSession
...	...

Response Timeout Time P2 & P2*



[1]Request correctly received response pending

Classification of Protocol Services

- Diagnostic Management Functional Unit
  - e.g.: DiagnosticSessionControl (0x10)
  - e.g.: ECUReset (0x11)
- Data Transmission Functional Unit
  - e.g.: ReadDataByIdentifier (0x22)
  - e.g.: WriteDataByIdentifier (0x2E)
- Stored Data Transmission Functional Unit
  - e.g.: ReadDTCInformation (0x19)
  - e.g.: ClearDiagnosticInformation (0x14)
- Input/Output Control Functional Unit
  - e.g.: InputOutputControlByIdentifier (0x2F)
- Routine Functional Unit
  - e.g.: RoutineControl (0x31)
- Upload/Download Functional Unit
  - e.g.: RequestDownload (0x34)

Overview protocol services

($10) DiagnosticSessionControl
($11) EcuReset
($14) ClearDiagnosticInformation
($19) ReadDTCInformation
($22) ReadDataByIdentifier
($23) ReadMemoryByAddress
($24) ReadScalingDataByIdentifier
($27) SecurityAccess
($28) CommunicationControl
($2A) ReadDataByPeriodicIdentifier
($2C) DynamicallyDefineDataIdentifier
($2E) WriteDataByIdentifier
($2F) InputOutputControlByIdentifier
($31) RoutineControl
($34) RequestDownload
($35) RequestUpload
($36) TransferData
($37) RequestTransferExit
($38) RequestFileTransfer
($3D) WriteMemoryByAddress
($3E) TesterPresent
($83) AccessTimingParameter
($84) SecuredDataTransmission
($85) ControlDTCSetting
($86) ResponseOnEvent
($87) LinkControl

Focus on

6/26

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VECTOR

Principle and Implementation

Tester

ECU

Tester can execute the services which are supported in the active session.

possible modes

Default Session	22 ...	Start state. Tester can be connected.
Extended Session	22 2E 2F ...	Tester is connected. Use cases for development are supported,
Programming Session	22 2E ...	Tester is connected. ECU is in boot loader and supports reprogramming.

Recommended Session(s) for Services

Service	defaultSession	non-defaultSession
DiagnosticSessionControl – 0x10	x	x
ECUReset – 0x11	x	x
SecurityAccess – 0x27	not applicable	x
CommunicationControl – 0x28	not applicable	x
TesterPresent – 0x3E	x	x
AccessTimingParameter – 0x83	not applicable	x
SecuredDataTransmission – 0x84	not applicable	x
ControlDTCSetting – 0x85	not applicable	x
ResponseOnEvent – 0x86	x[a]	x
LinkControl – 0x87	not applicable	x
ReadDataByIdentifier – 0x22	x[b]	x
ReadMemoryByAddress – 0x23	x[c]	x
ReadScalingDataByIdentifier – 0x24	x[b]	x
ReadDataByPeriodicIdentifier – 0x2A	not applicable	x
DynamicallyDefineDataIdentifier – 0x2C	x[d]	x
WriteDataByIdentifier – 0x2E	x[b]	x
WriteMemoryByAddress – 0x3D	x[c]	x
ClearDiagnosticInformation – 0x14	x	x
ReadDTCInformation – 0x19	x	x
InputOutputControlByIdentifier – 0x2F	not applicable	x
RoutineControl – 0x31	x[e]	x
RequestDownload – 0x34	not applicable	x
RequestUpload – 0x35	not applicable	x
TransferData – 0x36	not applicable	x
RequestTransferExit – 0x37	not applicable	X
RequestFileTransfer – 0x38	not applicable	X

[a] It is implementation specific whether the ResponseOnEvent service is also allowed during the defaultSession.

[b] Secured dataIdentifiers require a SecurityAccess service and therefore a non-default diagnostic session.

[c] Secured memory areas require a SecurityAccess service and therefore a non-default diagnostic session.

[d] A dataIdentifier can be defined dynamically in the default and non-default diagnostic session.

[e] Secured routines require a SecurityAccess service and therefore a non-default diagnostic session. A routine that requires to be stopped actively by the client also requires a non-default session.

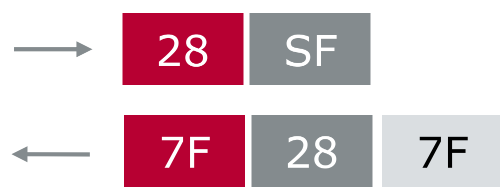
VECTOR

Negative Response Codes for Session Violation, NRC 7F

NRC	Description
31	requestOutOfRange
7E	subFunctionNotSupportedInActiveSession
7F	serviceNotSupportedInActiveSession

Example:

- Service 28 Communication Control is usually not supported in Default session but in Extended session.
- If the ECU is in Default session:



VECTOR

Negative Response Codes for Session Violation, NRC 7E

NRC	Description
31	requestOutOfRange
7E	subFunctionNotSupportedInActiveSession
7F	serviceNotSupportedInActiveSession

Example:

- 10 02 shall be prohibited in „Default“ session.
- 10 03 shall be allowed in „Default“ session.
- If the ECU is in Default session:

→ 10 02

← 7F 10 7E

VECTOR

Negative Response Codes for Session Violation, NRC 31

NRC	Description
31	requestOutOfRange
7E	subFunctionNotSupportedInActiveSession
7F	serviceNotSupportedInActiveSession

Example:

- „Hardware Version“ is readable in „Programming“ session.
- But „Vehicle Speed“ is not readable in „Programming“ session.
- If the ECU is in Programming session:

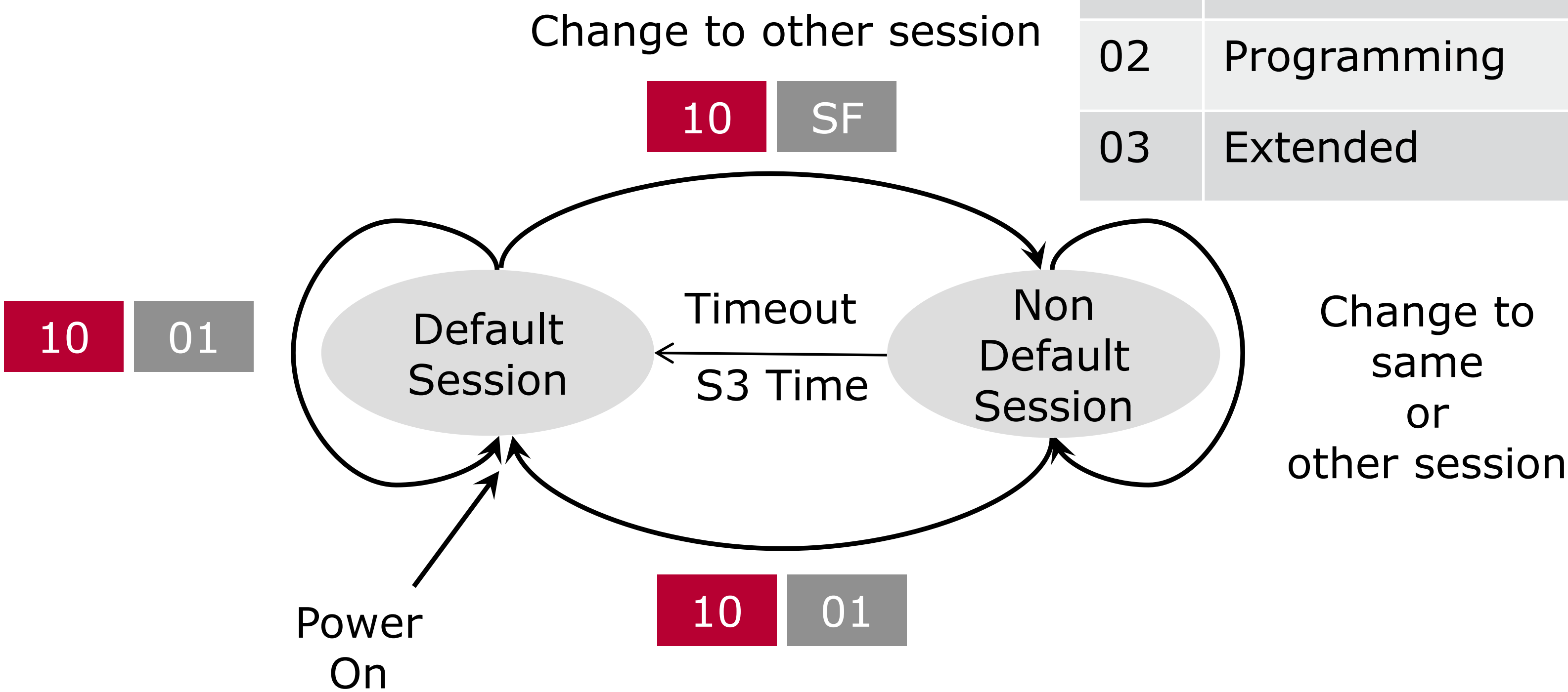
→	22	C0	01
←	7F	22	31

C0 01 is just an example value for vehicle speed.

So it depends on the service structure and the context which NRC is sent for a session violation.

Session State Handling

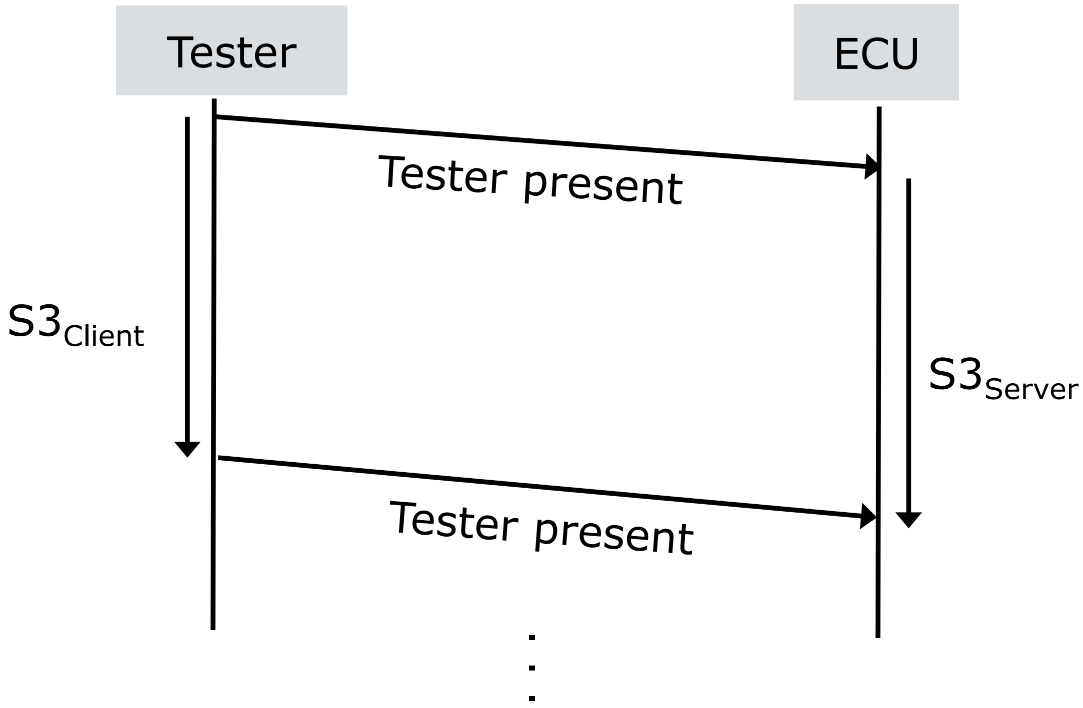
SF	Session
01	Default
02	Programming
03	Extended



Non Default Sessions are timeout controlled

Tester present 3E or another request gets session active

Session Timeout Time S3



$S3_{Cleint}$ < $S3_{Server}$ due to Gateways etc.

Request/Response Format

Request

Data byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	**DiagnosticSessionControl Request Service Id**	**M**	**10**	**DSC**
#2	sub-function = [diagnosticSessionType]	M	00-FF	LEV_ DS_

Hex	Description	Cvt	Mnemonic
00	**ISOSAEReserved**	M	ISOSAERESRVD
01	**defaultSession**	M	DS
02	**programmingSession**	U	PRGS
03	**extendedDiagnosticSession**	U	EXTDS
...	...	...	...

Response

Data byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	**DiagnosticSessionControl Response Service Id**	M	**50**	**DSCPR**
#2	diagnosticSessionType	M	00-FF	LEV_DS_
#3 : #6	sessionParameterRecord[] #1 = [data#1 : data#4]	M : M	00-FF : 00-FF	SPREC_ DATA_1 : DATA_4

Supported NRC

NRC	Description	Mnemonic
0x12	**sub-functionNotSupported** This NRC shall be sent if the sub-function parameter is not supported.	SFNS
0x13	**incorrectMessageLengthOrInvalidFormat** This NRC shall be sent if the length of the message is wrong.	IMLOIF
0x22	**conditionsNotCorrect** This NRC shall be returned if the criteria for the request DiagnosticSessionControl are not met.	CNC

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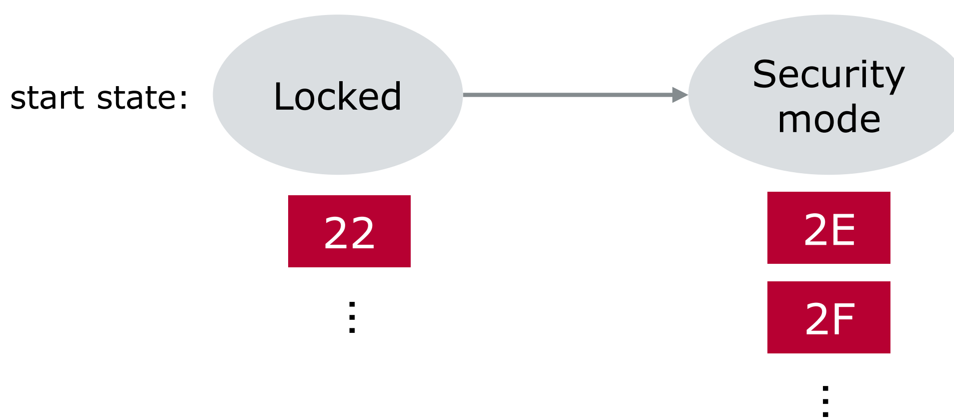
Security Access

Tester wants to execute a protected service.

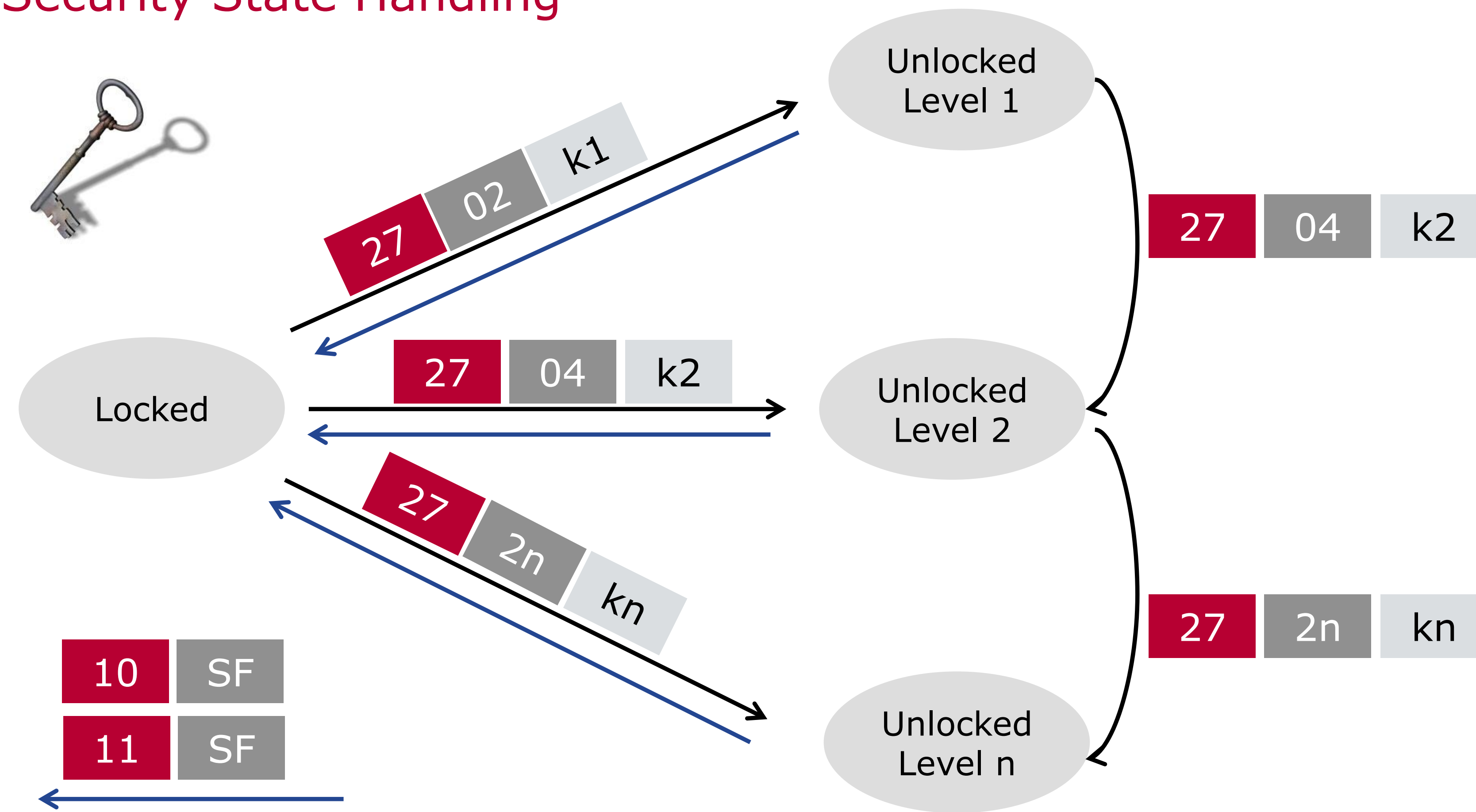
Examples for protected services:

- Write (2E)
- I/O Control (2F)
- Reprogramming services

Tester needs to unlock a security mode (often called level) of the ECU.



Security State Handling

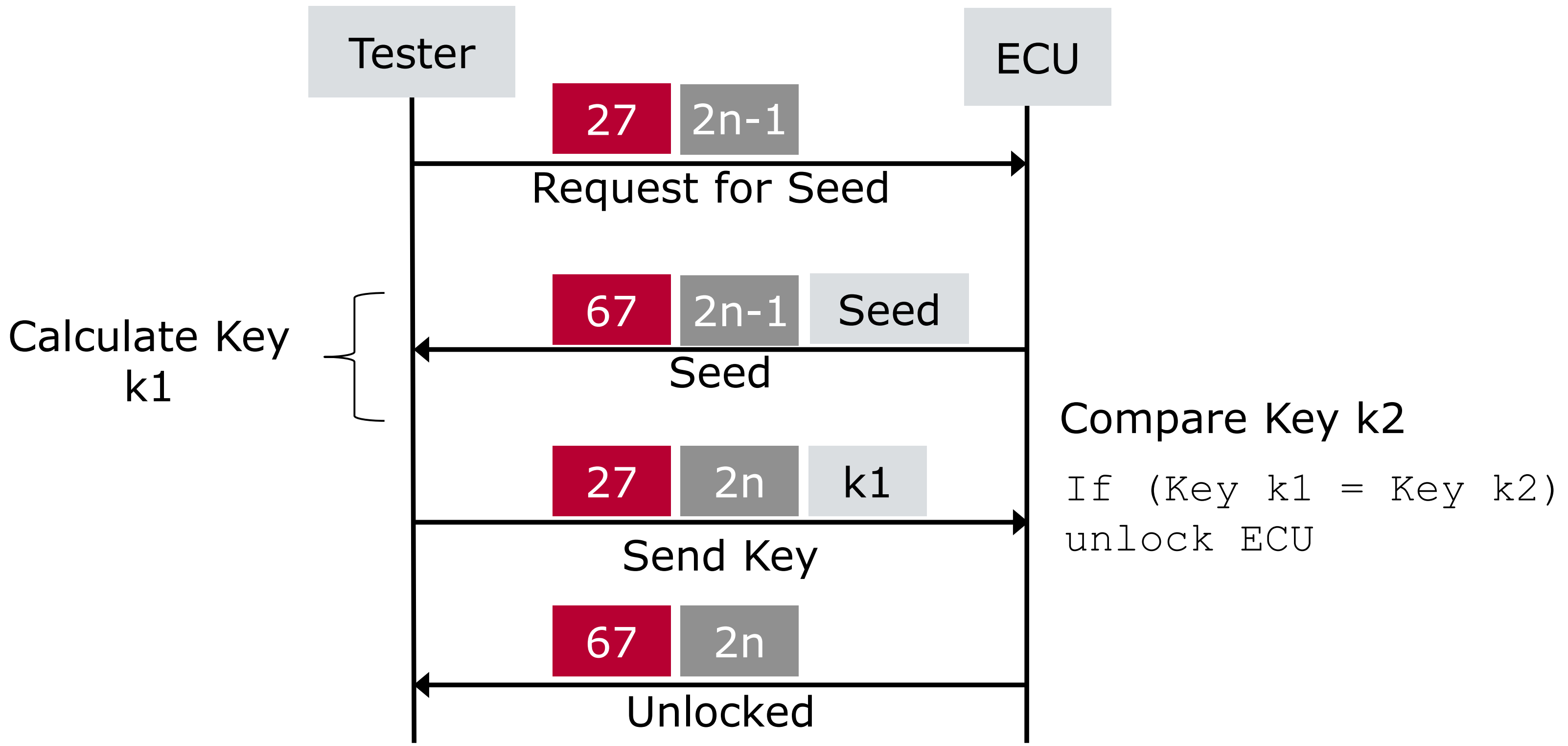


- Only one security level shall be active at any instant of time.
- After PowerOn/Reset the ECU is in locked state.
- If a server is already unlocked when a RequestSeed message is received, server shall respond a positive response message with a seed value 0x00.

VECTOR

Seed and Key

- Some diagnostic data/ -services can be restricted due to safety reasons. To get access to these ECUs a Seed&Key process is used for level n:



Step1 : Request Seed

Request seed

Data byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	**SecurityAccessRequest SId**	**M**	**27**	**SA**
#2	SecurityAccessType = [RequestSeed]	M	01	SAT_RSD

Response seed

Data byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	**SecurityAccess Response SId**	**M**	**67**	**SAPR**
#2	SecurityAccessType = [RequestSeed]	M	01	SAT_SK
#3	SecuritySeed [] = [seed#1(high byte)	M	AA	SEED1HB
#4	seed#2	M	BB	SEED2
#5	seed#3	M	CC	SEED3
#6	seed#4(low byte)]	M	DD	SEED4LB

Step2 : Send Key

Send key

Data byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	**SecurityAccess Request SId**	**M**	**27**	**SA**
#2	SecurityAccessType = [SendKey]	M	02	SAT_SK
#3	SecurityKey [] = [key#1(high byte)	M	11	KEY1HB
#4	key#2	M	22	KEY2
#5	key#3	M	33	KEY3
#6	key#4(low byte)]	M	44	KEY4LB

Response

Data byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	**Positive Response Service Id**	**M**	**67**	**SAPR**
#2	SecurityAccessType = [SendKey]	M	02	SAT_SK

Request Seed When Unlocked level 1

Request seed

Data byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	**SecurityAccessRequest SId**	**M**	**27**	**SA**
#2	SecurityAccessType = [RequestSeed]	M	01	SAT_RSD

Response seed

Data byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	**SecurityAccess Response SId**	**M**	**67**	**SAPR**
#2	SecurityAccessType = [RequestSeed]	M	01	SAT_SK
#3	SecuritySeed [] = [seed#1(high byte)	M	00	SEED1HB
#4	seed#2	M	00	SEED2
#5	seed#3	M	00	SEED3
#6	seed#4(low byte)]	M	00	SEED4LB

Supported NRC

NRC	Description	Mnemonic
0x12	**sub-functionNotSupported** This NRC shall be sent if the sub-function parameter is not supported.	SFNS
0x13	**incorrectMessageLengthOrInvalidFormat** This NRC shall be sent if the length of the message is wrong.	IMLOIF
0x22	**conditionsNotCorrect** This NRC shall be returned if the criteria for the request SecurityAccess are not met.	CNC
0x24	**requestSequenceError** Send if the ‘sendKey’ sub-function is received without first receiving a ‘requestSeed’ request message.	RSE
0x31	**requestOutOfRange** This NRC shall be sent if the user optional securityAccessDataRecord contains invalid data.	ROOR
0x35	**invalidKey** Send if an expected 'sendKey' sub-function value is received and the value of the key does not match the server's internally stored/calculated key.	IK
0x36	**exceededNumberOfAttempts** Send if the delay timer is active due to exceeding the maximum number of allowed false access attempts.	ENOA
0x37	**requiredTimeDelayNotExpired** Send if the delay timer is active and a request is transmitted.	RTDNE

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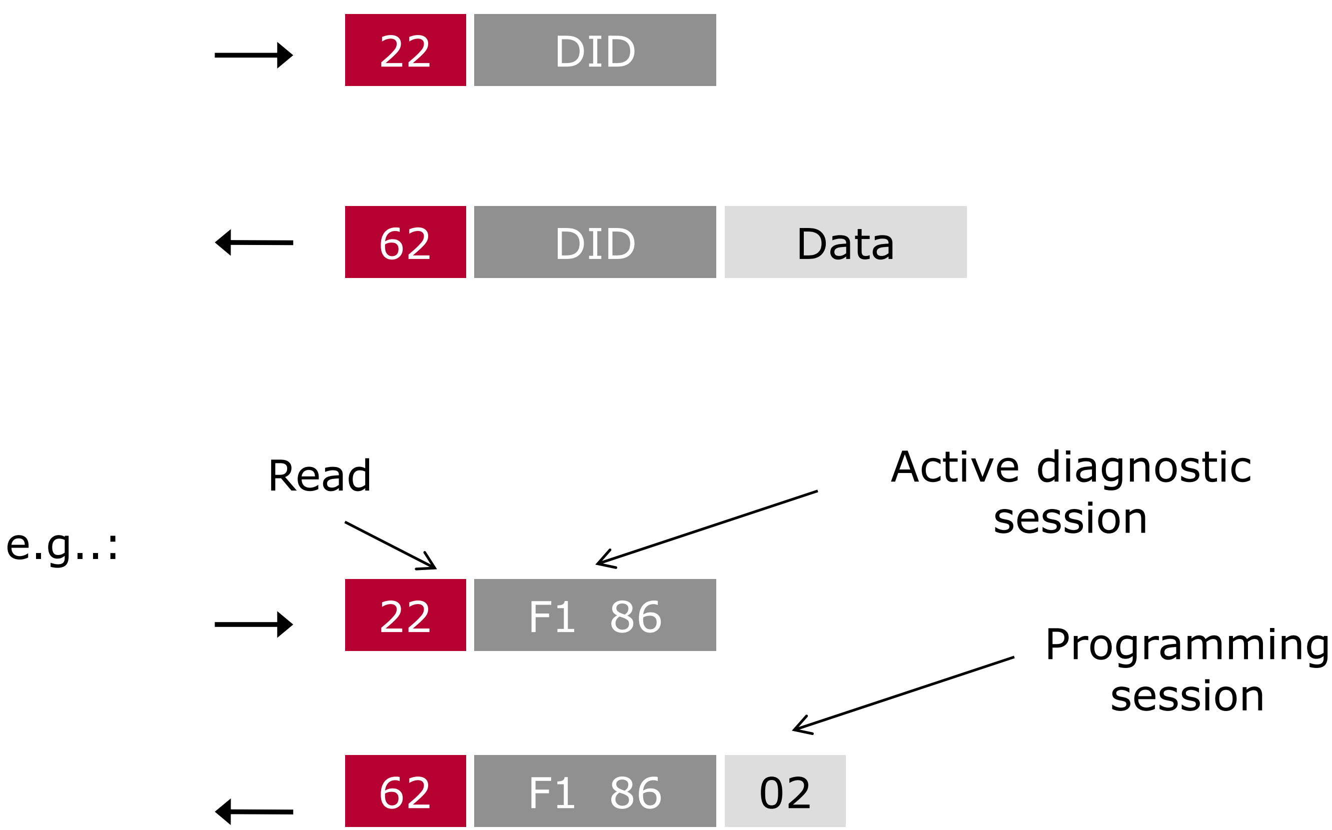
Request/Response Format



VECTOR

DID data-parameter definitions

0xF186	**ActiveDiagnosticSessionDataIdentifier** This value shall be used to report the active diagnostic session in the server. The values are defined by the diagnosticSessionType subfunction parameter in the DiagnosticSessionControl service.	U	ADSDID
0xF187	**vehicleManufacturerSparePartNumberDataIdentifier** This value shall be used to reference the vehicle manufacturer spare part number. Record data content and format shall be server specific and defined by the vehicle manufacturer.	U	VMSPNDID
0xF188	**vehicleManufacturerECUSoftwareNumberDataIdentifier** This value shall be used to reference the vehicle manufacturer ECU (server) software number. Record data content and format shall be server specific and defined by the vehicle manufacturer.	U	VMECUSNDID
0xF189	**vehicleManufacturerECUSoftwareVersionNumberDataIdentifier** This value shall be used to reference the vehicle manufacturer ECU (server) software version number. Record data content and format shall be server specific and defined by the vehicle manufacturer.	U	VMECUSVNDID
0xF18A	**systemSupplierIdentifierDataIdentifier** This value shall be used to reference the system supplier name and address information. Record data content and format shall be server specific and defined by the system supplier.	U	SSIDDID

For more information:

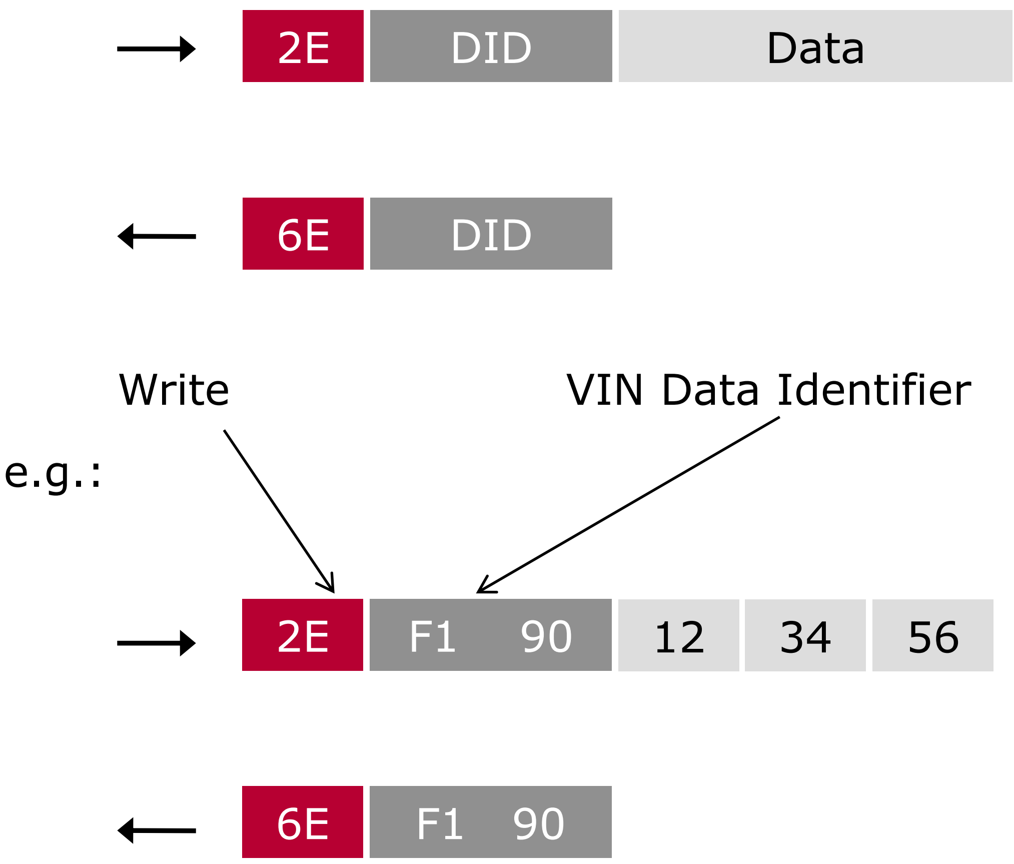
ISO 14229-1:2013(E) Annex C Data transmission functional unit data-parameter definitions C.1 DID parameter definitions

VECTOR

Supported NRC

NRC	Description	Mnemonic
0x13	**incorrectMessageLengthOrInvalidFormat** This NRC shall be sent if the length of the request message is invalid or the client exceeded the maximum number of dataIdentifiers allowed to be requested at a time.	IMLOIF
0x14	**responseTooLong** This NRC shall be sent if the total length of the response message exceeds the limit of the underlying transport protocol (e.g., when multiple DIDs are requested in a single request).	RTL
0x22	**conditionsNotCorrect** This NRC shall be sent if the operating conditions of the server are not met to perform the required action.	CNC
0x31	**requestOutOfRange** This NRC shall be sent if none of the requested dataIdentifier values are supported by the device; none of the requested dataIdentifiers are supported in the current session; the requested dynamicDefinedDataIdentifier has not been assigned yet;	ROOR
0x33	**securityAccessDenied** This NRC shall be sent if at least one of the dataIdentifiers is secured and the server is not in an unlocked state.	SAD

Request/Response Format



Supported NRC

NRC	Description	Mnemonic
0x13	**incorrectMessageLengthOrInvalidFormat** This NRC shall be sent if the length of the message is wrong.	IMLOIF
0x22	**conditionsNotCorrect** This NRC shall be sent if the operating conditions of the server are not met to perform the required action.	CNC
0x31	**requestOutOfRange** This NRC shall be sent if: — The dataIdentifier in the request message is not supported in the server or the dataIdentifier is supported for read only purpose (via ReadDataByIdentifier service); — Any data transmitted in the request message after the dataIdentifier is invalid (if applicable to the node);	ROOR
0x33	**securityAccessDenied** This NRC shall be sent if the dataIdentifier, which reference a specific address, is secured and the server is not in an unlocked state.	SAD
0x72	**generalProgrammingFailure** This NRC shall be returned if the server detects an error when writing to a memory location.	GPF

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VECTOR

Concept--What is a fault?

- If a system detects a fault, it is stored by means of a **Diagnostic Trouble Code** ——DTC
- A DTC represents:
  - a Malfunction, which is observable
  - And/Or: A Repair instruction is available (e.g. indication for maintenance or wearing)
  - a legal Fault
    - Emission related fault
    - in future evtl. safety relevant faults
- A fault always refers to a ECU, a fault location and a fault type.

Technical coding schemes for a Fault Path

Fault Path: ECU → Fault location → Fault type

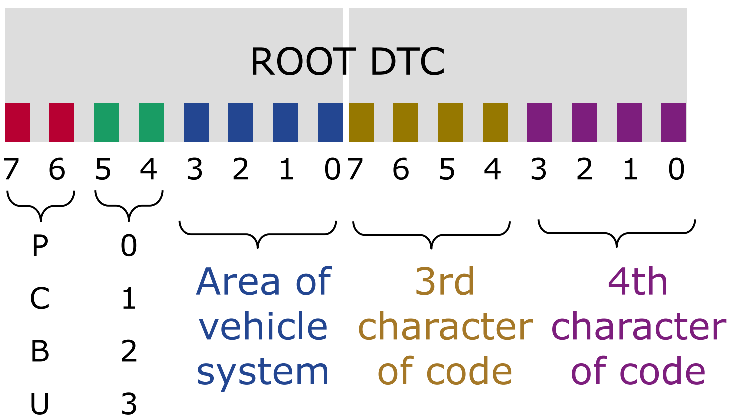
Coding schemes for a Fault Path:

UDS general	DTC	
UDS with ISO 15031-6 / SAE J2012-DA	Root DTC	FTB
WWH-OBD in ISO 27145-2	Root DTC	FTB
OBD II	DTC	
J1939-73	SPN	FMI

95% of the cases for UDS

Root DTC in UDS with ISO 15031-6 / SAE J2012-DA

Fault location



Value	Description
P00-P02	Fuel and air metering
P03	Ignition system or misfire
P04	Auxiliary emission controls
P05	Vehicle speed, idle control, and auxiliary inputs
P06	Computer and auxiliary outputs
P07-P09	Transmission
P0A-P0E	Hybrid Propulsion

Value	Description
00	Powertrain (P)
01	Chassis (C)
10	Body (B)
11	Network (U)

Value	Description
00	ISO/SAE controlled
01	Manufacturer controlled
10	ISO/SAE controlled
11	ISO/SAE controlled

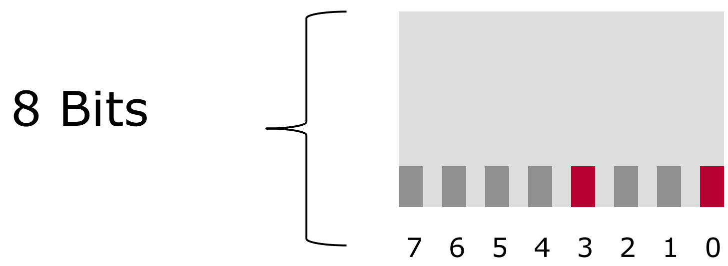
0x19 ReadDTCInformation

- A selection of fault memory services

4/28

19	02	Reading of DTCs (via DTC status mask)
19	04	Reading of Snapshot data (environmental data when the error occured)
19	06	Reading of ExtendedDataRecords (environmental data when the service is executed)
19	0A	Reading of supported DTCs

DTC Status Byte



Bit	Description
0	TestFailed
1	TestFailedThisOperationsCycle
2	PendingDTC
3	ConfirmedDTC
4	TestNotCompletedSinceLastClear
5	TestFailedSinceLastClear
6	TestNotCompletedThisOperationCycle
7	WarningIndicatorRequested

For more information:

ISO 14229-1:2013(E) Annex D Stored data transmission functional unit data-parameter definitions D.2 DTCStatusMask and statusOfDTC bit definitions

VECTOR

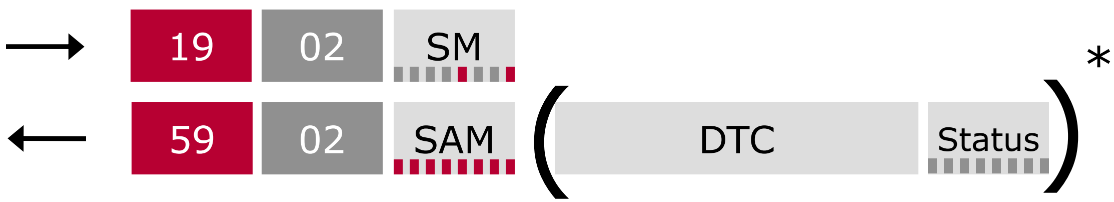
DTC Status

- DTC Status is 1 Byte which describes states for a DTC
  - Error occured („test failed...“)
  - Error was confirmed („confirmed“)
  - ...

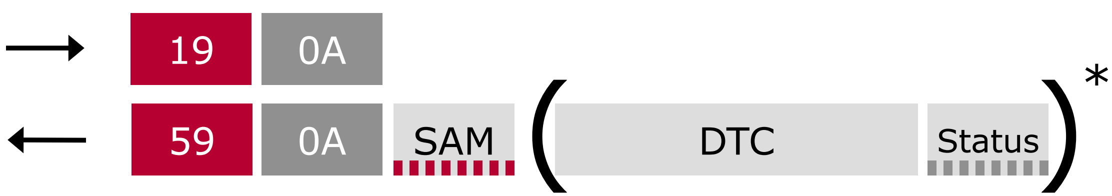
- **Note: Not all DTC Status bits are always supported**
- 19 02 uses 3 different types
  - In the Request: DTC Status Mask (default values for the test device)
  - In the Positive Response: DTC Status Availability Mask (for ECU specification)
  - For each DTC in Positive Reponse: To interprete DTC Status

Reading of DTCs

- Service to read the DTCs according the status mask



- Service to read all supported DTCs



Value	Description
SM	StatusMask
SAM	StatusAvailabilityMask

Example for reportDTCByStatusMask(0x19 0x02)

- Assumptions:
  - DTC P0A9B-17 Hybrid Battery Temperature Sensor - circuit voltage above threshold (0x0A9B17), statusOfDTC 0x24 (0010 0100 b).
  - DTC P2522-1F A/C Request "B" - circuit intermittent (0x25221F), statusOfDTC of 0x60 (0110 0000 b).
  - DTC P0805-11 Clutch Position Sensor - circuit short to ground (0x080511), statusOfDTC of 0x2F (0010 1111 b).

Request:

Message direction	**client → server**		
Message Type	**Request**		
A_Data Byte	**Description (all values are in hexadecimal)**	**Byte Value**	**Mnemonic**
#1	ReadDTCInformation Request SID	0x19	RDTCI
#2	sub-function = reportDTCByStatusMask, suppressPosRspMsgIndicationBit = FALSE	0x02	RDTCBSM
#3	DTCStatusMask	0x84	DTCSM

Example for reportDTCByStatusMask(0x19 0x02)

Response:

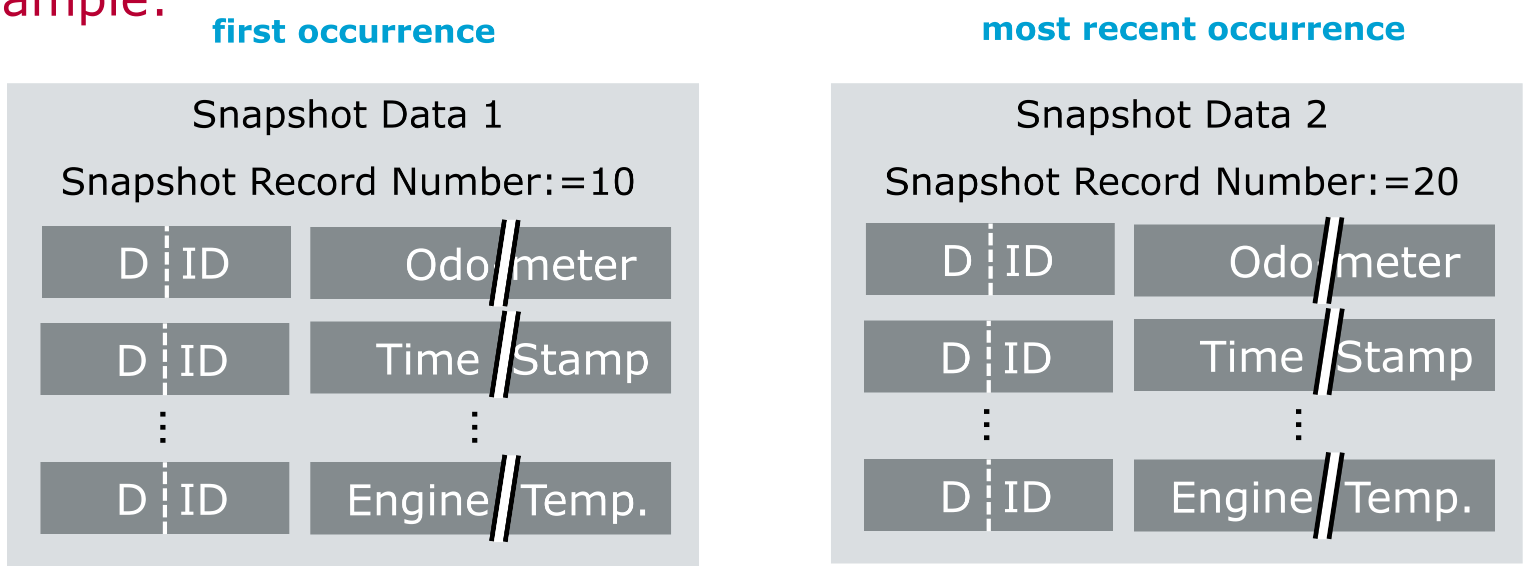
Message direction	**server → client**		
Message Type	**Response**		
A_Data Byte	**Description (all values are in hexadecimal)**	**Byte Value**	**Mnemonic**
#1	ReadDTCInformation Response SID	0x59	RDTCIPR
#2	reportType = reportDTCByStatusMask	0x02	RDTCBSM
#3	DTCStatusAvailabilityMask	0x7F	DTCSAM
#4	DTCAndStatusRecord#1 [DTCHighByte]	0x0A	DTCHB
#5	DTCAndStatusRecord#1 [DTCMiddleByte]	0x9B	DTCMB
#6	DTCAndStatusRecord#1 [DTCLowByte]	0x17	DTCLB
#7	DTCAndStatusRecord#1 [statusOfDTC]	0x24	SODTC
#8	DTCAndStatusRecord#2 [DTCHighByte]	0x08	DTCHB
#9	DTCAndStatusRecord#2 [DTCMiddleByte]	0x05	DTCMB
#10	DTCAndStatusRecord#2 [DTCLowByte]	0x11	DTCLB
#11	DTCAndStatusRecord#2 [statusOfDTC]	0x2F	SODTC

VECTOR

Environment Data in UDS: Snapshots → 19 04

For each DTC FTB **exactly** one DID list for **all** UDS Snapshots (SRNr != 0) of the DTCs is defined.

Example:



Only DIDs are allowed in snapshot records.

Additional snapshot records can be defined as well. **But they all need to store the same parameters (except snapshot record 0)**

Reading of Snapshots (Freeze Frames) in UDS

- Reading of DTC Snapshot Record By DTC Number

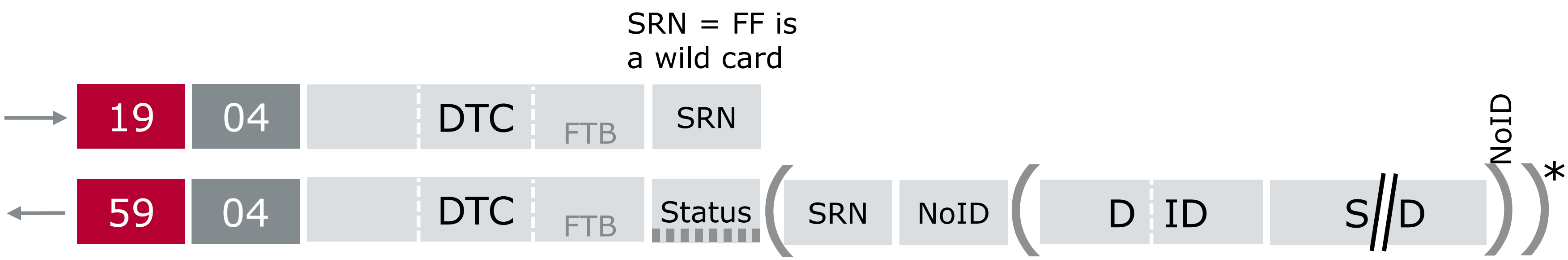


Abb.	Description
DTC	Diagnostic Trouble Code
SRN	Snapshot Record Number (usually means instance of time)
NoID	Number of Identifiers, i.e. number of DIDs
DID	Data ID
SD	Snapshot Data

If an invalid SRN is requested NRC 31 is sent. If FF is requested and there are no snapshots the list is empty.

Statistic Data in UDS: Extended Data Records → 19 06

For an ECU for each EDR No. one data layout is defined globally (usually independent of DTCs)

Example:

Extended Data Record 1

Extended Data Record Number:=10

Occurren-ce Counter

Aging Counter

Extended Data Record 2

Extended Data Record Number:=20

Fault Detec-tion Counter

For each DTC FTB a list of supported extended data record numbers (and therefore the extended data records) is defined.

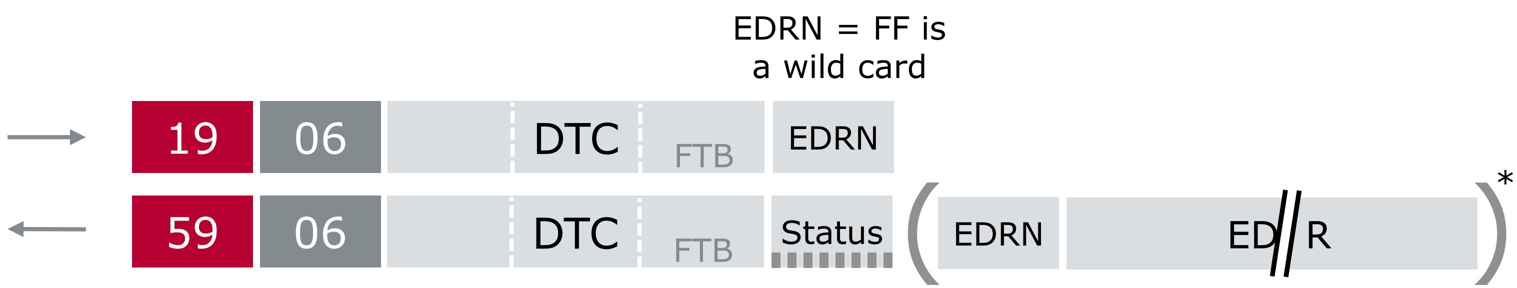
Several extended data records can be defined.

Different extended data records have different parameters.

The parameters do not need to be part of a DID.

Reading of Extended Data Records in UDS

- Reading of DTC Ext Data Record By DTC Number

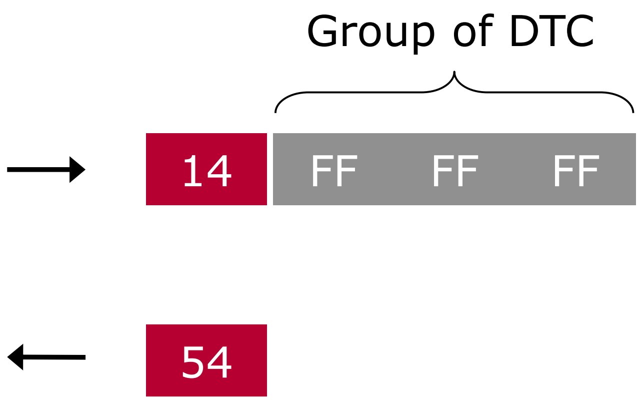


If an invalid EDRN is requested NRC 31 is sent. If FF is requested and there are no extended data records the list is empty.

Abb.	Description
DTC	Diagnostic Trouble Code
EDRN	Extended Data Record Number
EDR	Extended Data Record

0x14 ClearDiagnosticInformation

- The service 14 ClearDiagnosticInformation resets the DTCs (it changes the DTC Status)



- The 3 bytes „Group of DTC" are a mask to choose a subgroup of DTCs
  - FF FF FF means all DTCs

Group Of DTC

Byte Value	Description	Cvt	Mnemonic
0x000000 – 0x0000FF	This range of values is reserved for future legislative requirements.	M	RFLU
to be determined by vehicle manufacturer	Powertrain Group: engine and transmission	U	PG
	Powertrain DTCs	U	PDTC_
	Chassis Group	U	CG
	Chassis DTCs	U	CDTC_
	Body Group	U	BG
	Body DTCs	U	BDTC_
	Network Communication Group	U	NCG
	Network Communication DTCs	U	NCDTC_
0xFFFF00 – 0xFFFFFE	The lower byte shall always be the FunctionalGroupIdentifier as defined in Table D.15. For example, a value of 0xFFFF33 shall equal the Emissions Group and a value of 0xFFFFD0 shall equal the Safety Group.	M	ISOSAERESRVD
0xFFFFFF	All Groups (all DTCs)	M	AG

Agenda

Diagnostic Overview

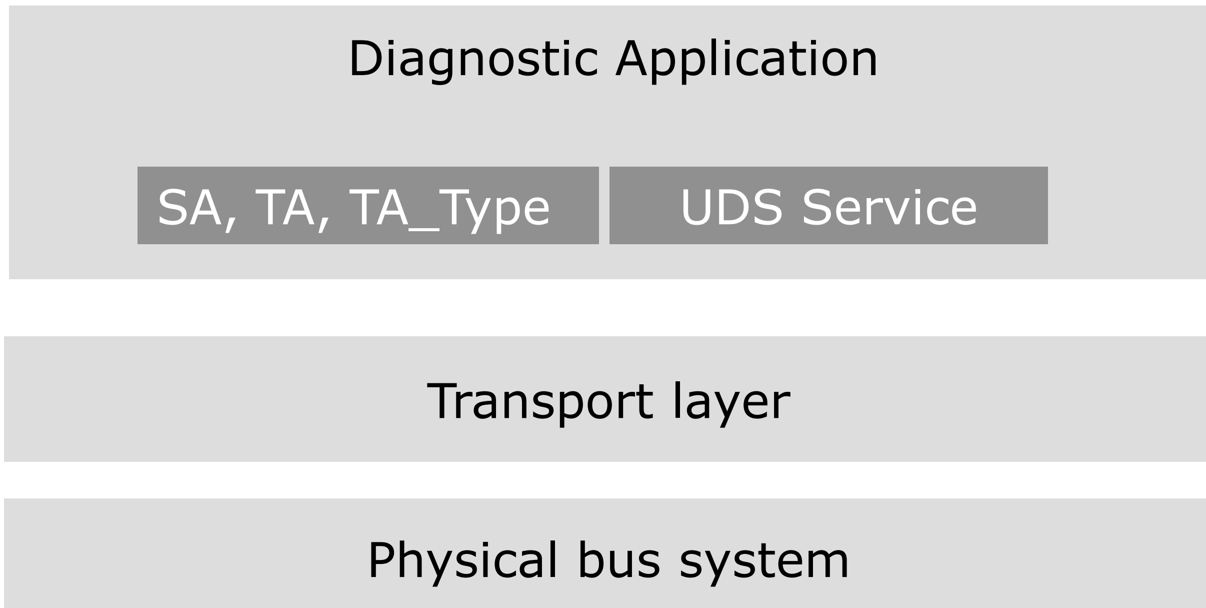
Diagnostic Protocol UDS

- Services Overview
- Diagnostic Session Control
- Security Access
- Read/Write Data By Identifier
- Fault Memory
- **Communication Method in Diagnostics**
- Review/Summary

Vector Diagnostic Solution

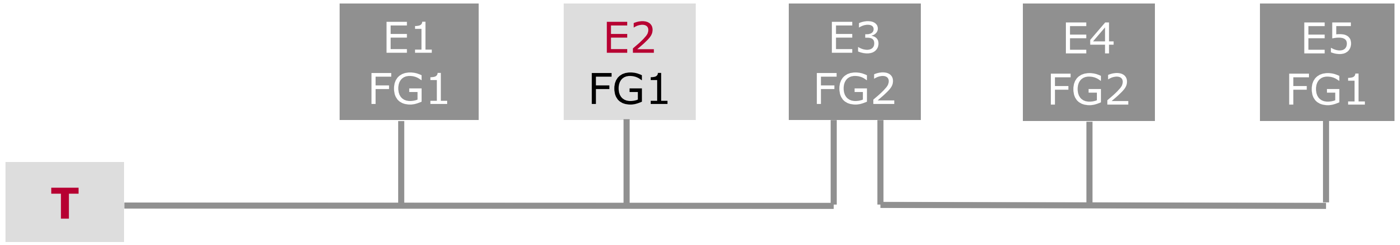
- Solution
- Tool-chain
- Summary

Communication Method



Physical Addressing

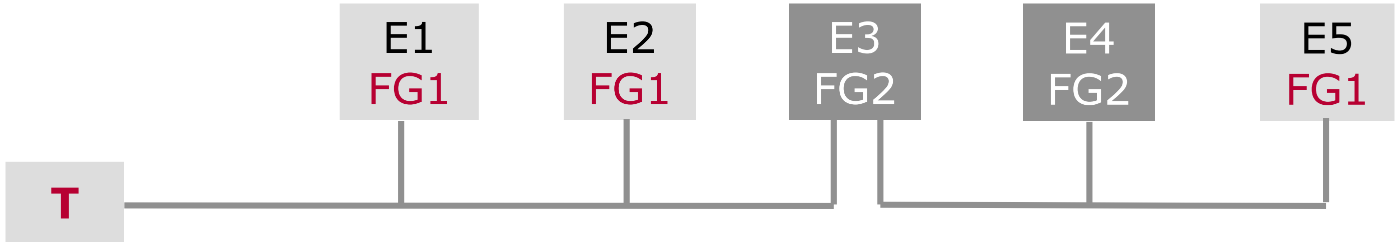
- Tester T → ECU



(SA, TA, TA_Type) = (T, E2, physical)

Functional Addressing

- Tester → Functional Group (broadcast/groupcast)



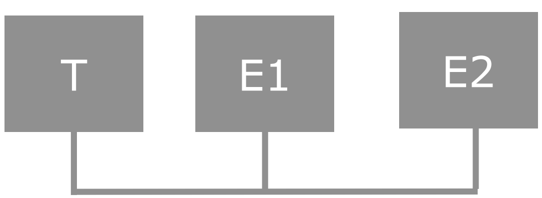
(SA, TA, TA_Type) = (T, FG1, functional)

VECTOR

Normal Addressing

- Addressing information (SA, TA, TA_Type) is mapped to 1 CAN-ID

- Two CAN IDs per ECU for physical addressing: Request/Response
- One CAN ID per ECU for functional addressing: Request

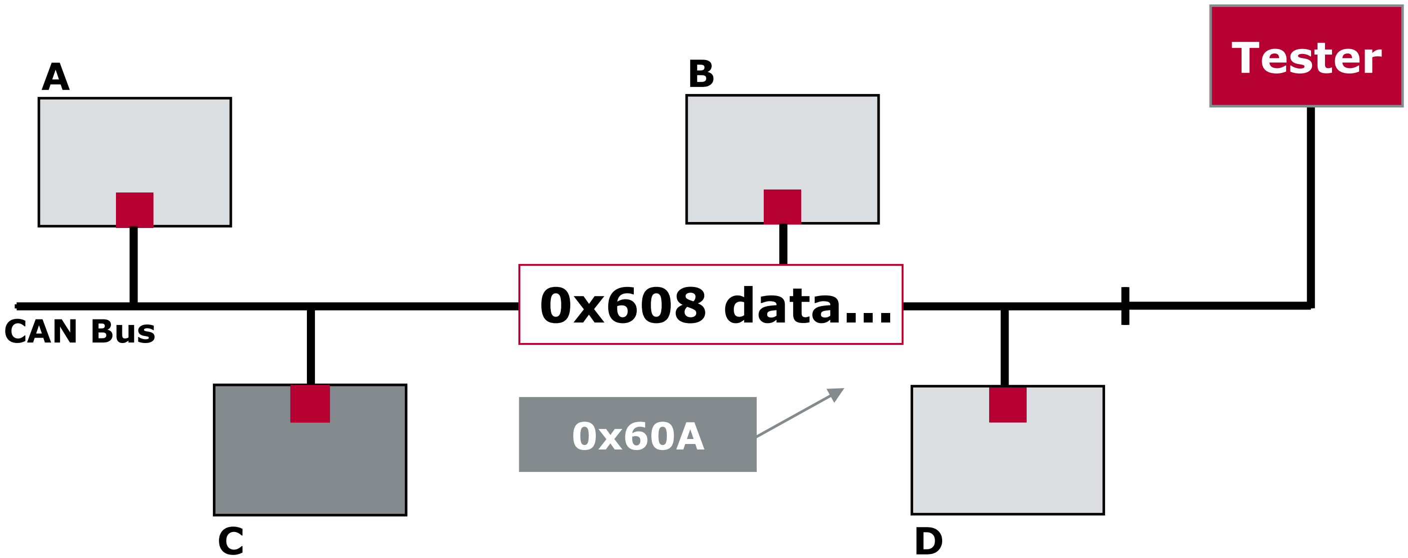


CAN-IDs
T, E1, phys
E1,T, phys
T, E2, phys
E2,T, phys
T, FG, func

Physical — Normal Addressing

PeerToPeer connections from Tester to an ECU on a broadcast based bus protocol

Name	ID	Sender	Receiver
DiagRequest_A	0x600	Tester	ECU_A
DiagRespone_A	0x602	ECU_A	Tester
DiagRequest_B	0x604	Tester	ECU_B
DiagResponse_B	0x606	ECU_B	Tester
DiagRequest_C	0x608	Tester	ECU_C
DiagResponse_C	0x60A	ECU_C	Tester
DiagRequest_D	0x60C	Tester	ECU_D
DiagResponse_D	0x60E	ECU_D	Tester

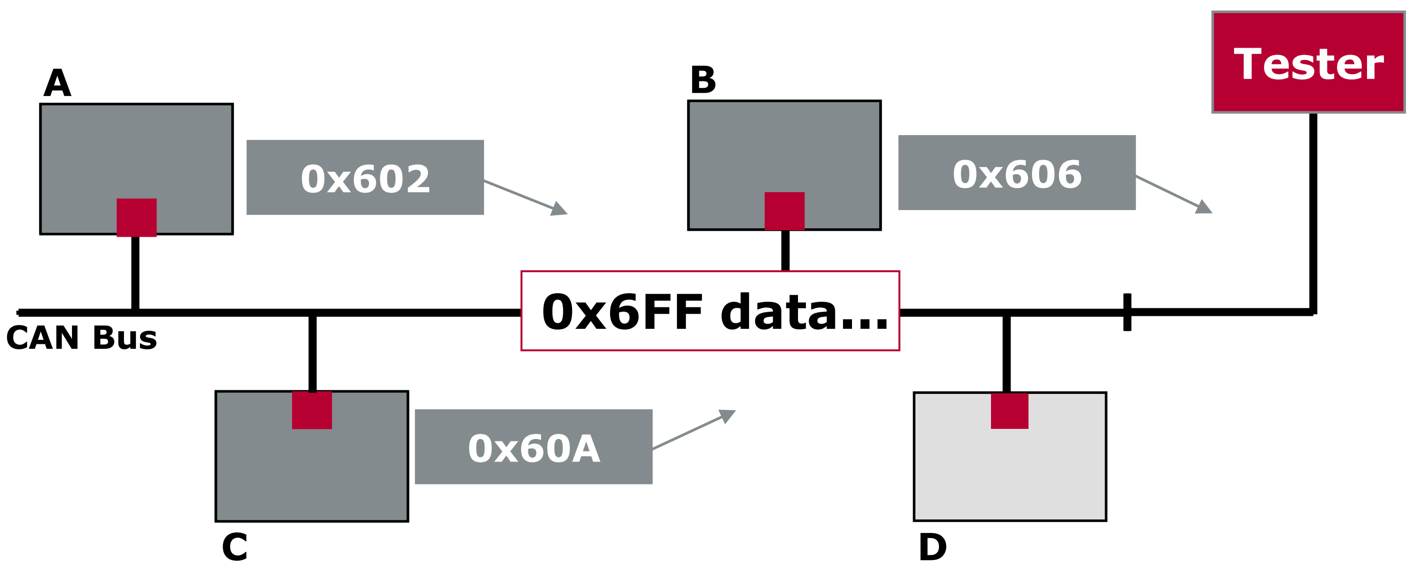


VECTOR

Functional — Normal Addressing

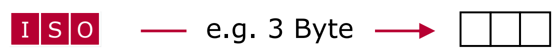
From Tester to
more ECUs on a broadcast bus protocol

Name	ID	Sender	Receiver
DiagRangeRequest	0x6FF	Tester	**ECU_A, B, C**
DiagResponse_A	0x602	ECU_A	**Tester**
DiagResponse_B	0x606	ECU_B	**Tester**
DiagResponse_C	0x60A	ECU_C	**Tester**
DiagResponse_D	0x60E	ECU_D	**Tester**
			Tester

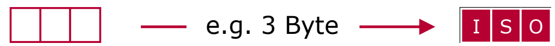


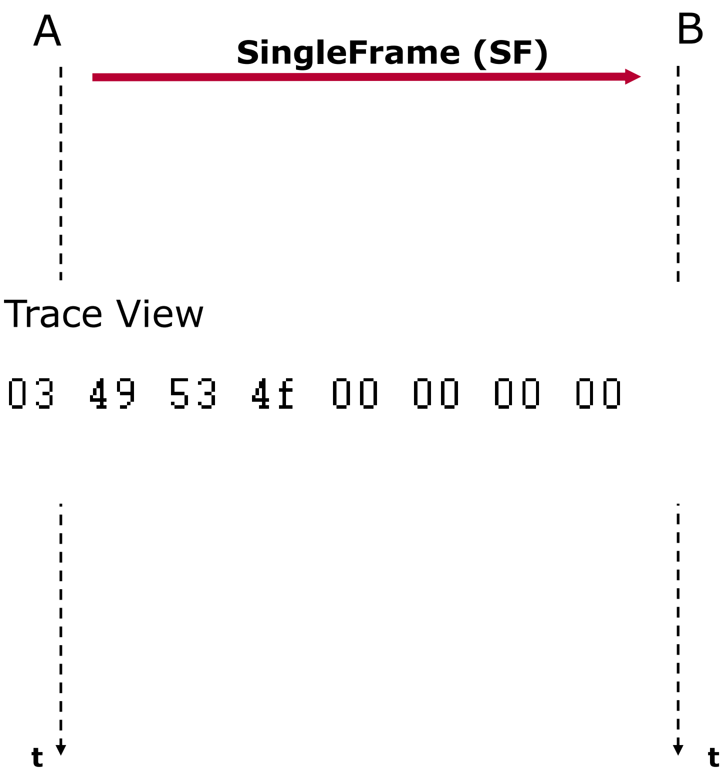
Unsegmented Message Transmission

Unsegmented message transmission

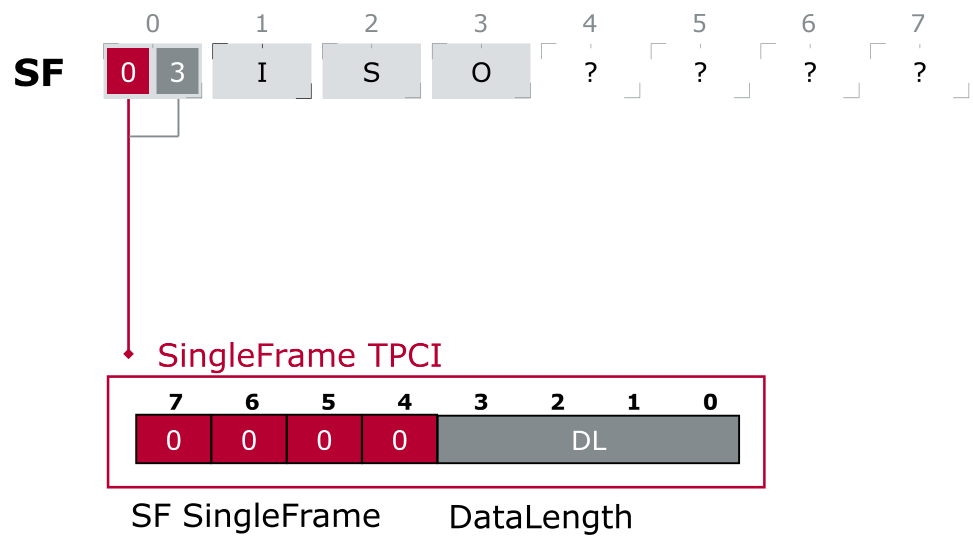


SingleFrame – Data Fits in One Message



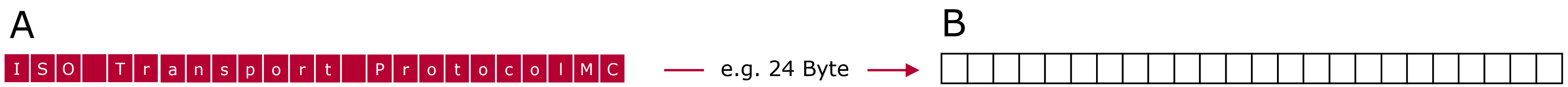


SingleFrame (SF)



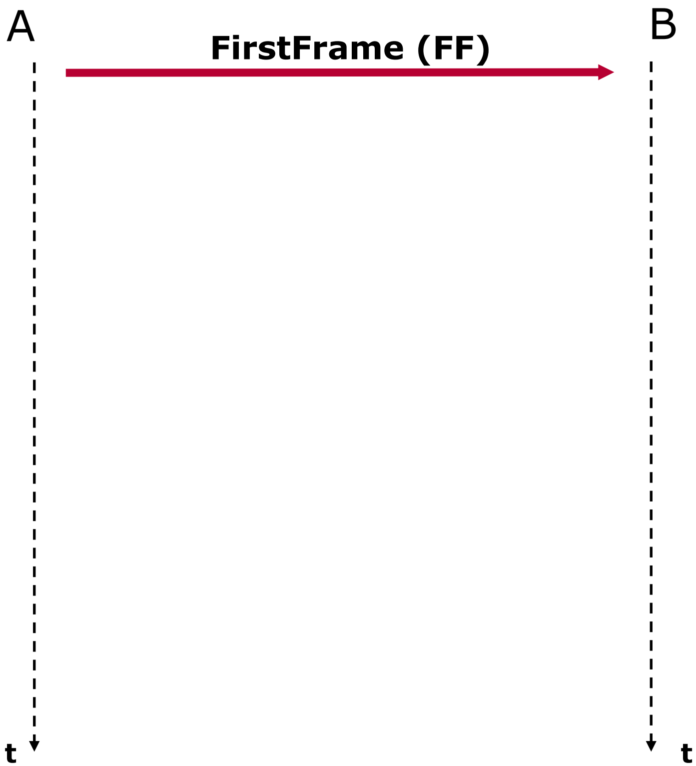
Segmented Message Transmission

Segmented message transmission

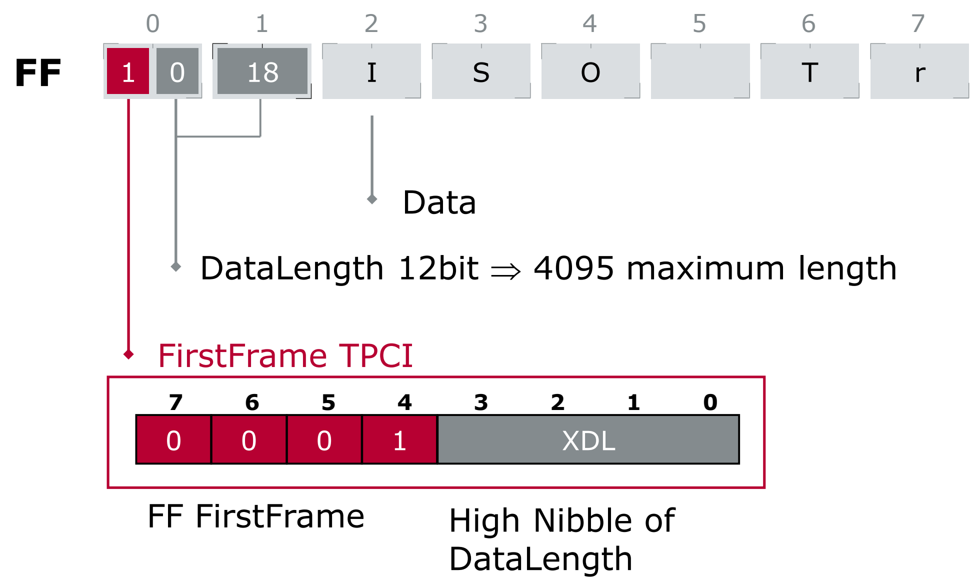


VECTOR

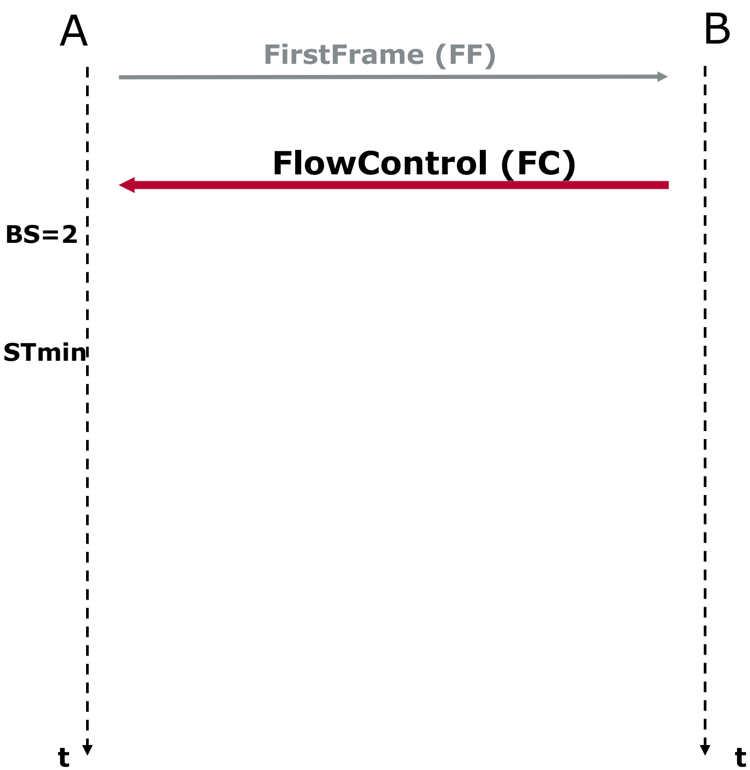
First Frame – Ignition of Transmission



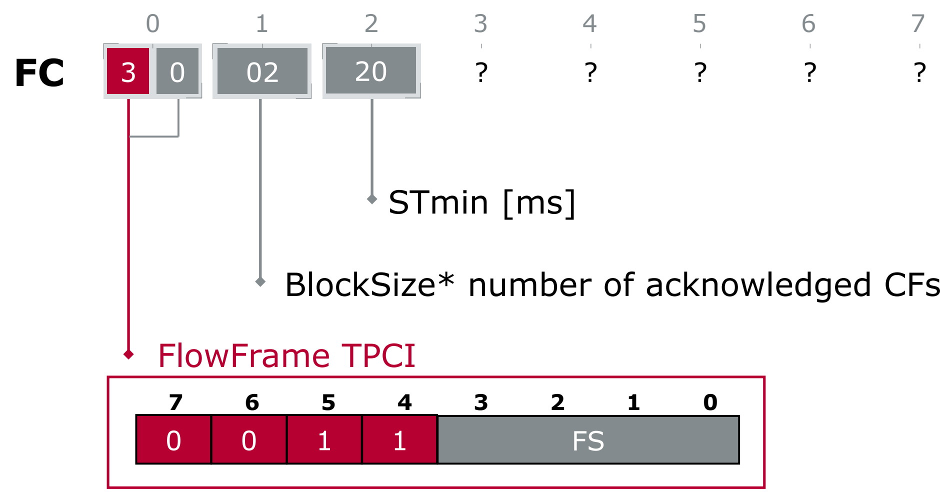
FirstFrame (FF)



Flow Control – Handshake



FlowControl (FC)



FC FlowControl

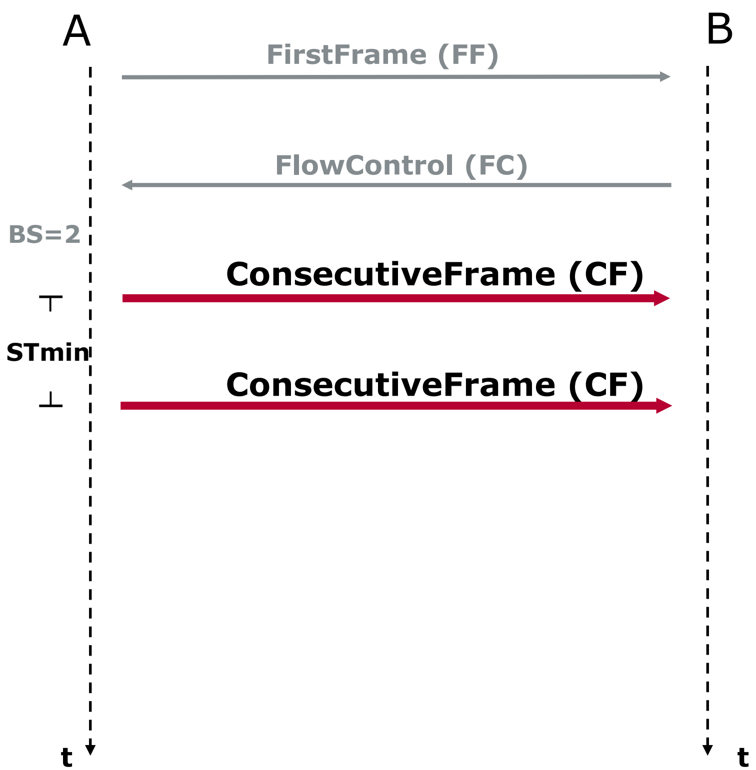
Flow State

0 clear to send
1 wait
2 overflow

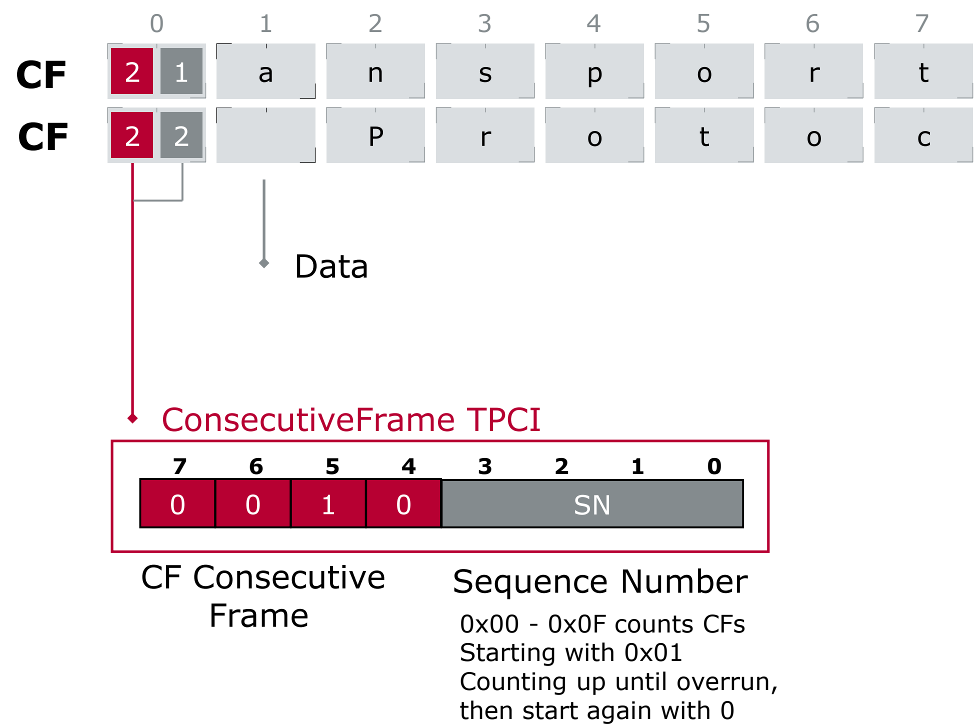
*Blocksize = 0: no further FlowControl frame

VECTOR

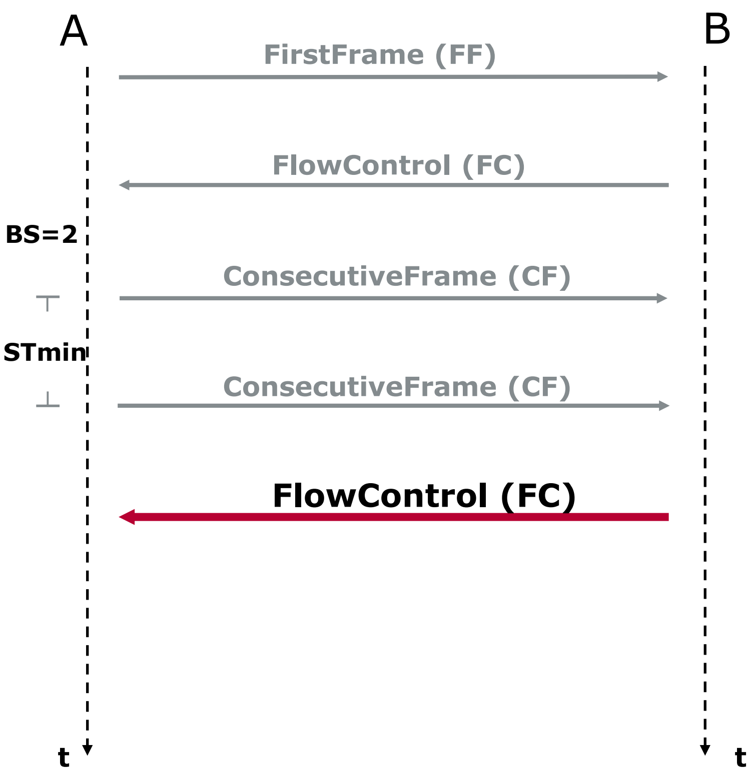
Consecutive Frames – Data Transmission



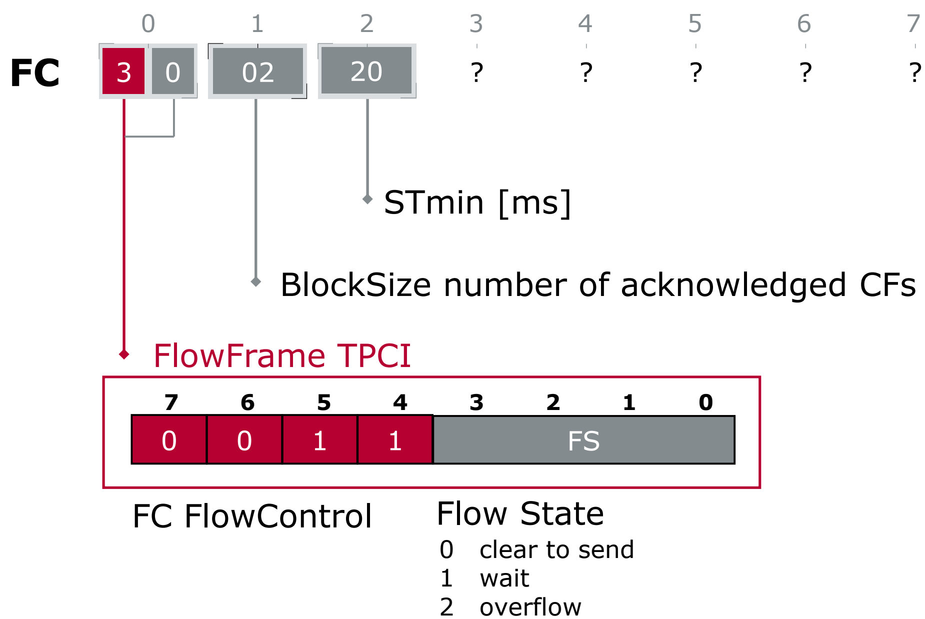
ConsecutiveFrame (CF)



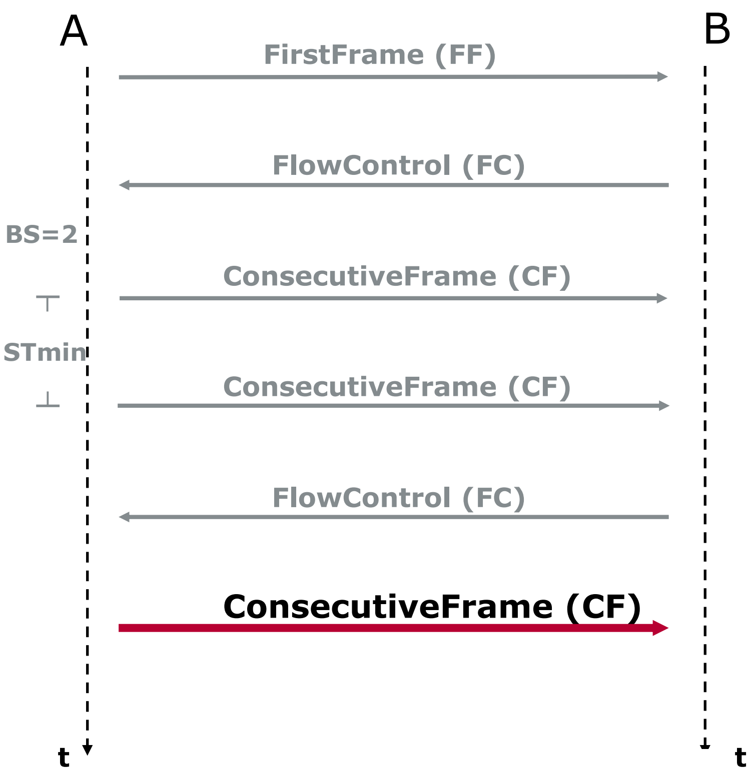
Flow Control – Acknowledge of Data Package



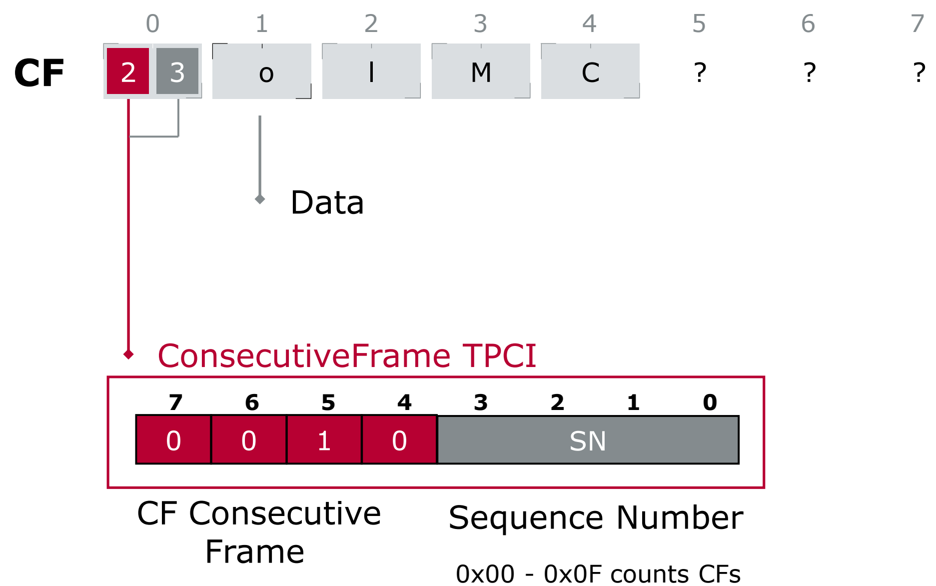
FlowControl (FC)



Consecutive Frames – Data Transmission



ConsecutiveFrame (CF)

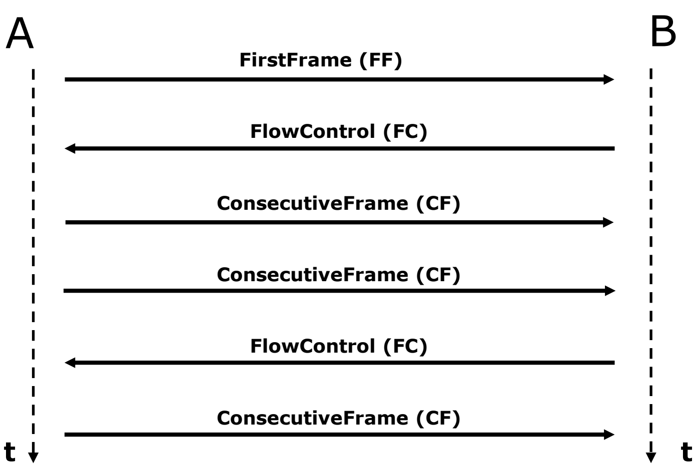


unacknowledged

VECTOR

Trace View – The Complete Transmission

	Time	Data
A →	39.4615	10 18 49 53 4f 20 54 72
B →	0.0002	30 02 14 00 00 00 00 00
A →	0.0202	21 61 6e 73 70 6f 72 74
A →	0.0203	22 20 50 72 6f 74 6f 63
B →	0.0002	30 02 14 00 00 00 00 00
A →	0.0203	23 6f 6c 4d 43 00 00 00



Trace View

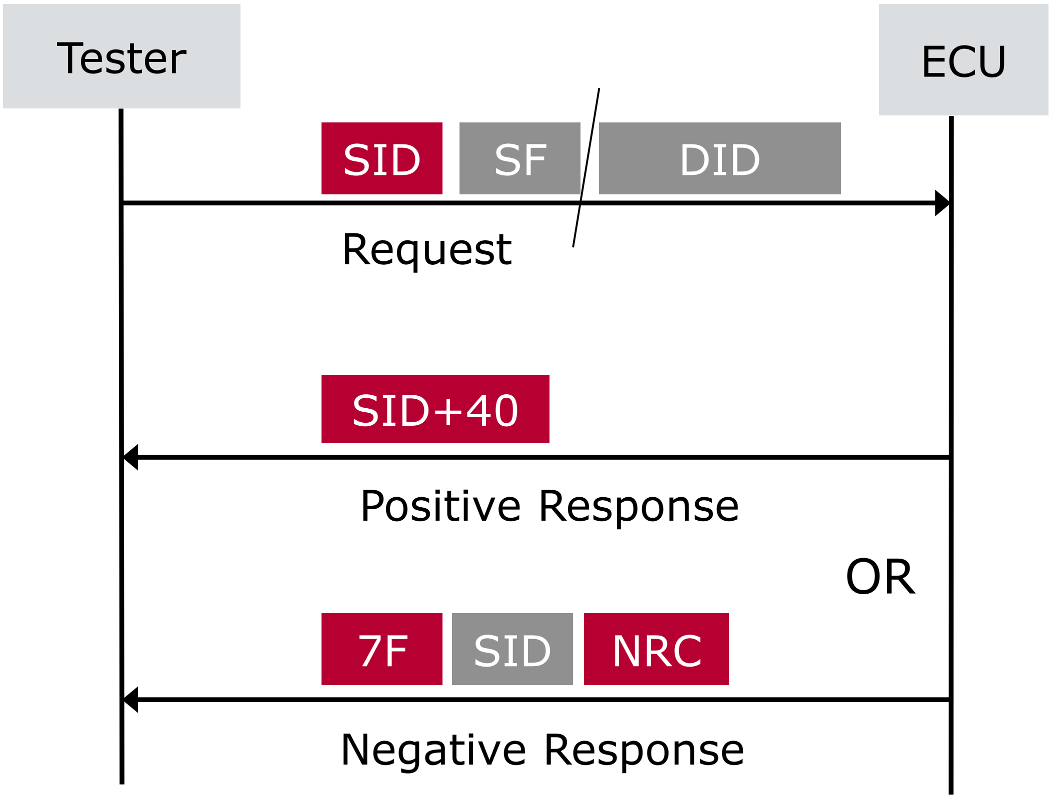
	0		1	2	3	4	5	6	7
FF	1	0	18	I	S	O		T	r
FC	3	0	02	20					
CF	2	1	a	n	s	p	o	r	t
CF	2	2		P	r	o	t	o	c
FC	3	0	02	20					
CF	2	3	o	l	M	C	?	?	?

Agenda

- Diagnostic Overview
- Diagnostic Protocol UDS
  - Services Overview
  - Diagnostic Session Control
  - Security Access
  - Read/Write Data By Identifier
  - Fault Memory
  - Communication Method in Diagnostics
  - ▶ **Review/Summary**
- Vector Diagnostic Solution
  - Solution
  - Tool-chain
  - Summary

VECTOR

Services Request / Response Behavior



SID = Service Identifier

SF = Subfunction

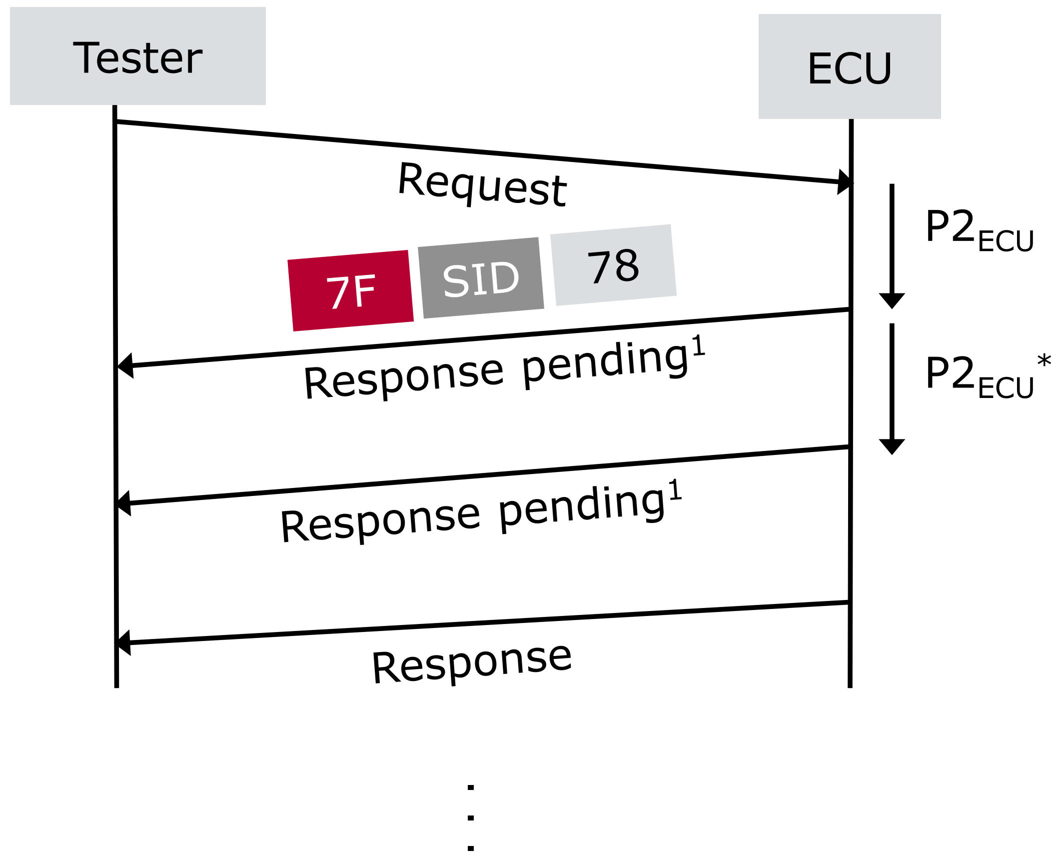
DID = Data Identifier

Six services

6/26

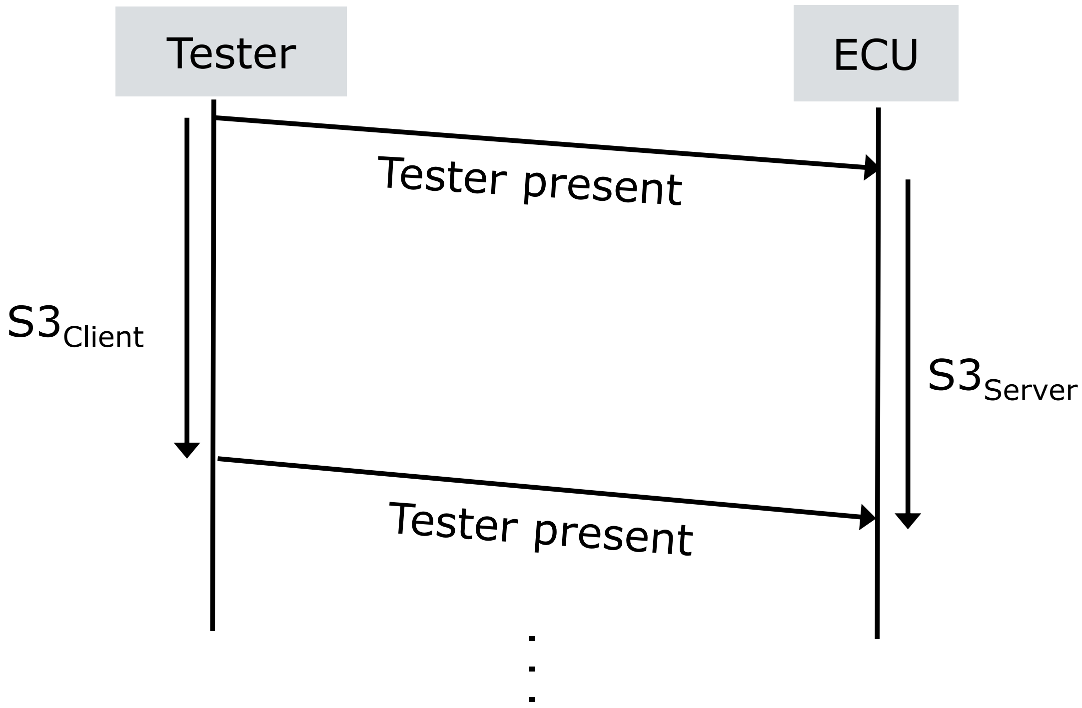
- Diagnostic Management
  - $10 Diagnostic Session Control
  - $27 Security Access
- Data Transmission Functional Unit
  - $22 Read Data By Identifier
  - $2E Write Data By Identifier
- Stored Data Transmission Functional Unit
  - $19 Read DTC Information
  - $14 Clear Diagnostic Information

Response Timeout Time P2 & P2*



[1]Request correctly received response pending

Session Timeout Time S3



$S3_{Cleint} < S3_{Server}$ due to Gateways etc.

Communication Method in Diagnostics

- Addressing
  - Physical Addressing 1:1
  - Functional Addressing 1:n
  - 3 CAN IDs per ECU for physical request & response, functional request

- Data Transmission
  - Unsegmented message transmission – Single Frame
  - Segmented message transmission – Multiple Frames

Q&A

Question Time

Agenda

- Diagnostic Overview
- Diagnostic Protocol UDS
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  - ▶ **Solution**
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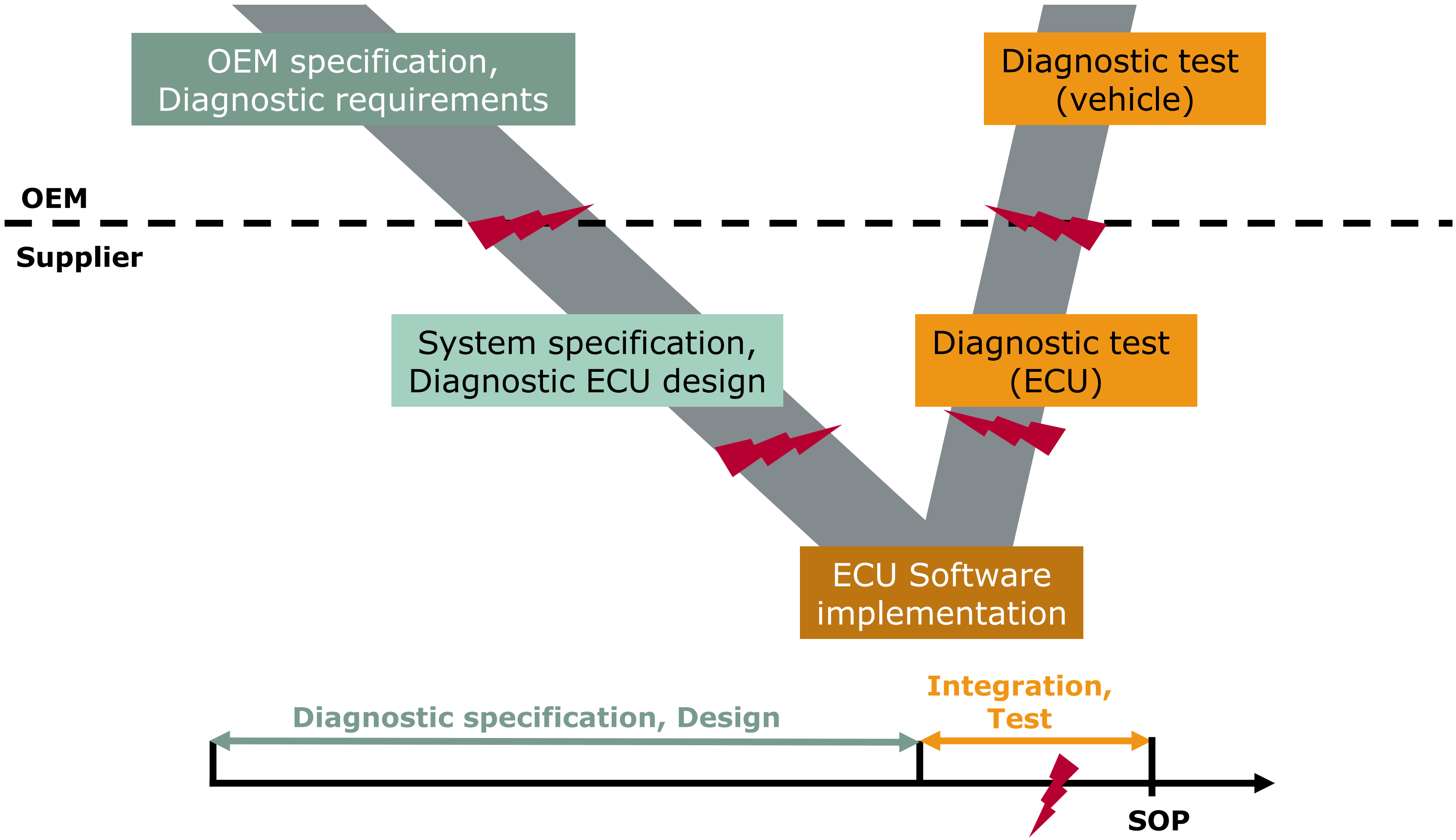
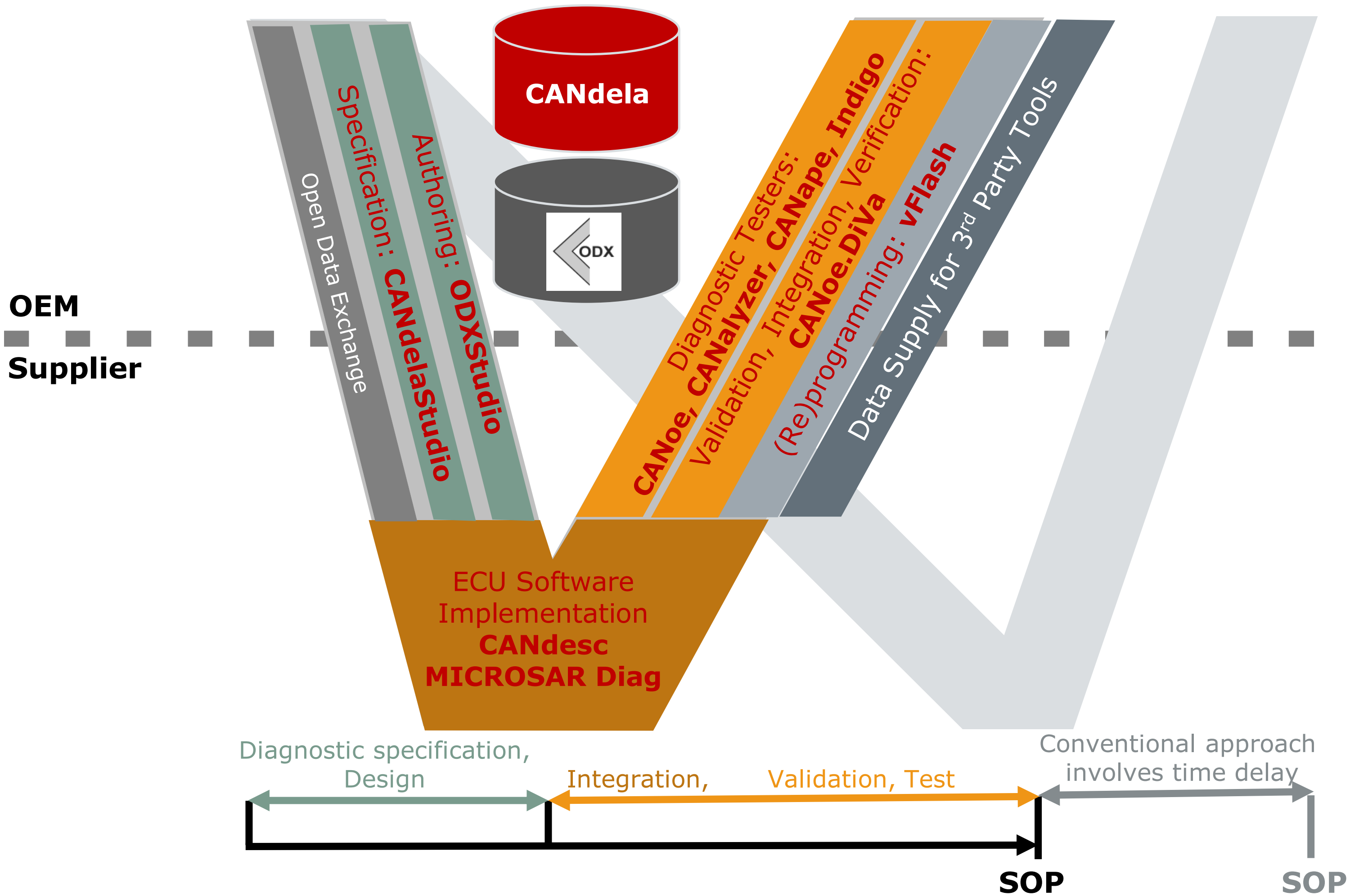
  - Summary

Conventional Diagnostic Development Process



Focus on Frontloading* with the Vector Diagnostic Tool Chain



Agenda

Diagnostic Overview

Diagnostic Protocol UDS

- Services Overview
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- Security Access
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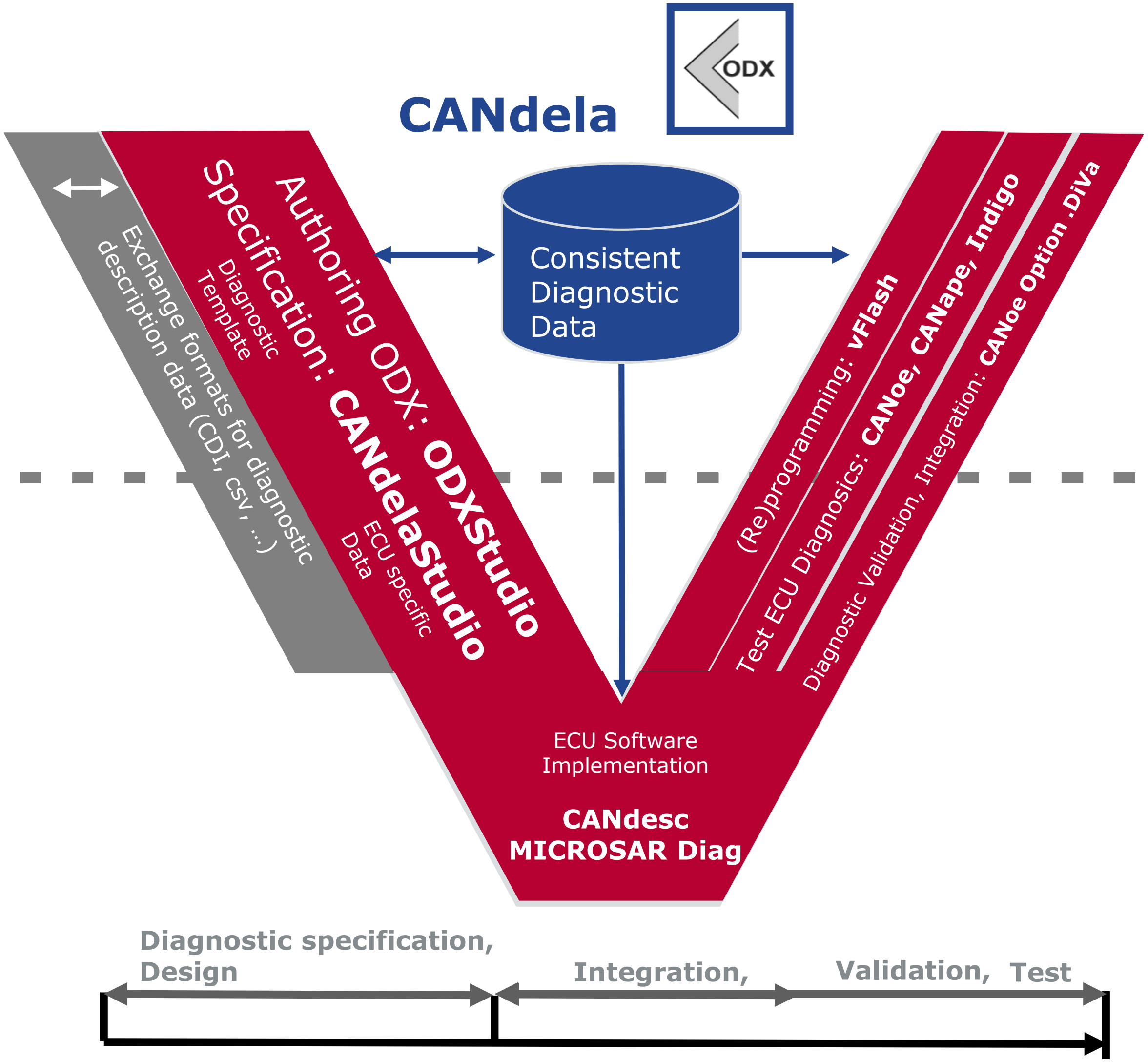
- Fault Memory
- Communication Method in Diagnostics
- Review/Summary

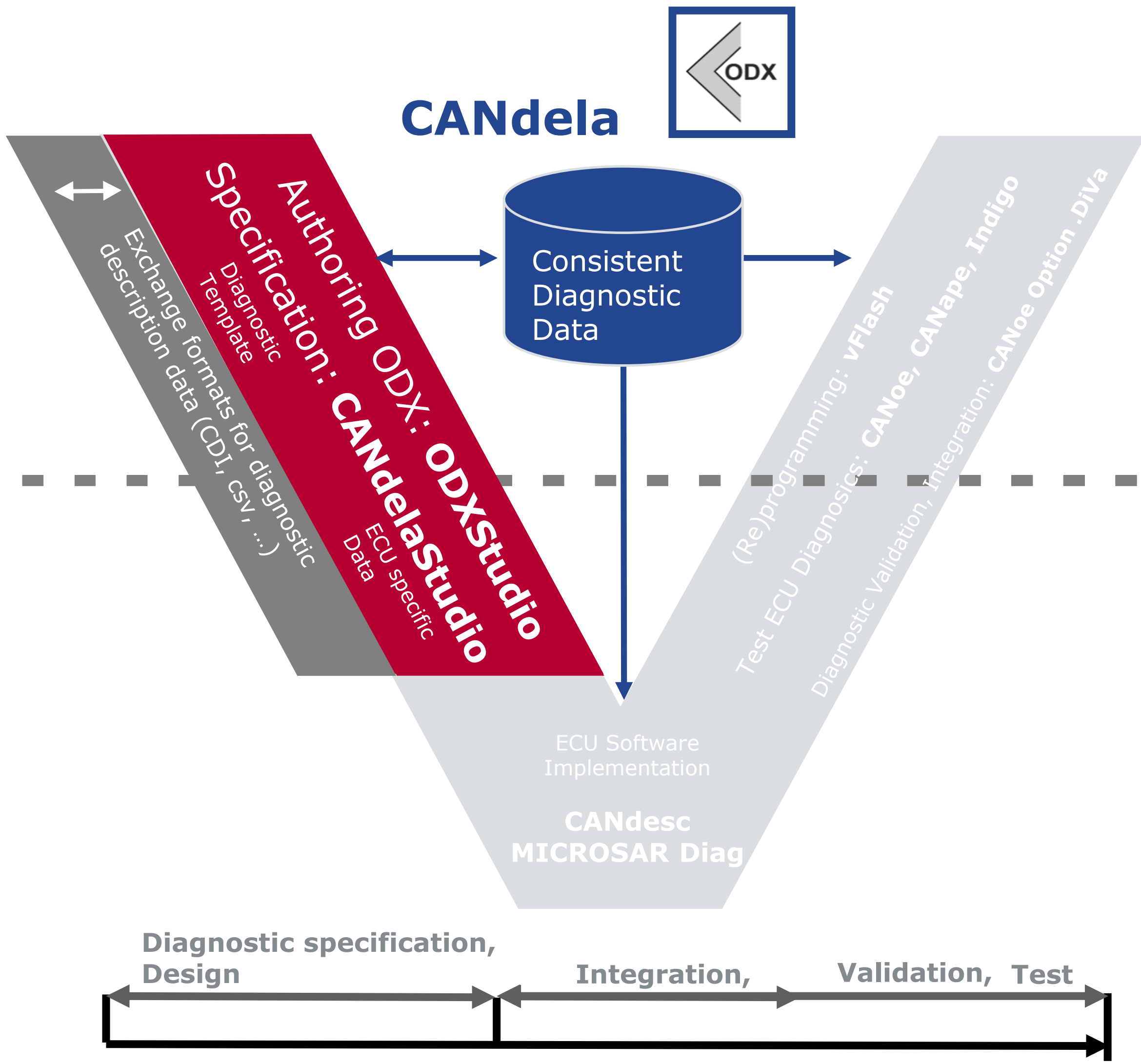
Vector Diagnostic Solution

- Solution
- ▶ **Tool-chain**
- Summary

Overview

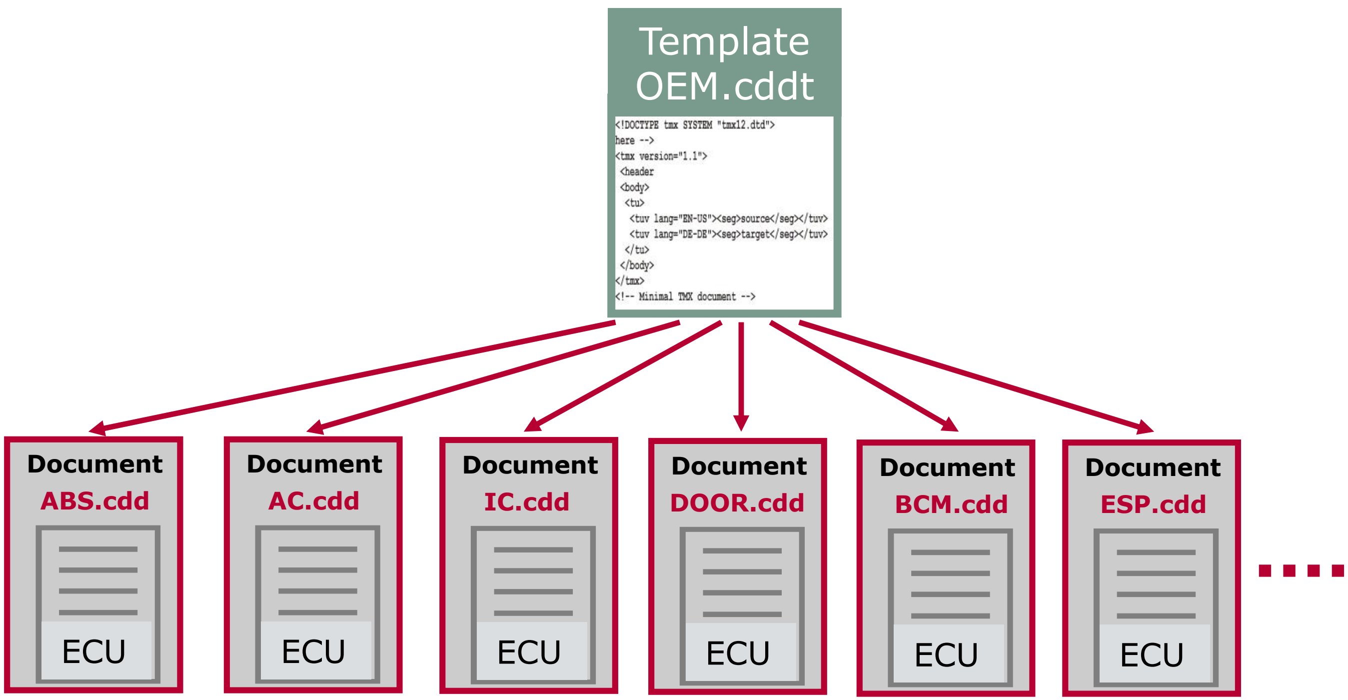

CANdela
ODX
Consistent Diagnostic Data
Exchange formats for diagnostic description data (CDI, csv, ...)
Specification: CANdelaStudio
Diagnostic Template
ECU specific Data
Authoring ODX: ODXStudio
(Re)programming: vFlash
Test ECU Diagnostics: CANoe, CANape, Indigo
Diagnostic Validation, Integration: CANoe Option .DiVa
ECU Software Implementation
CANdesc
MICROSAR Diag
Diagnostic specification, Design
Integration,
Validation, Test

Diagnostic Database Authoring Tools



Template concept

- The CANdela template represents the generic OEM diagnostic specification
- CANdela templates are available for most known OEMs
- CANdela documents are instances of the template



Program Window

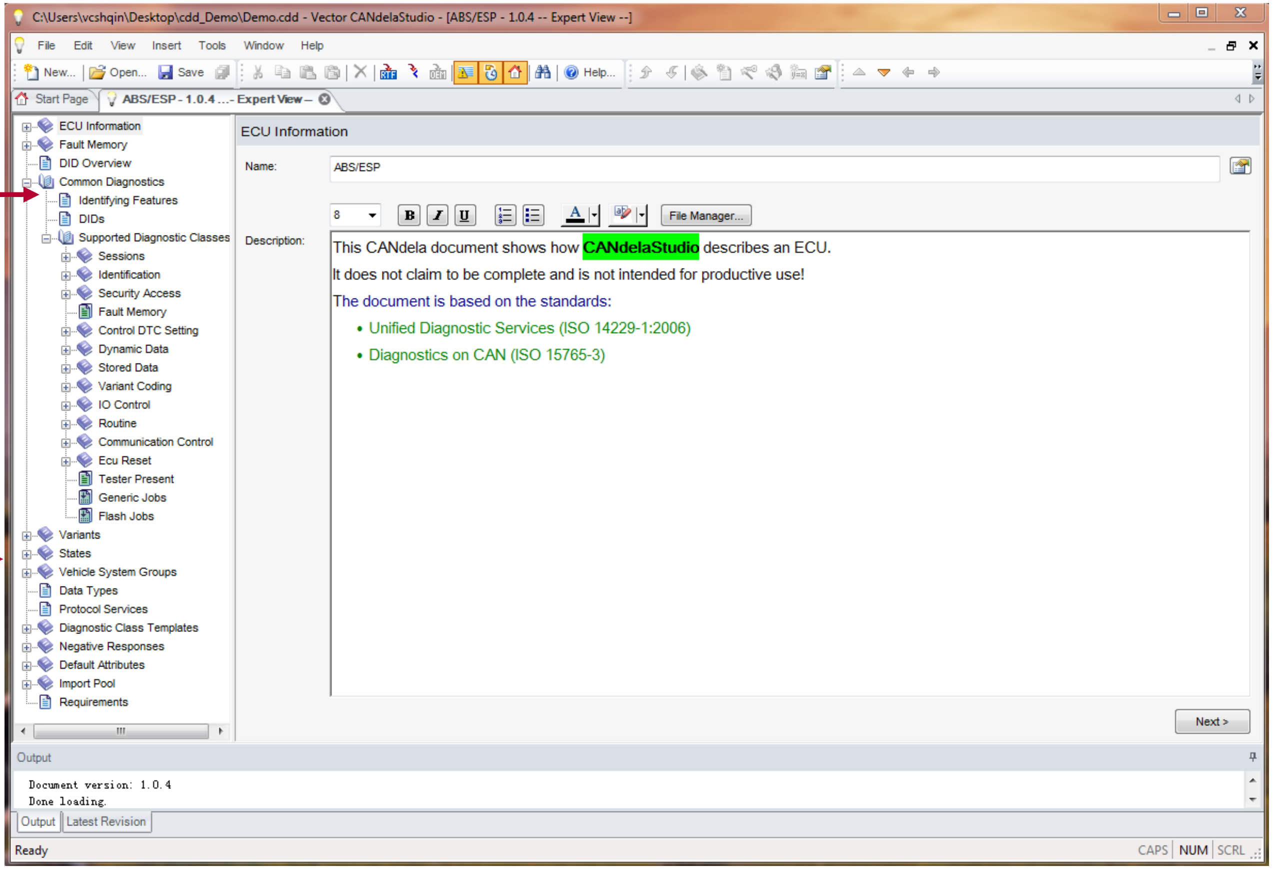
ECU Information

Common Diagnostic

Diag. instance

States

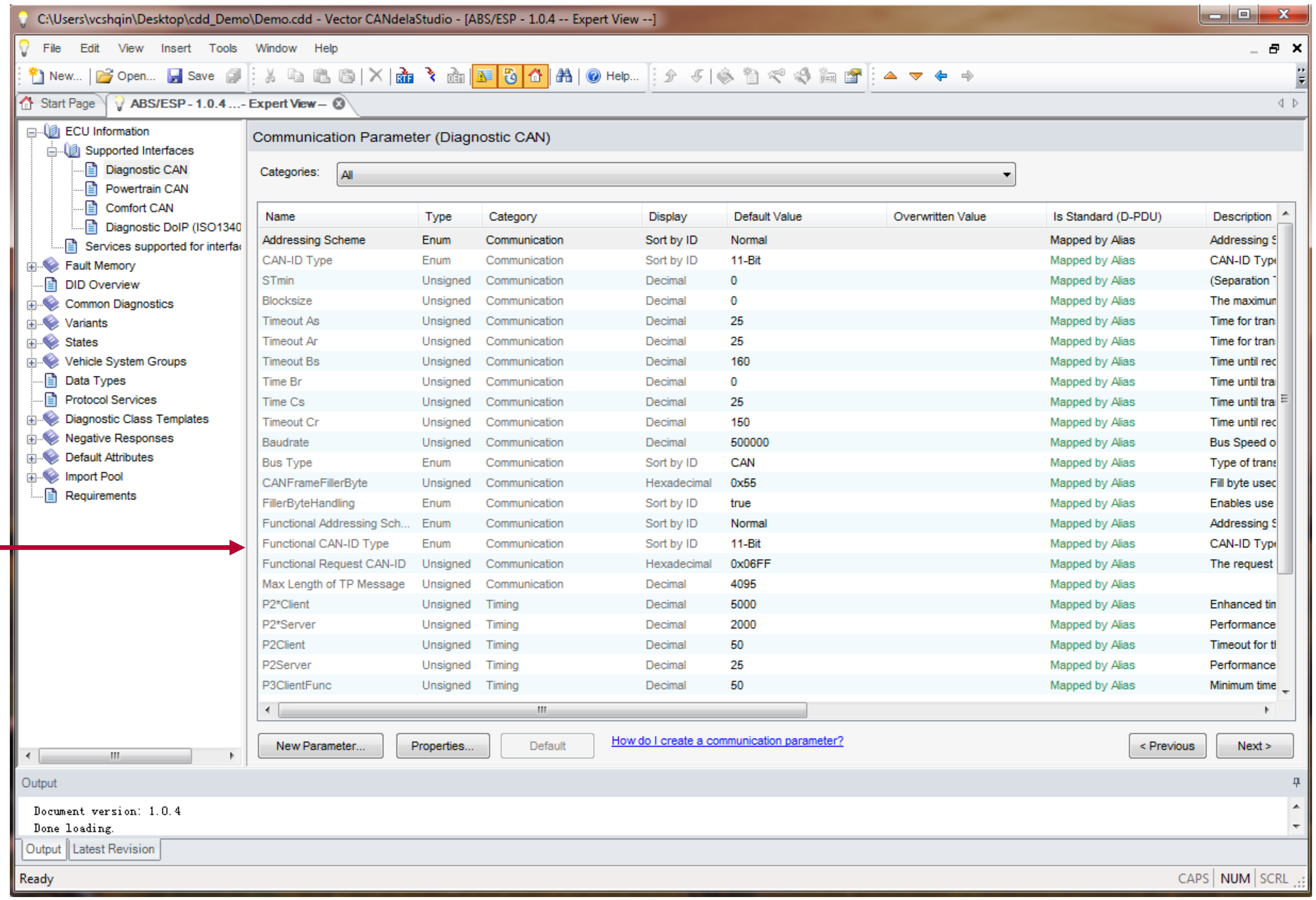
Output window



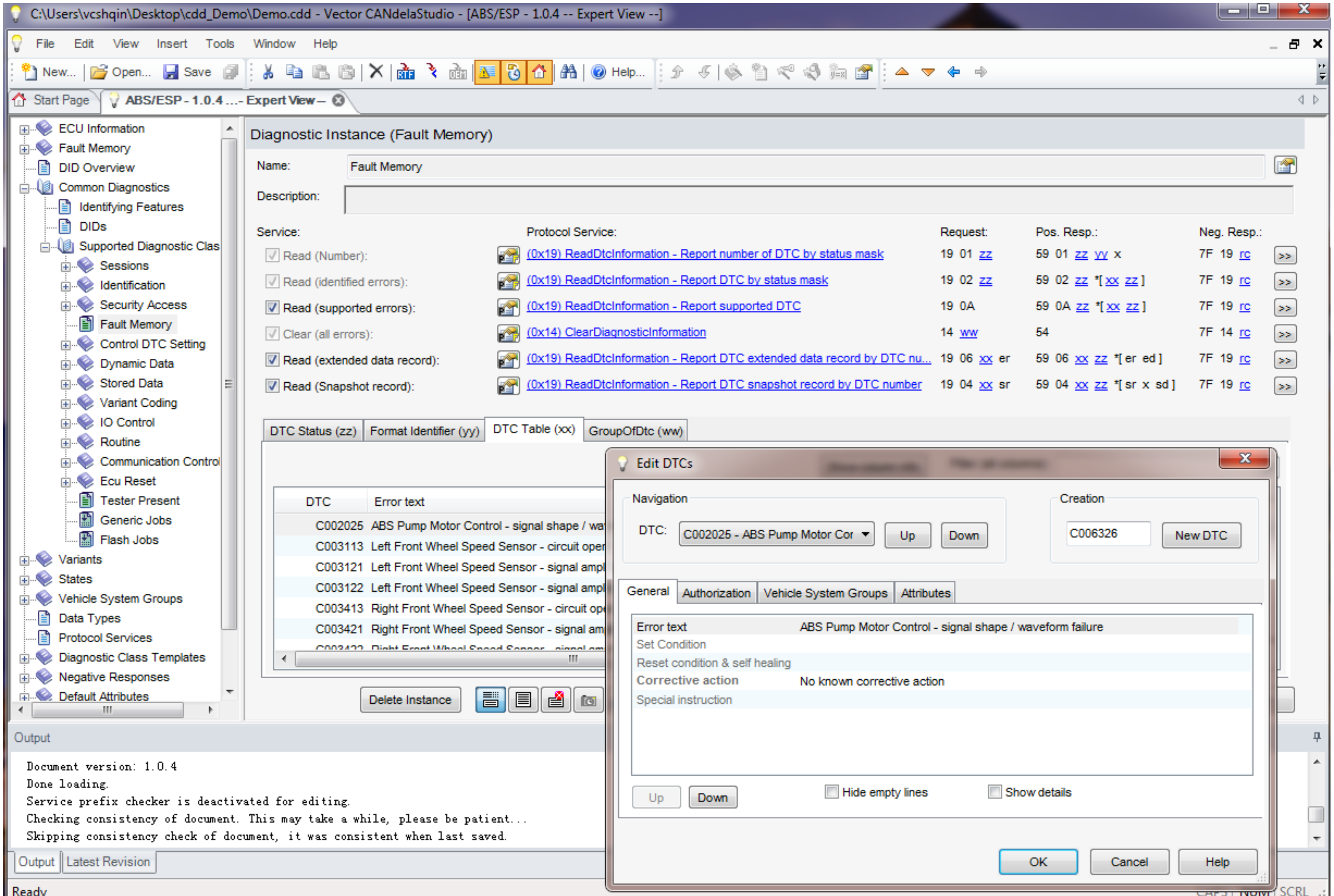
ECU Information

Interfaces

Communication Parameter



Authoring Diagnostics with CANdelaStudio



VECTOR

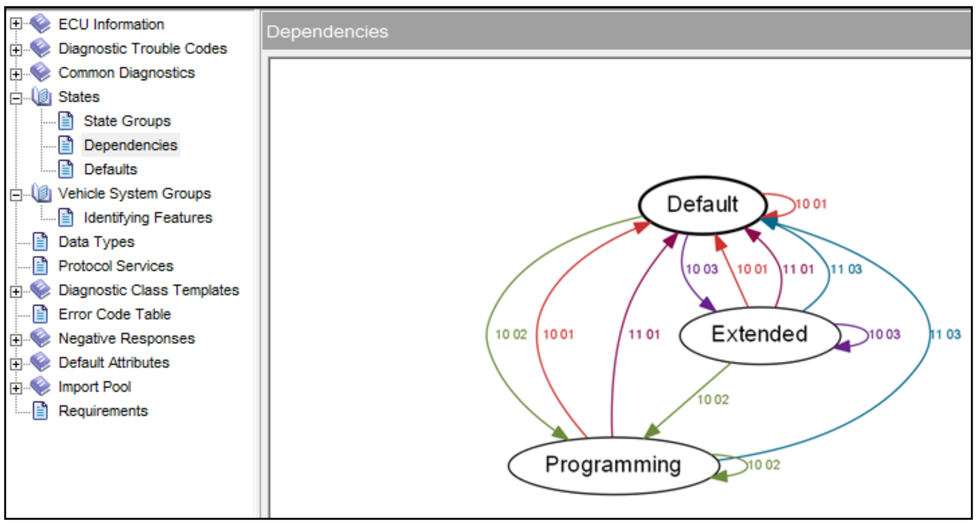
Session/Security States

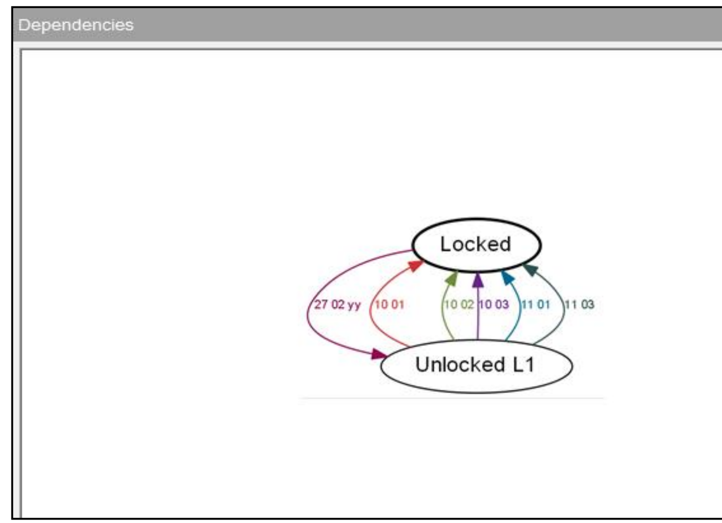
Prefix	Default	Programming	Extended
10 01	Default	Default	Default
10 02	(no)	Programming	Programming
10 03	Extended	Extended	Extended
11 01	(yes)	(yes)	(yes)
11 03	(yes)	(yes)	(yes)
14 ww	(yes)	(yes)	(yes)
19 01 zz	(yes)	(yes)	(yes)
19 02 zz	(yes)	(yes)	(yes)
19 04 xx sr	(yes)	(yes)	(yes)
19 06 xx er	(yes)	(yes)	(yes)
19 0A	(yes)	(yes)	(yes)
22 01 00	(yes)	(yes)	(yes)
22 01 07	(yes)	(yes)	(yes)
22 01 09	(yes)	(yes)	(yes)
22 01 0A	(yes)	(yes)	(yes)
22 01 0B	(yes)	(yes)	(yes)
22 01 0C	(yes)	(yes)	(yes)
22 01 0D	(yes)	(yes)	(yes)
22 01 0E	(yes)	(yes)	(yes)
22 01 0F	(yes)	(yes)	(yes)
22 01 10	(yes)	(yes)	(yes)

State group: Session < Previous Next >

Prefix	Locked	Unlocked L1
10 01	(yes)	Locked
10 02	(yes)	Locked
10 03	(yes)	Locked
11 01	(yes)	Locked
11 03	(yes)	Locked
14 ww	(yes)	(yes)
19 01 zz	(yes)	(yes)
19 02 zz	(yes)	(yes)
19 04 xx sr	(yes)	(yes)
19 06 xx er	(yes)	(yes)
19 0A	(yes)	(yes)
22 01 00	(yes)	(yes)
22 01 07	(yes)	(yes)
22 01 09	(yes)	(yes)
22 01 0A	(yes)	(yes)
22 01 0B	(yes)	(yes)
22 01 0C	(yes)	(yes)
22 01 0D	(yes)	(yes)
22 01 0E	(yes)	(yes)
22 01 0F	(yes)	(yes)
22 01 10	(yes)	(yes)

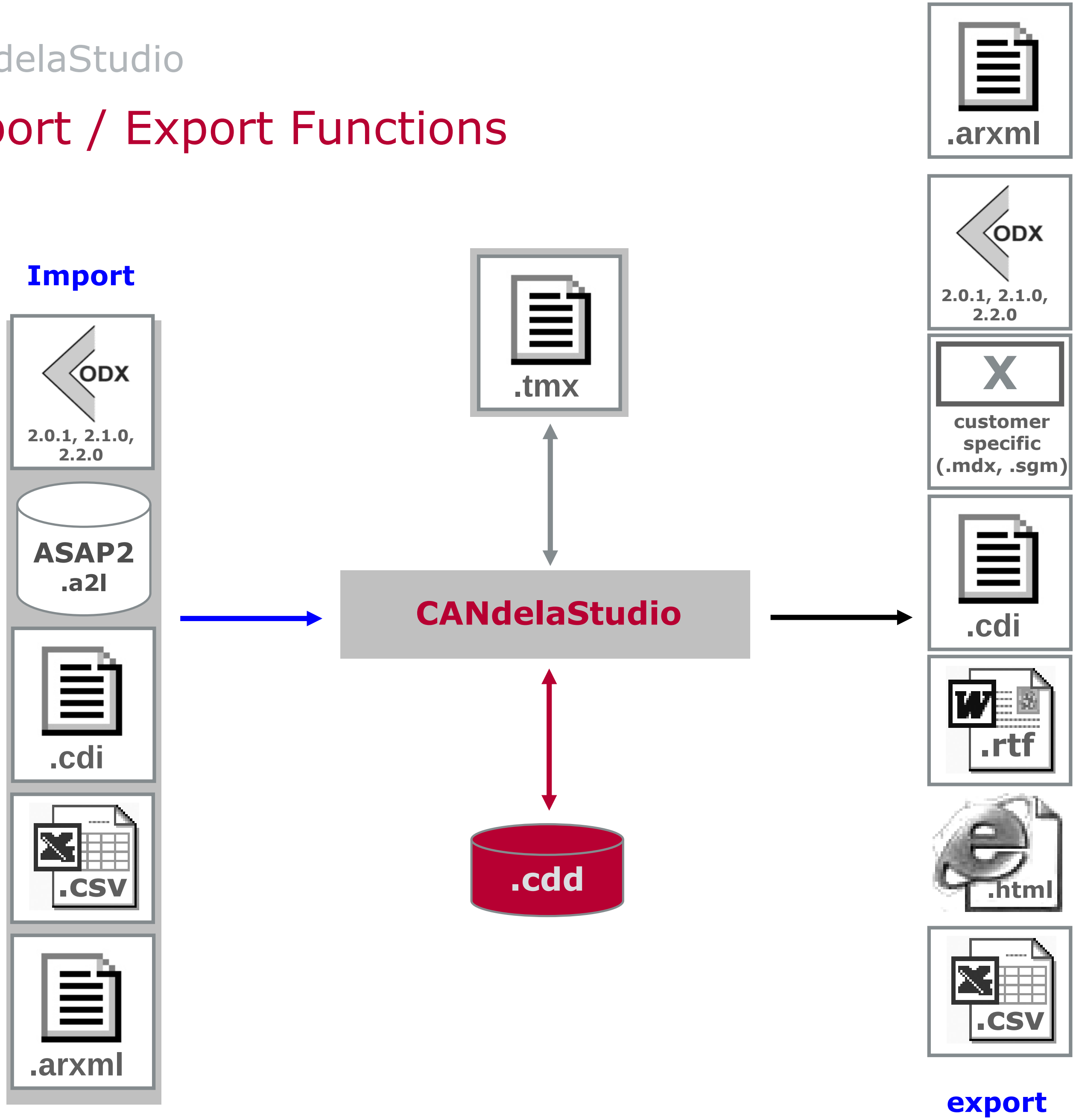
State group: SecurityAccess





Easily set and review session/security states of diagnostic services.

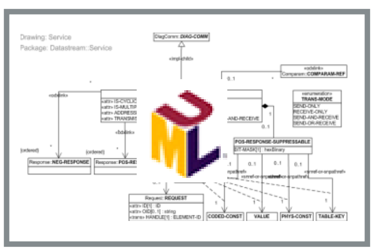
Import / Export Functions



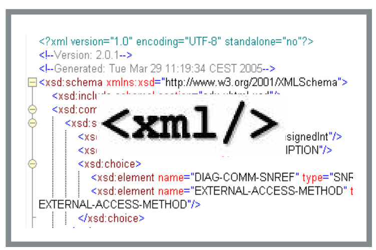
Components of the standard

- Joint ASAM/ISO working group

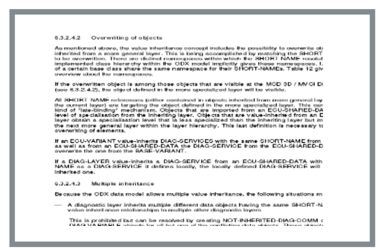
Components of the standard



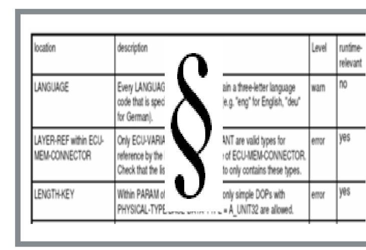
UML Model



XML Schema

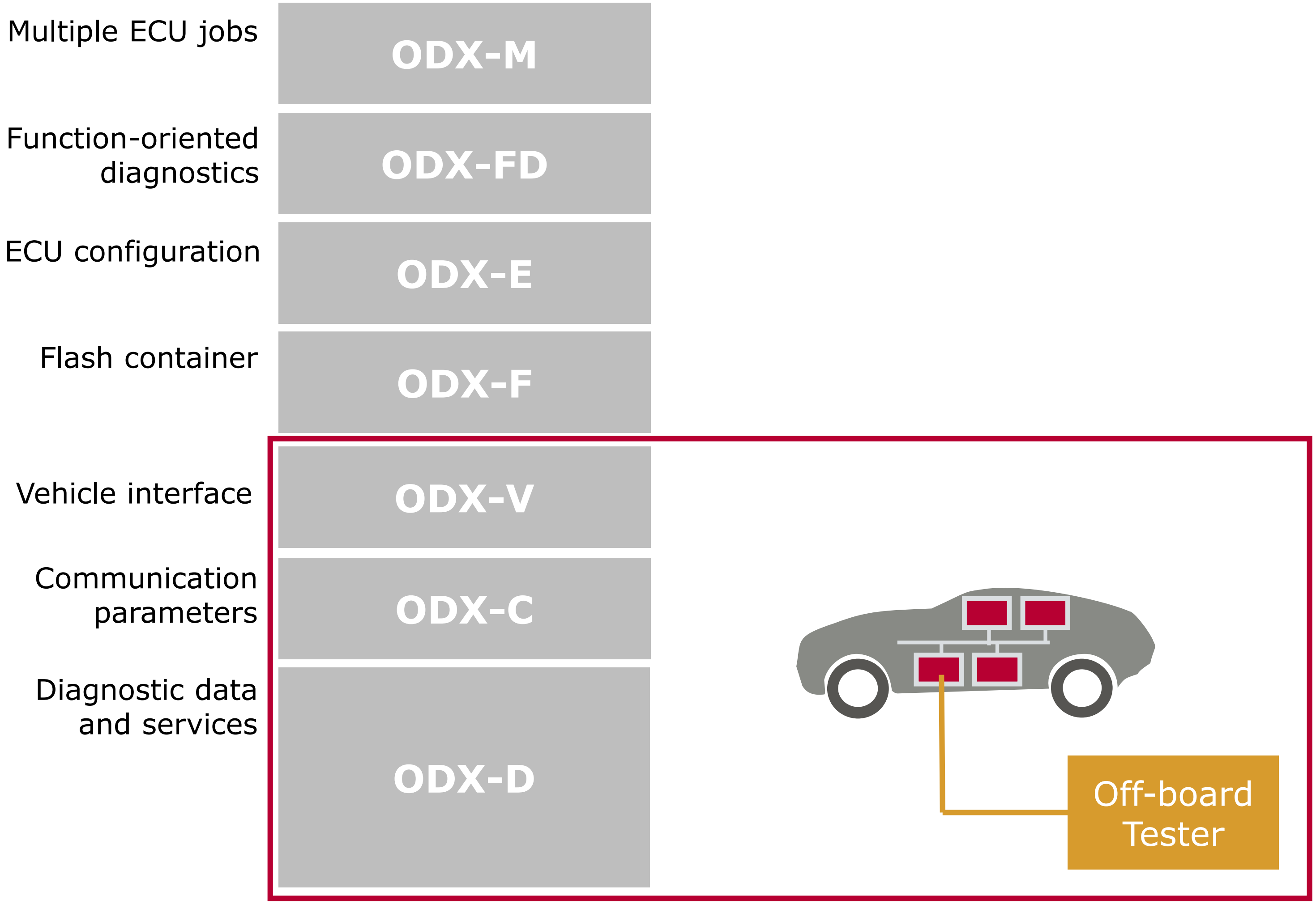


Textual Description



Checker Rules

Subject

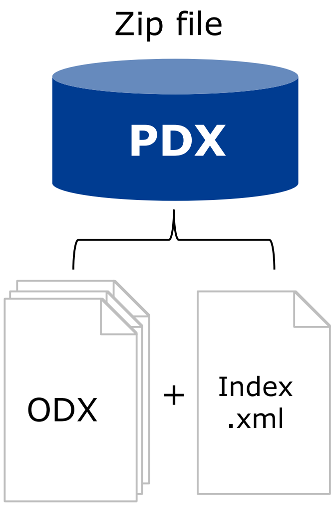


ODX does not only cover diagnostic use-cases, but they are the main focus!

ODX/PDX

- ODX files
  - In any case one ODX file contains exactly one ODX-CATEGORY
  - File extensions for each category are suggested (odx-d, odx-f, odx-e, …)

- PDX files
  - Contains one or several ODX files.
  - Intention: PDX represents ECU or vehicle
  - Zip file, but with extension .pdx
  - May contain additional files:
    Picture, text, java code …
  - Must contain a file index.xml which contains the content of the package.



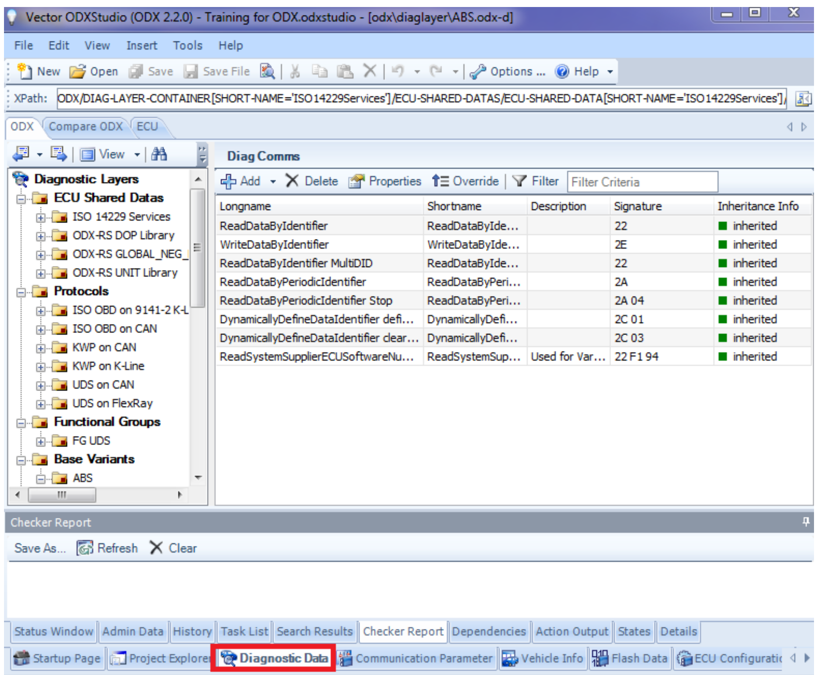
ODXStudio

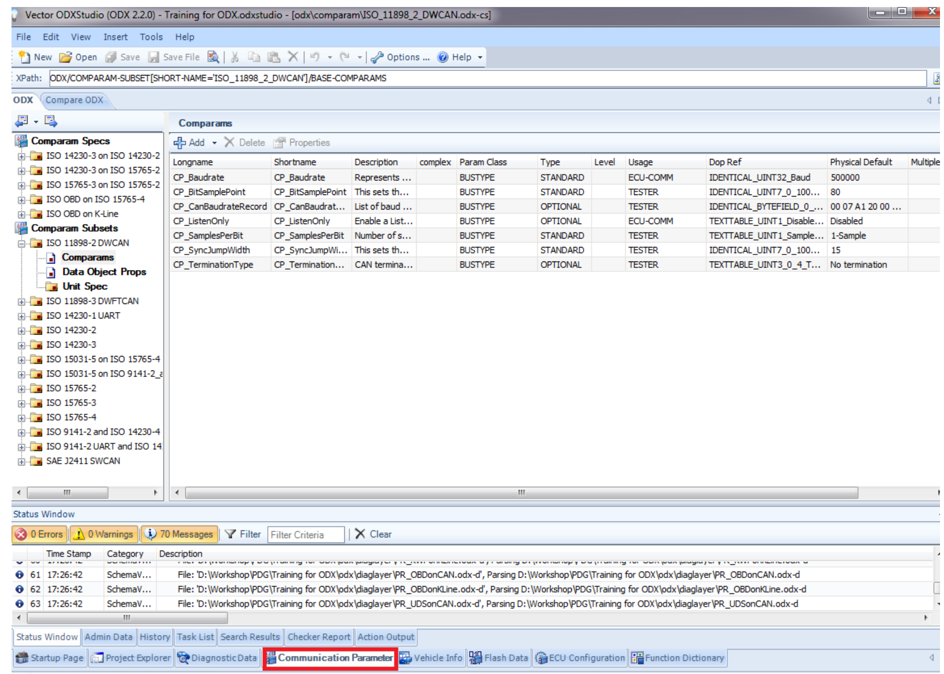
- is a user oriented authoring tool for diagnostic data in ODX format
- covers each ODX category in a separate view
- is extensible through Plugins and Actions
- supports custom perspectives to adapt to OEM specific ODX authoring guidelines and to simplify editing
- is smoothly scalable from ODX data for a single ECU up to complete vehicle or model series projects
- utilizes the native ODX data model to allow an optimal roundtrip

- supports both ODX 2.0.1
  and ODX 2.2.0 natively

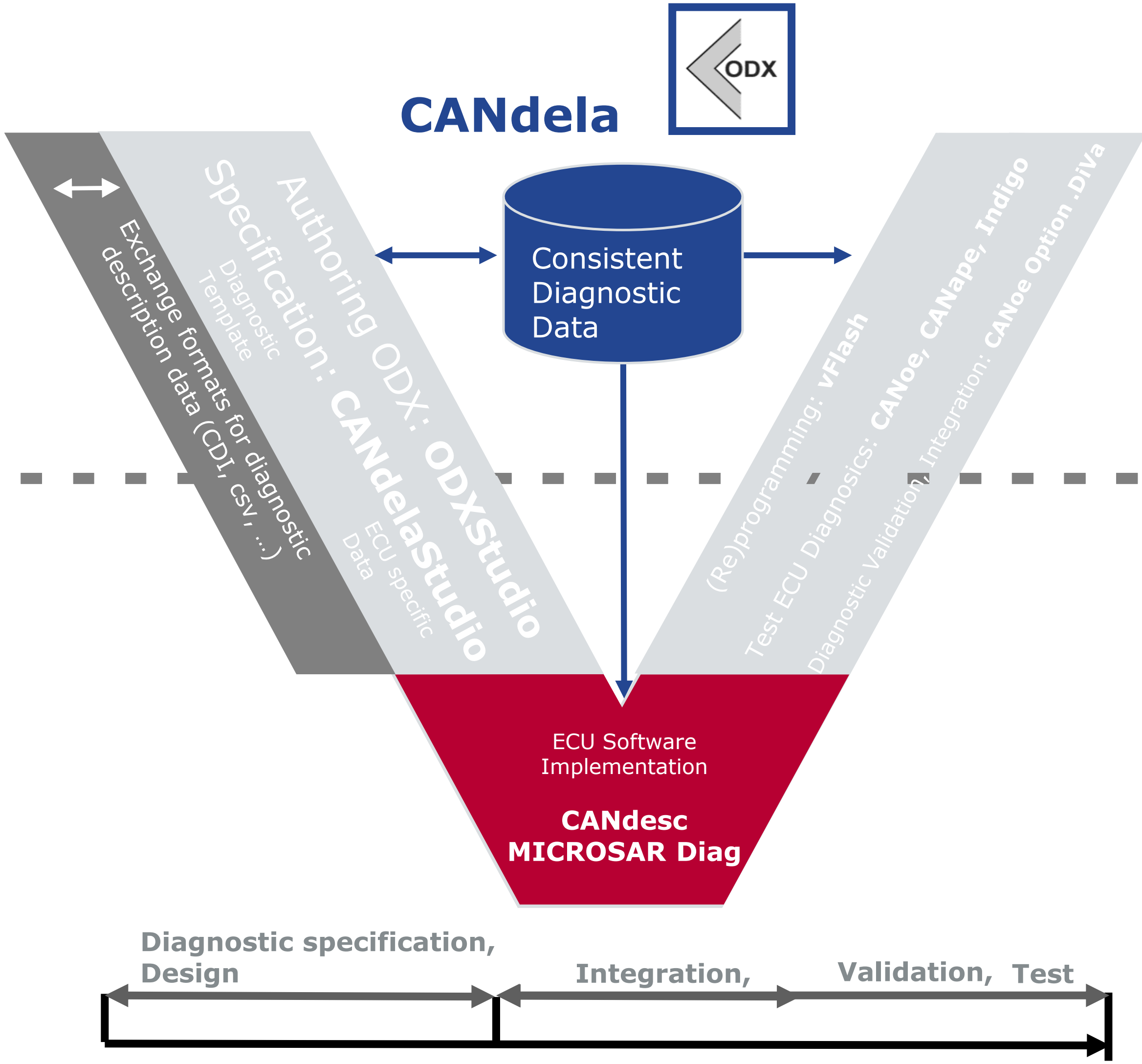
Program Window

- One separate view for each ODX category
- Show every detail of ODX **and/*or*** show customized perspective

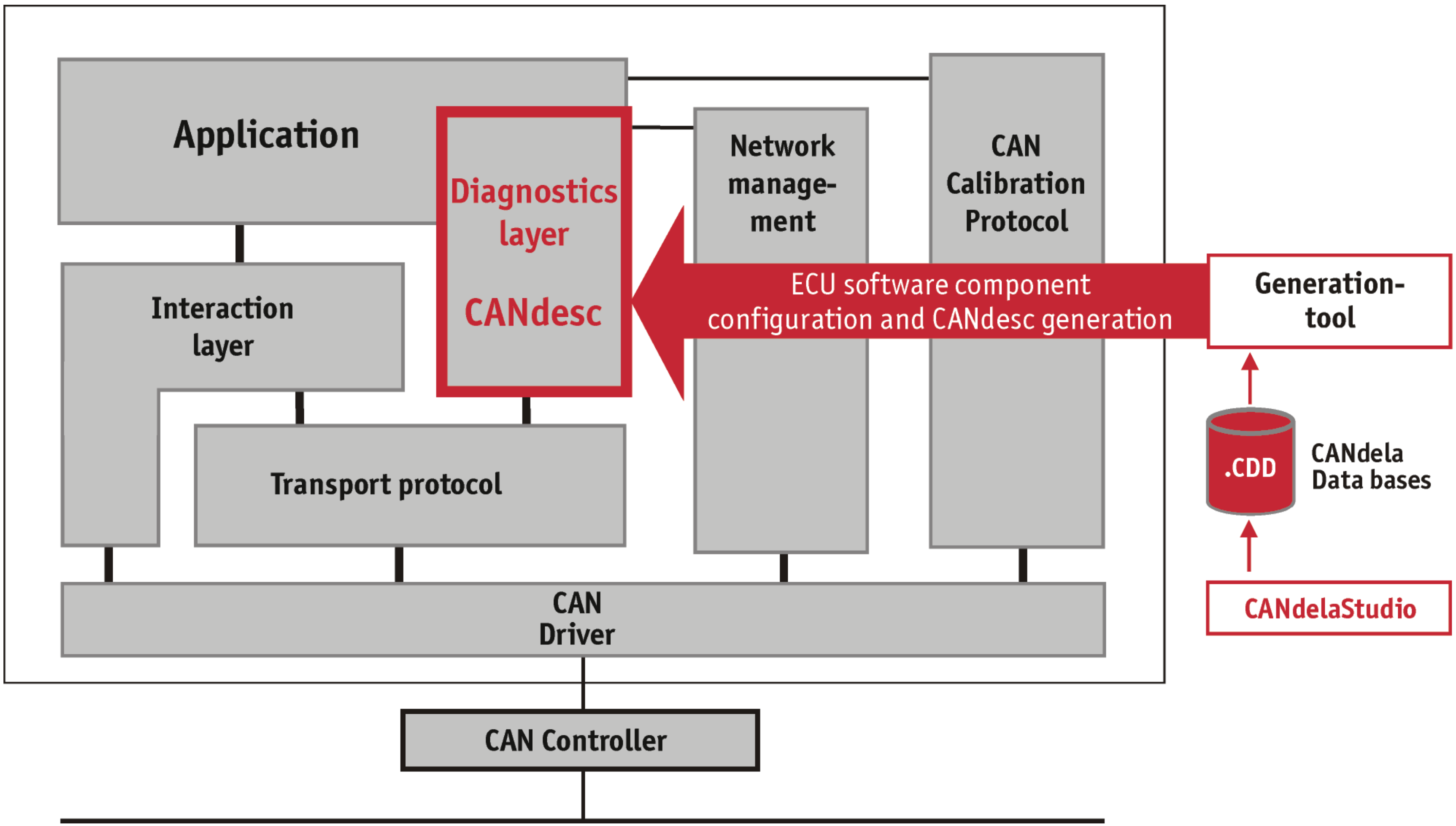




Embedded Diagnostics

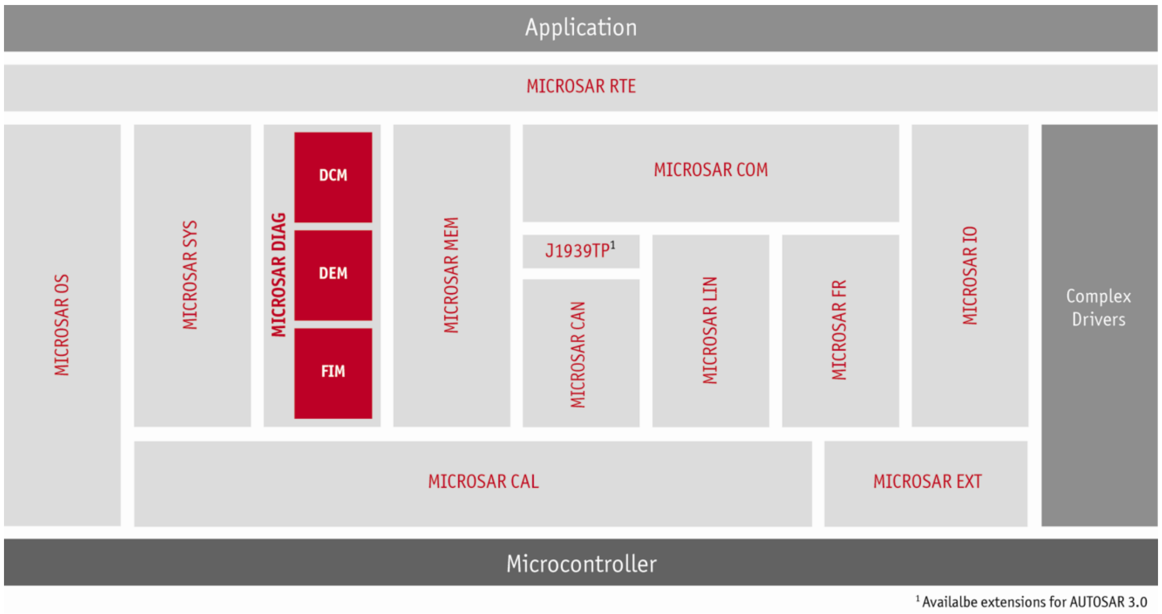


CANbedded/CANdesc

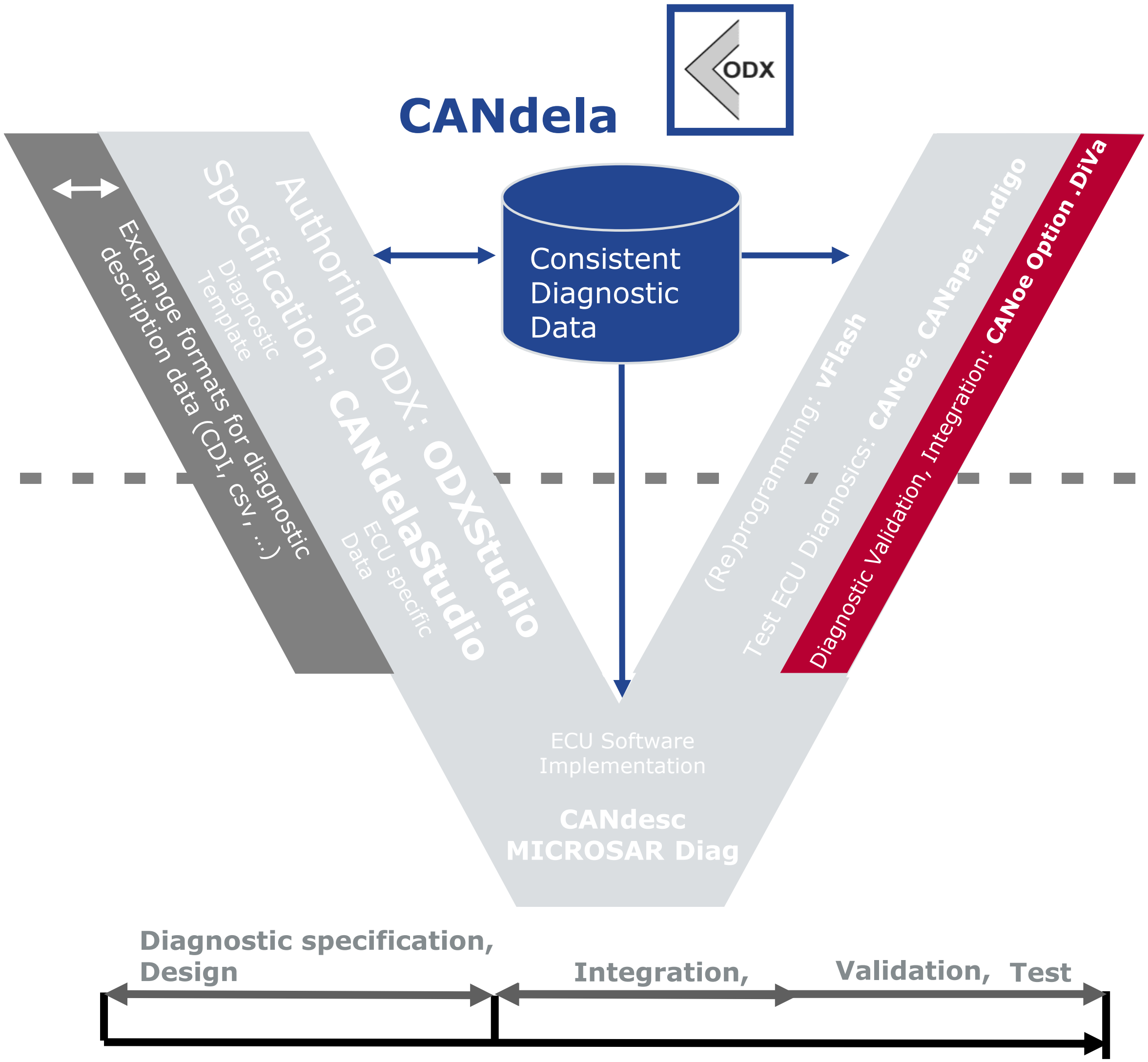


MICROSAR Diag

- AUTOSAR is an open and standardized automotive software architecture, jointly developed by automobile manufacturers, suppliers and tool developers.
- MICROSAR = Vector implementation of AUTOSAR standard
- AUTOSAR DIAG BSW are available for different OEMs and different versions 2.0, 2.1, 3.x, 4.x



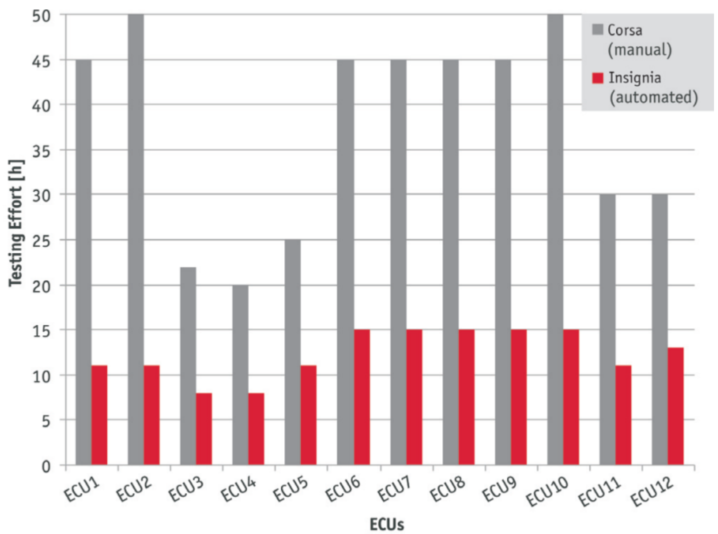
Automated Generated Tests



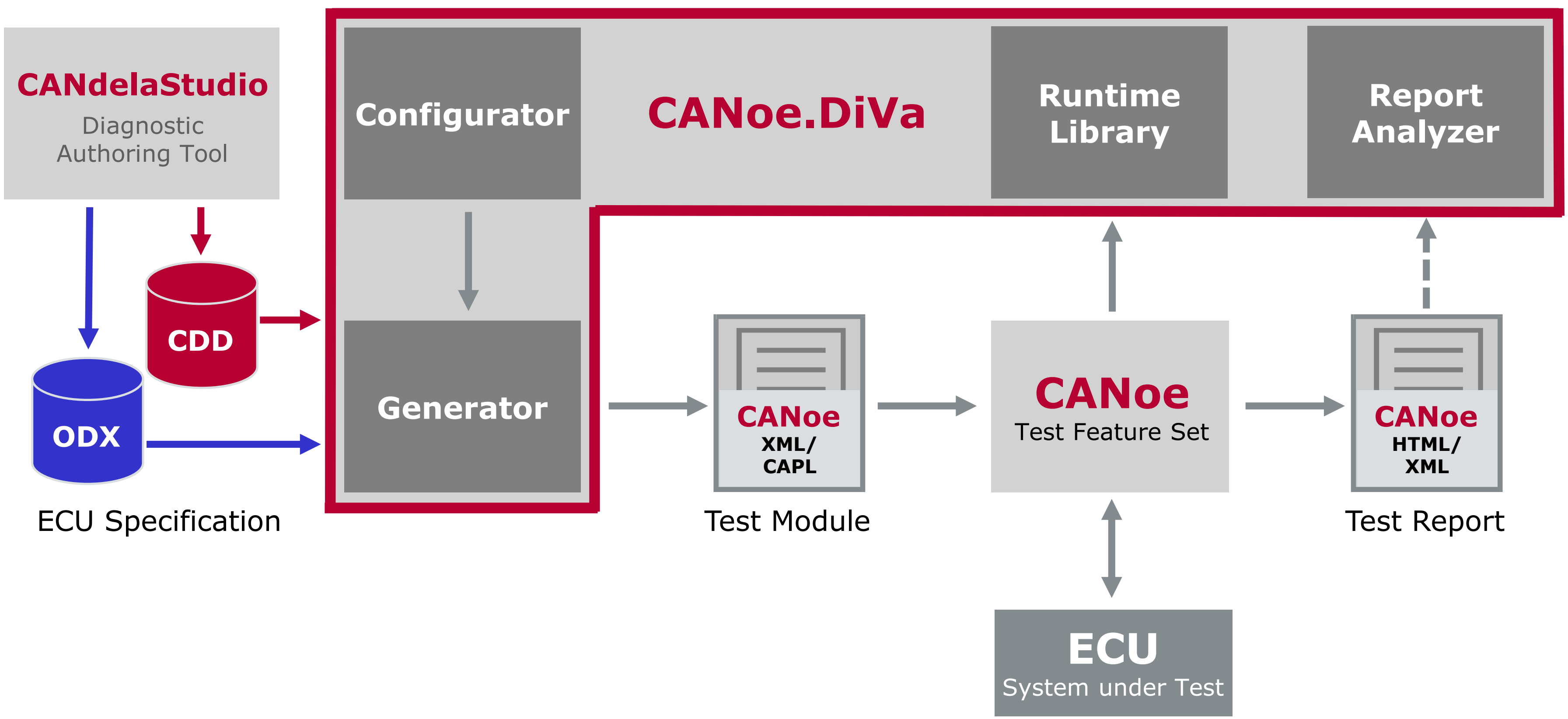
DiVa Features Overview

- Provides an automated generation and execution of diagnostic software tests with a broad test coverage.
- Generate ECU specific test cases out of the diagnostic specification
- Execute these tests and generate a test report (XML).
- Assist to classify, comment and follow-up issues in the report.
- Supports UDS, KWP (many brands), GMW3110, OBDII, WWH-OBD
- Supports ODX 2.0.1, 2.2.0, cdd-files

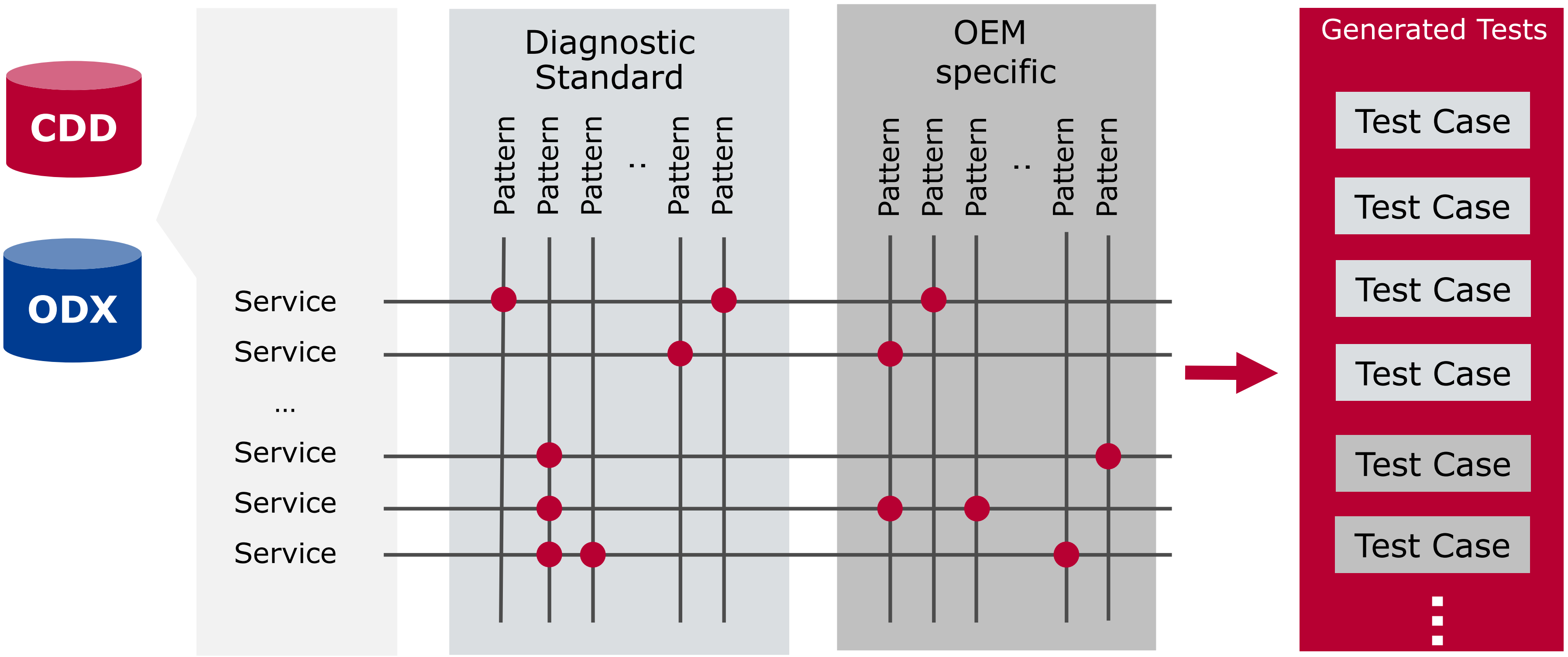
Enhance quality, reduce cost!



CANoe.DiVa Approach



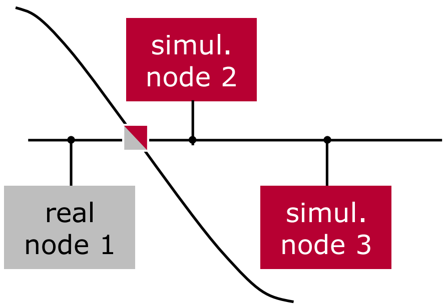
Test Generation



Test coverage CANoe.DiVa

- Diagnostic message flow
  - Physical and functional addressing
- Diagnostic Protocol Timing
  - P2, P2*, S3
- Diagnostic Protocol Format
  - Valid/Invalid Request
  - Response (single, none, multiple)
  - Multiple Id requests
- Data Content
- Diagnostic ECU Application
- Sessions and Security Levels
  - Service execution in the different sessions and security levels
  - Session and security state transitions

DiVa Features - Fault Memory Testing



- Provoke network signal failures
  - Communication timeouts
  - Data consistency failures

- Provoke hardware failures using the VT System:
  - Short-circuits (Ground, UBatt, Pins)
  - I/O failures (interruption, resistance, voltage)
  - Individual error settings

- Any other failures using user scripts

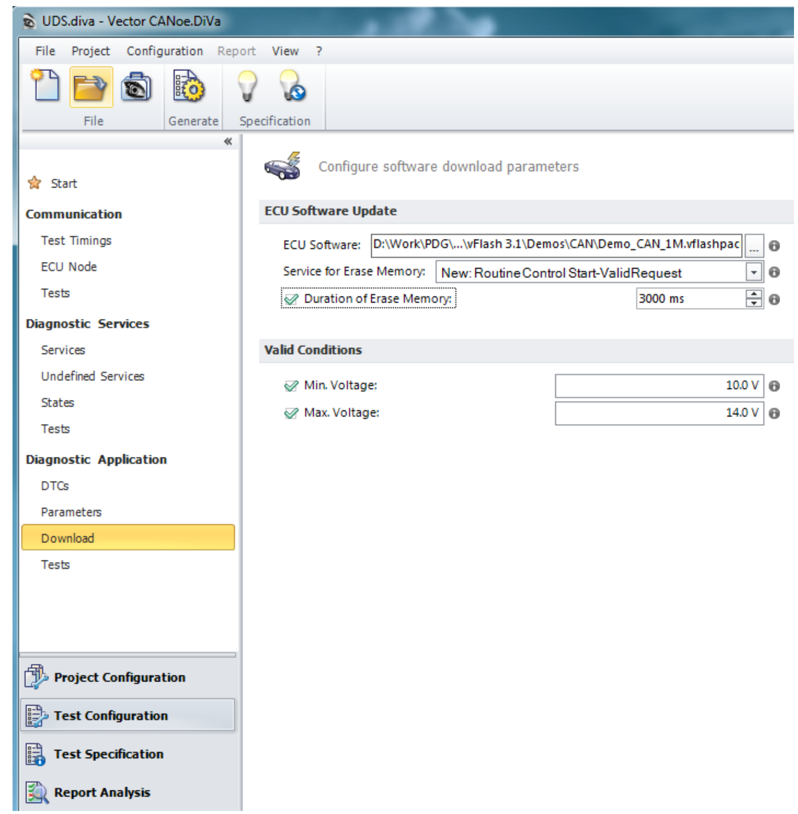
Software Download Tests

- The Download Tests …
  - …since CANoe.DiVa 4.0
  - …require an installed vFlash (version 2.7 or higher)
  - …configure the Flash Container (vFlashPack)

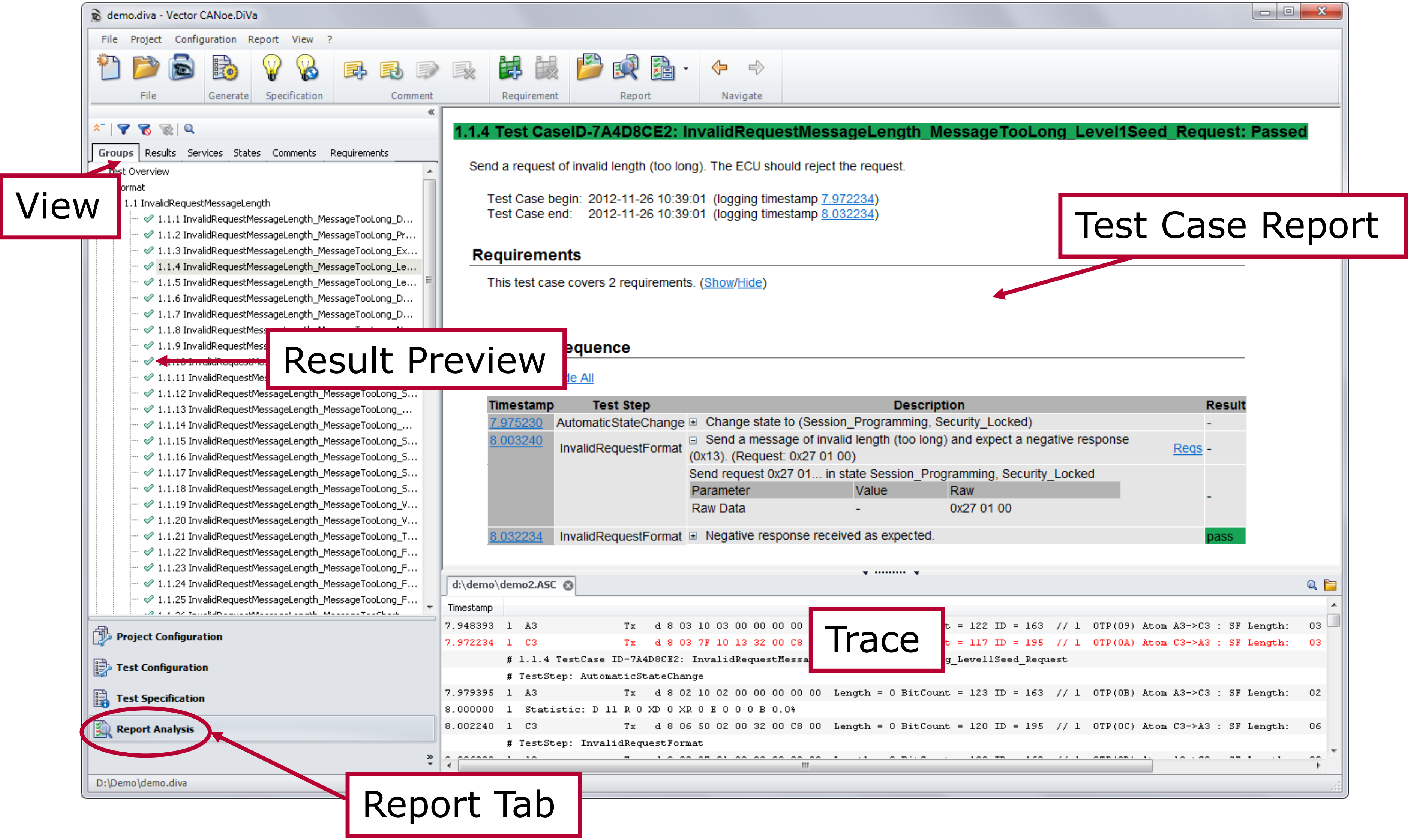
- Valid Flash execution
- Valid Flash execution at min. and max. voltage
- Errors during transfer data
  - Cancel by clamp reset
  - Cancel by stop transmission

- Erase Memory Errors
  - Clamp reset during EraseMemory
  - Cancel sequence after EraseMemory
  - Skip erase

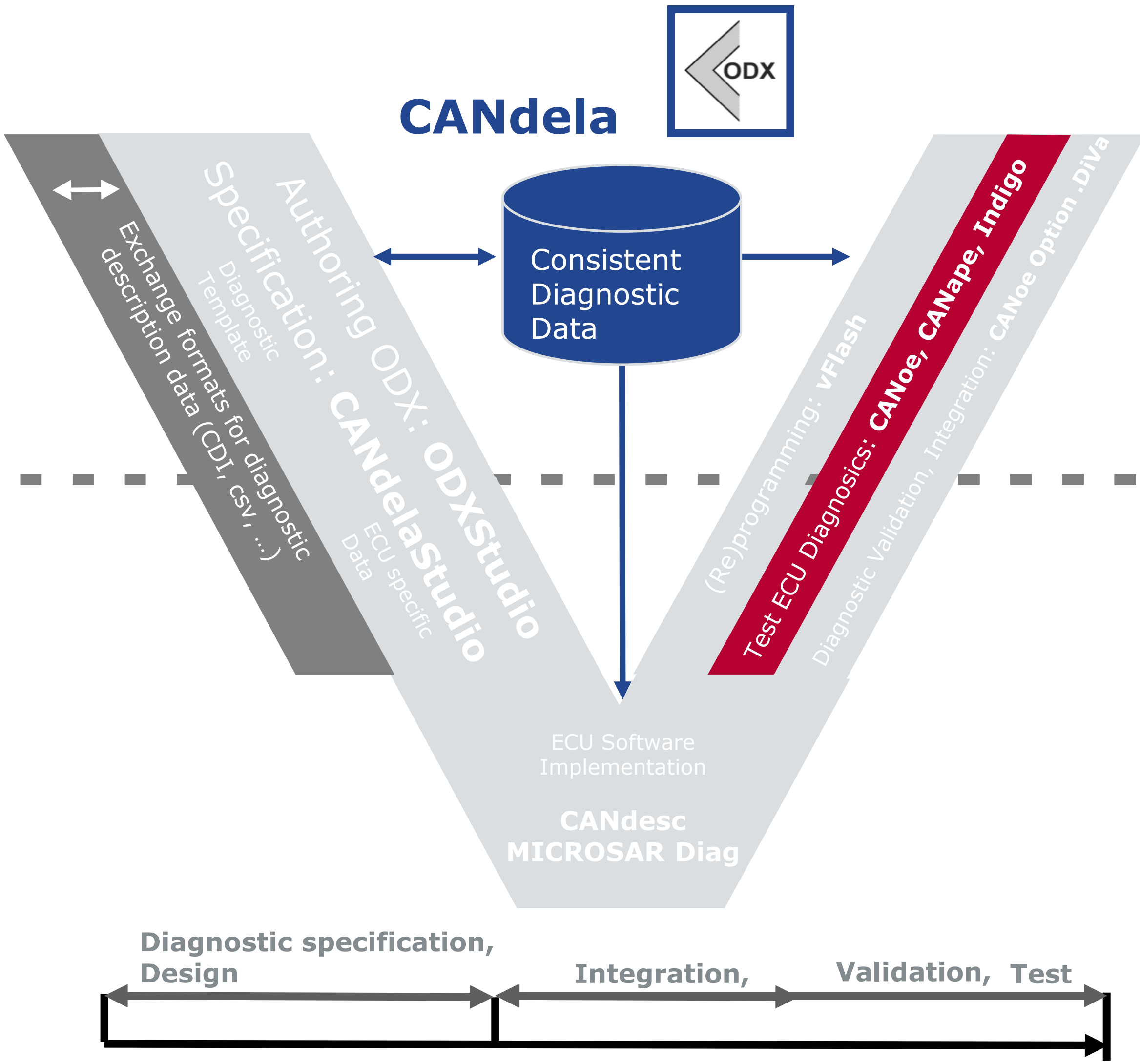
- Data Transfer
  - Transmit wrong CRC
  - Transmit wrong Signature



Report Analysis



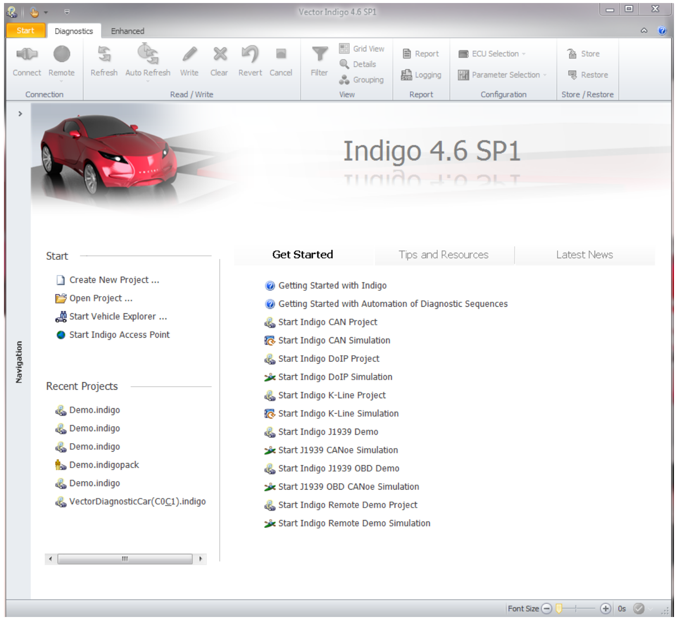
Manually Testing Supports



VECTOR

Indigo

- Self-configuring: Indigo uses database semantic information, reads ECU information and applies heuristics to configure the diagnostic capabilities and the behavior of the tester.
- Use-case driven: Indigo addresses the daily diagnostic tasks with specialized use-case views.
- Vehicle oriented: Indigo allows to view and modify vehicle-wide parameters and serves as a starting point to dive into the ECUs.
- Supports easy-to-use Vector Diagnostic Scripting (.NET)
- Supports UDS, KWP (many brands), GMW3110, WWH-OBD, OBDII
- Supports DoIP, CAN, K-Line, J1939
- Supports ODX 2.0.1, 2.2.0, cdd-files

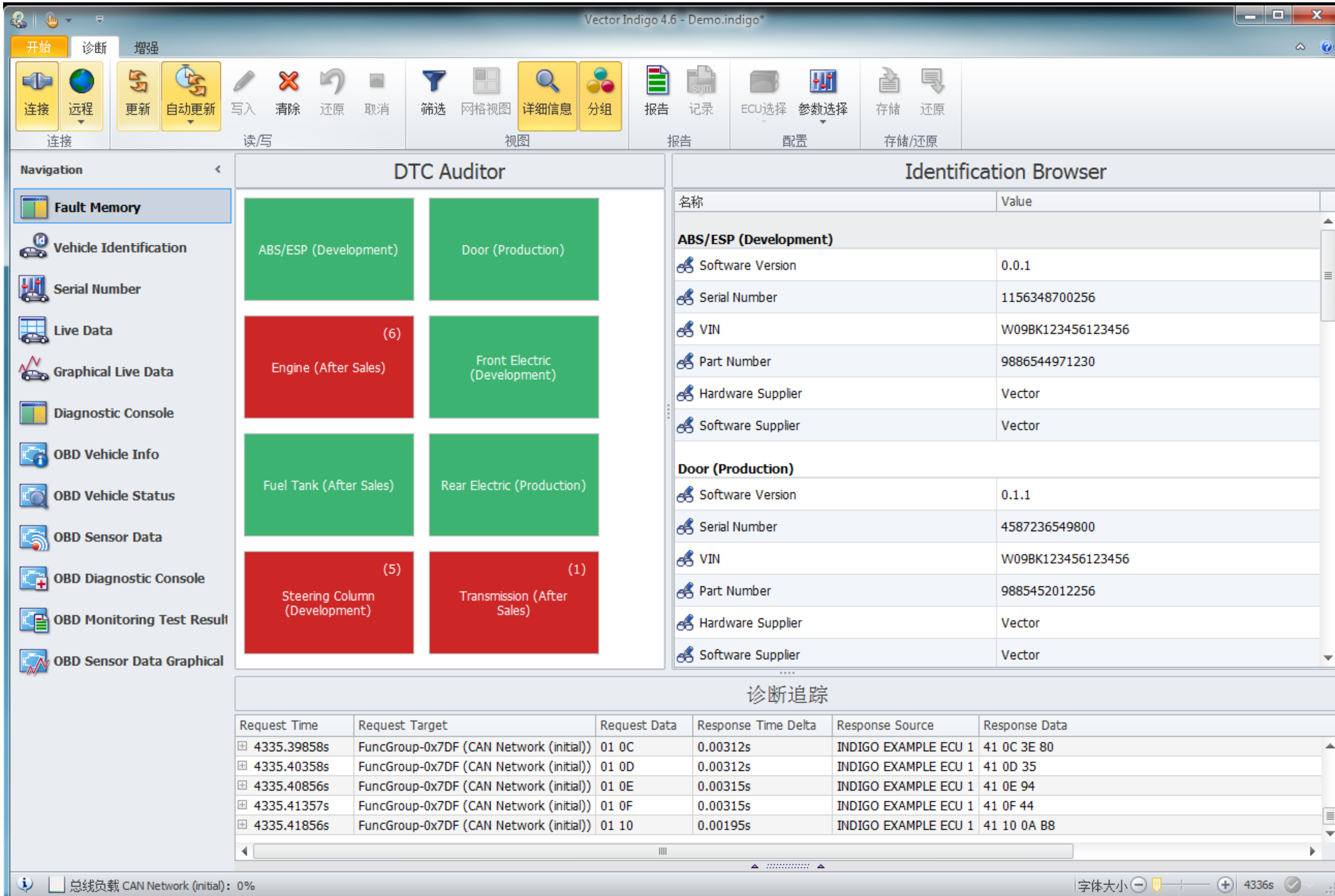


Diagnostic tool to be used intuitively

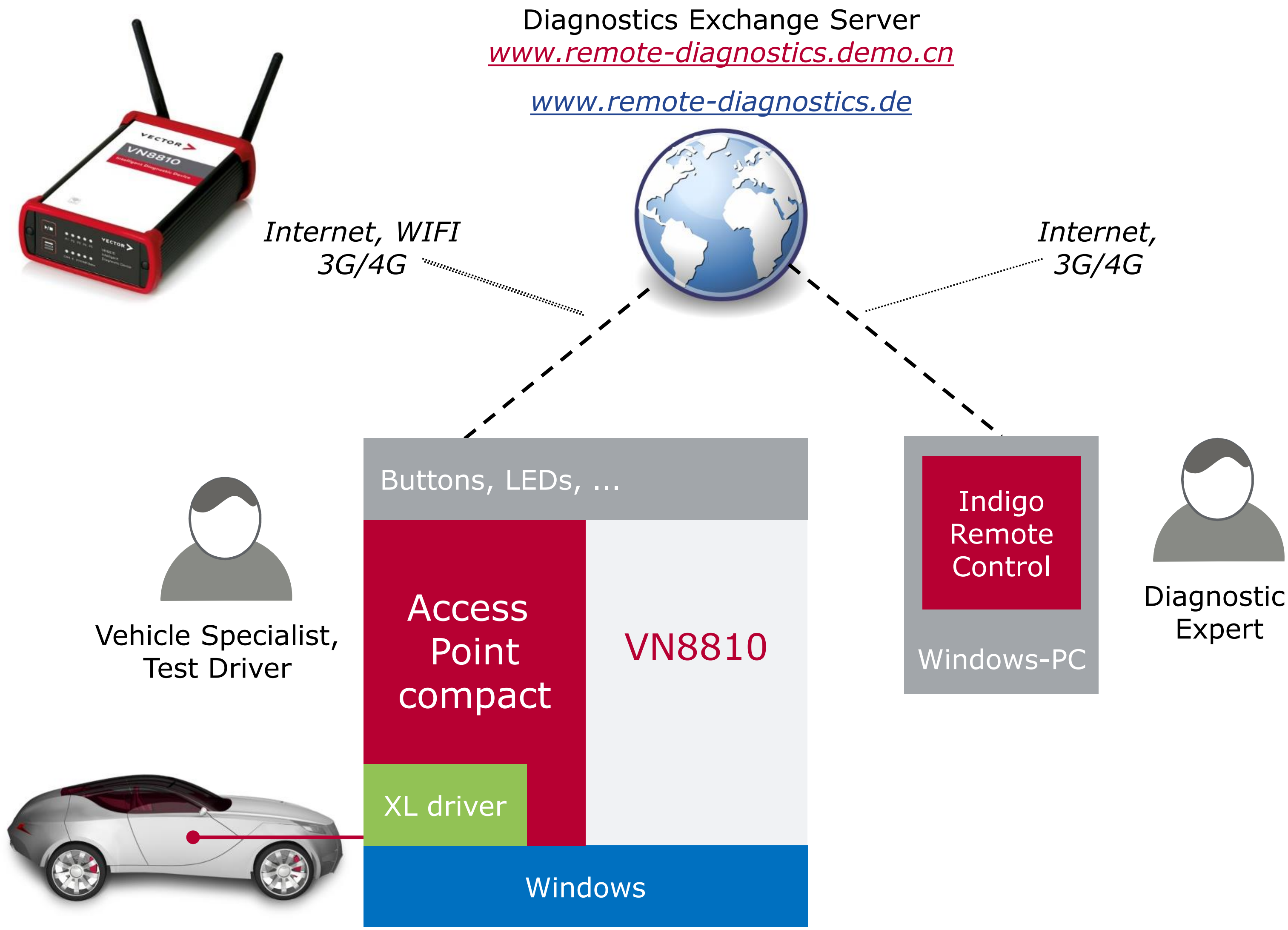
– without diagnostic protocol knowledge.

VECTOR

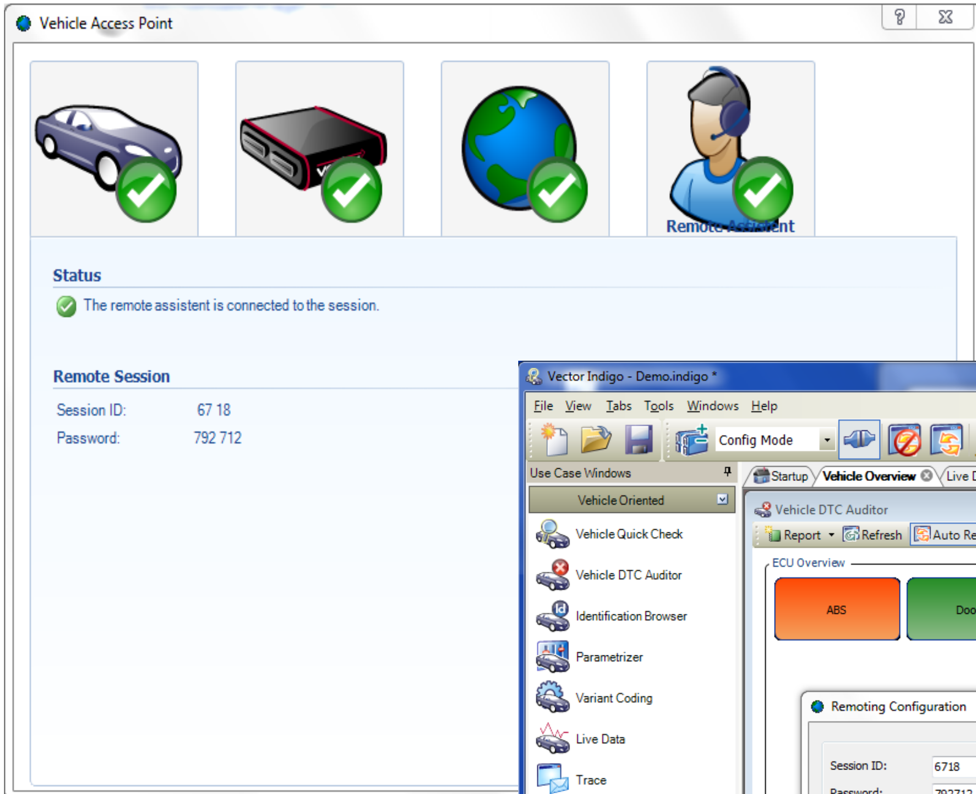
Indigo GUI

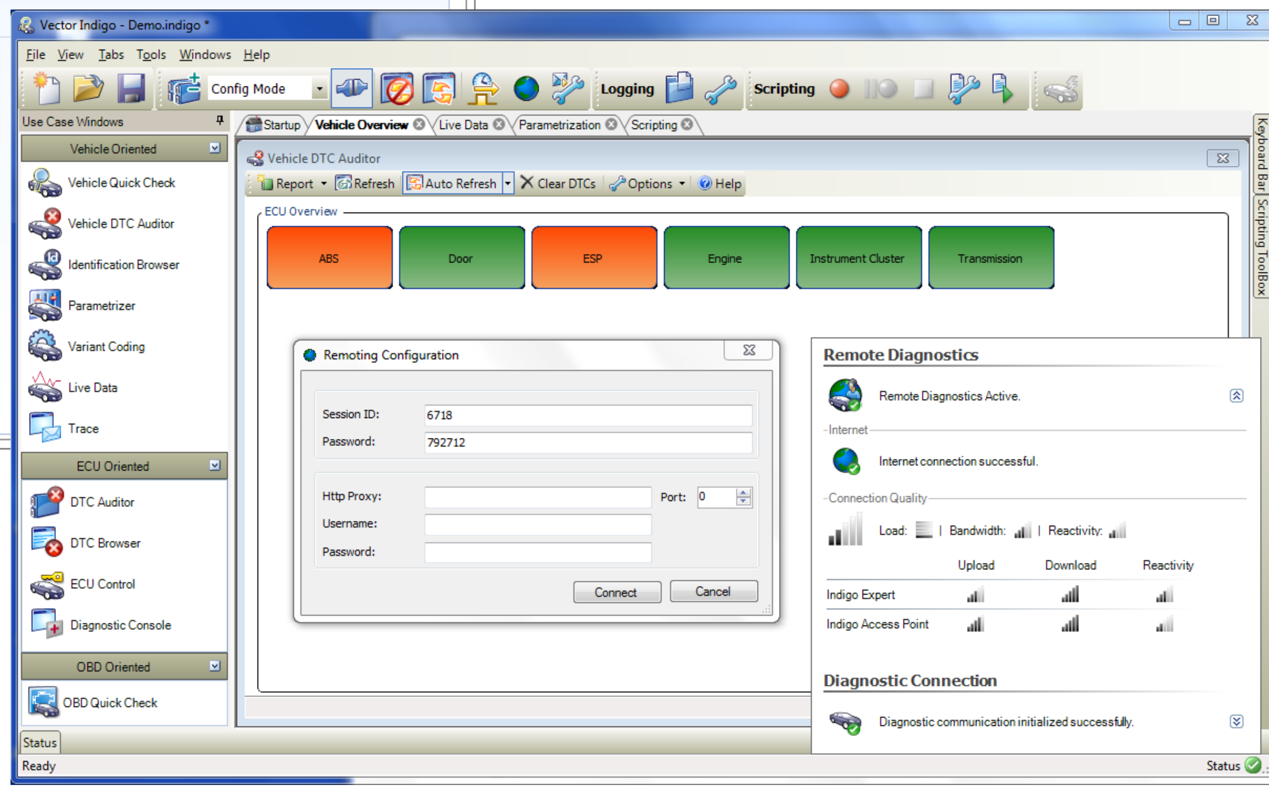


Indigo Remote System -- Structure



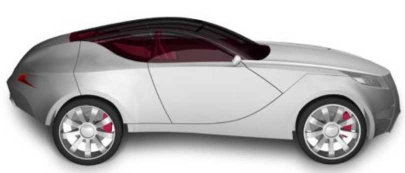
Indigo Remote System - Look and Feel



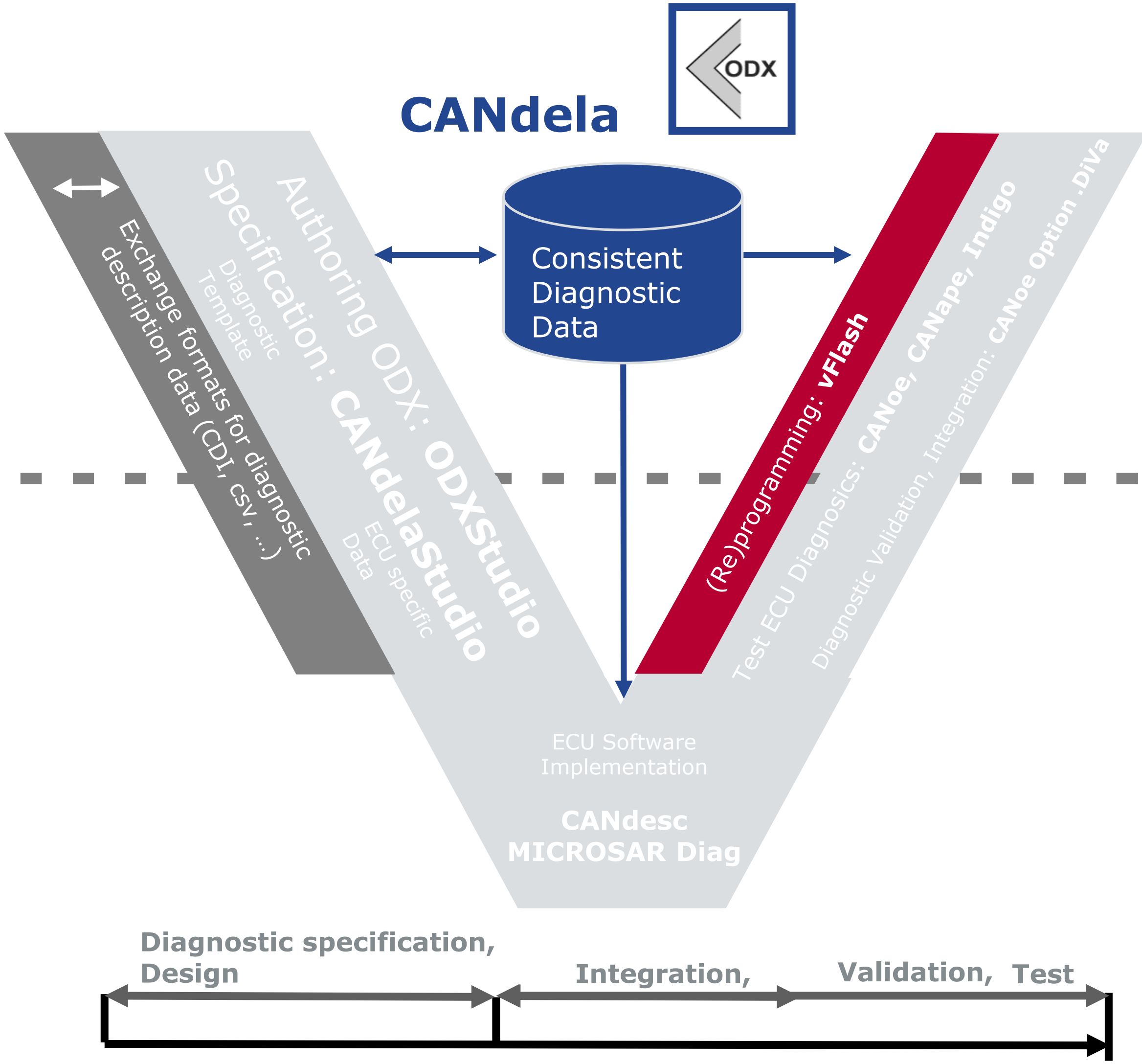


Remote Expert View

Access Point

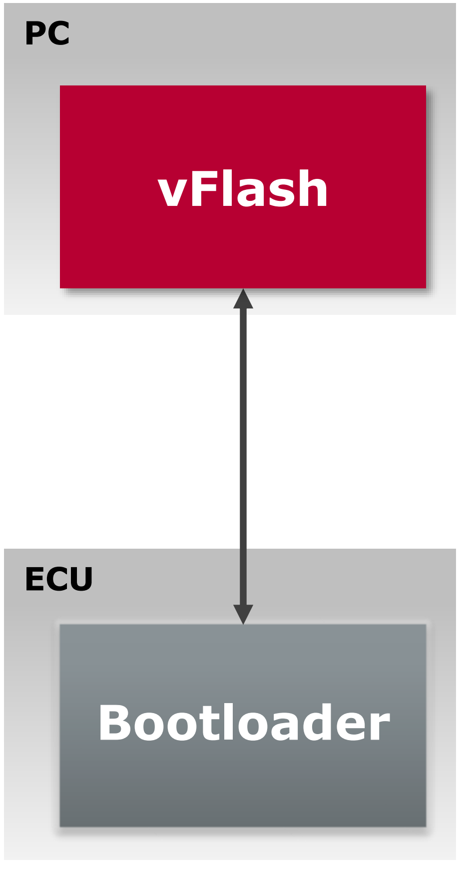


Flashing Tool

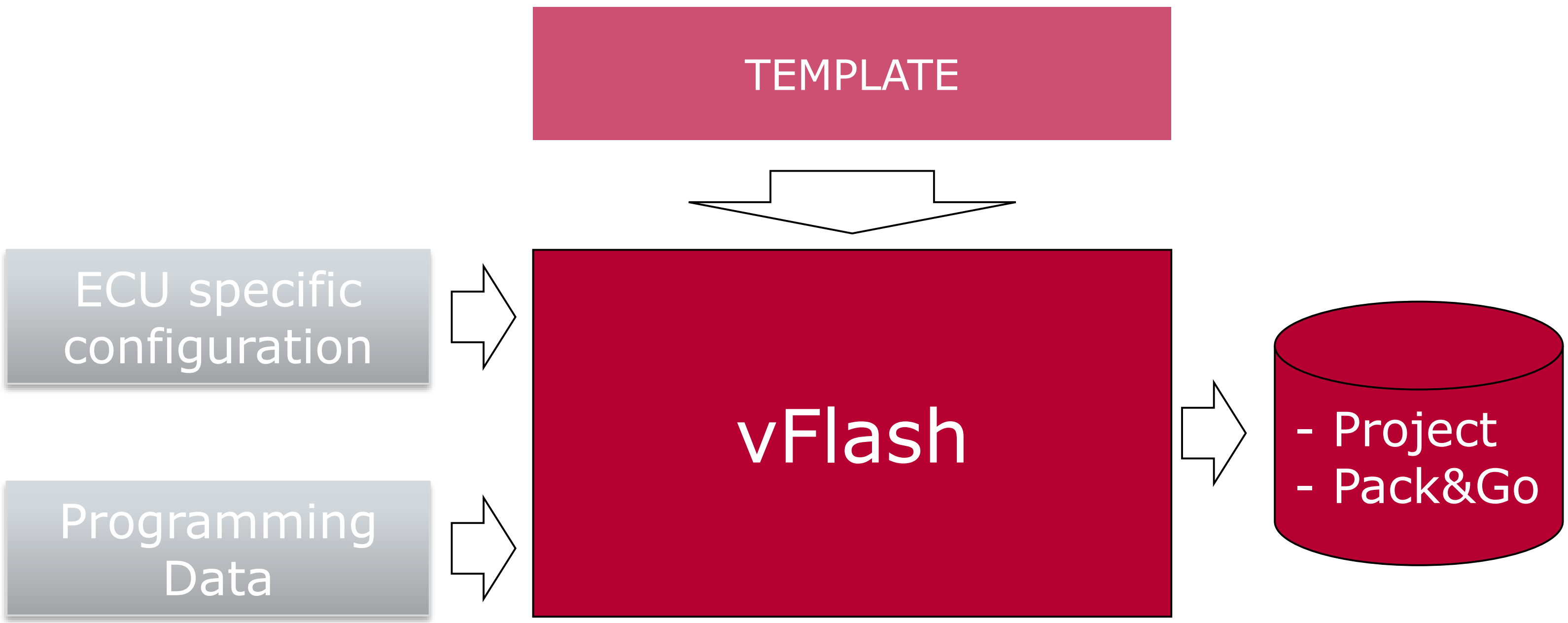


Flashing - vFlash

- Flash over different bus systems
  - CAN/CANFD
  - DoIP
  - LIN
  - FlexRay
- Consume different data formats
  - Native (Intel-Hex, Motorola-S, Binary)
  - ODX-F based flash containers
- Flash according to different OEM flash specifications/sequences
- Efficient and user-oriented
  - One-click flashing *OR* batch processing
  - Minimized flashing times
  - Quick start-up: Project is defined based upon a pre-defined template

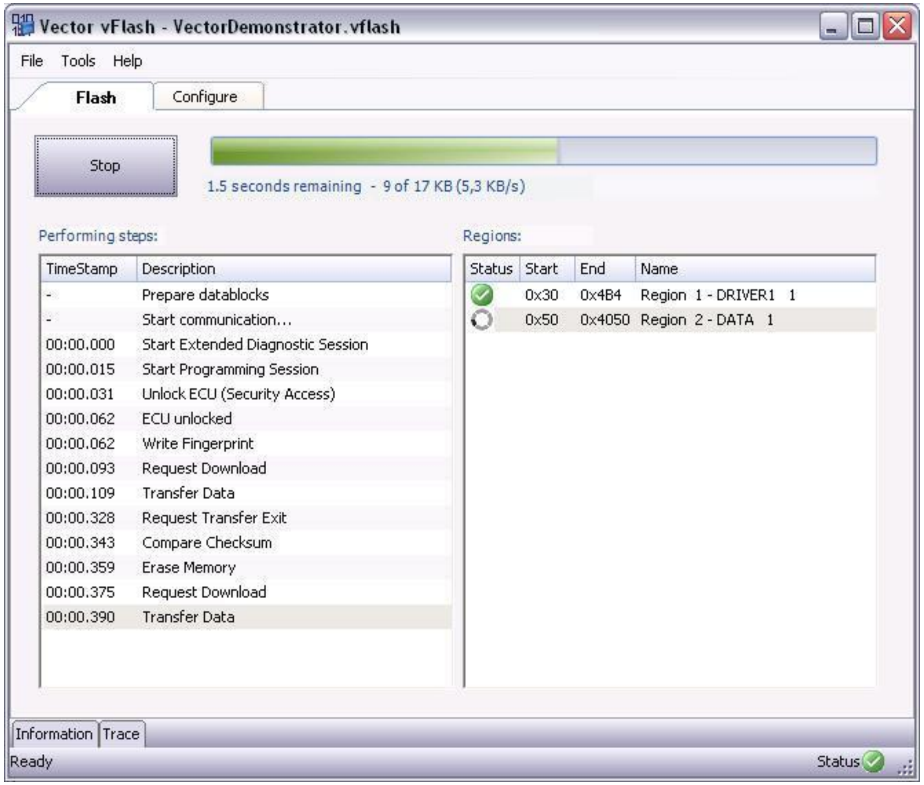


vFlash Concept



vFlash Look & Feel

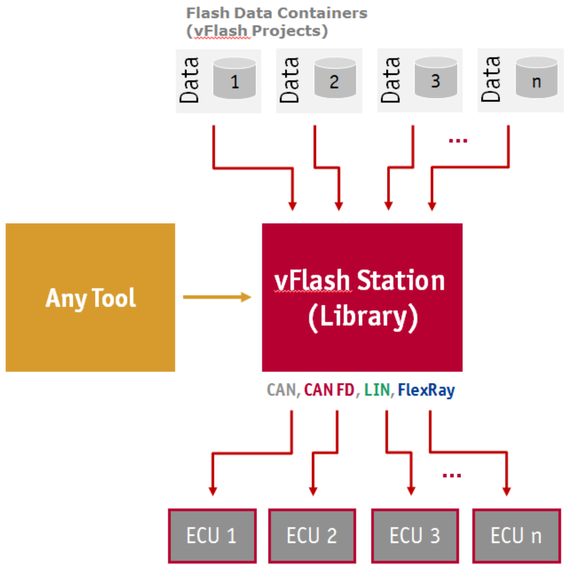
- Load existing project
  - Project file
    - All required files are just referenced
  - Pack&Go
    - All required files are packed together (Project, Seed&Key.dll, flashware, …)
- Start flashing immediately



vFlash Station

- Features
  - Parallel reprogramming of up to 8 ECUs
  - Programming each ECU over a separate bus
  - Flashing different bus types (CAN, CAN FD, FlexRay, LIN) at the same time
  - Controlling each flash process independently

- Availability
  - Is provided as library
  - Comes with C- and C#-API
  - Is available as additional tool edition
    (vFlash Station license also covers vFlash)



Agenda

Diagnostic Overview

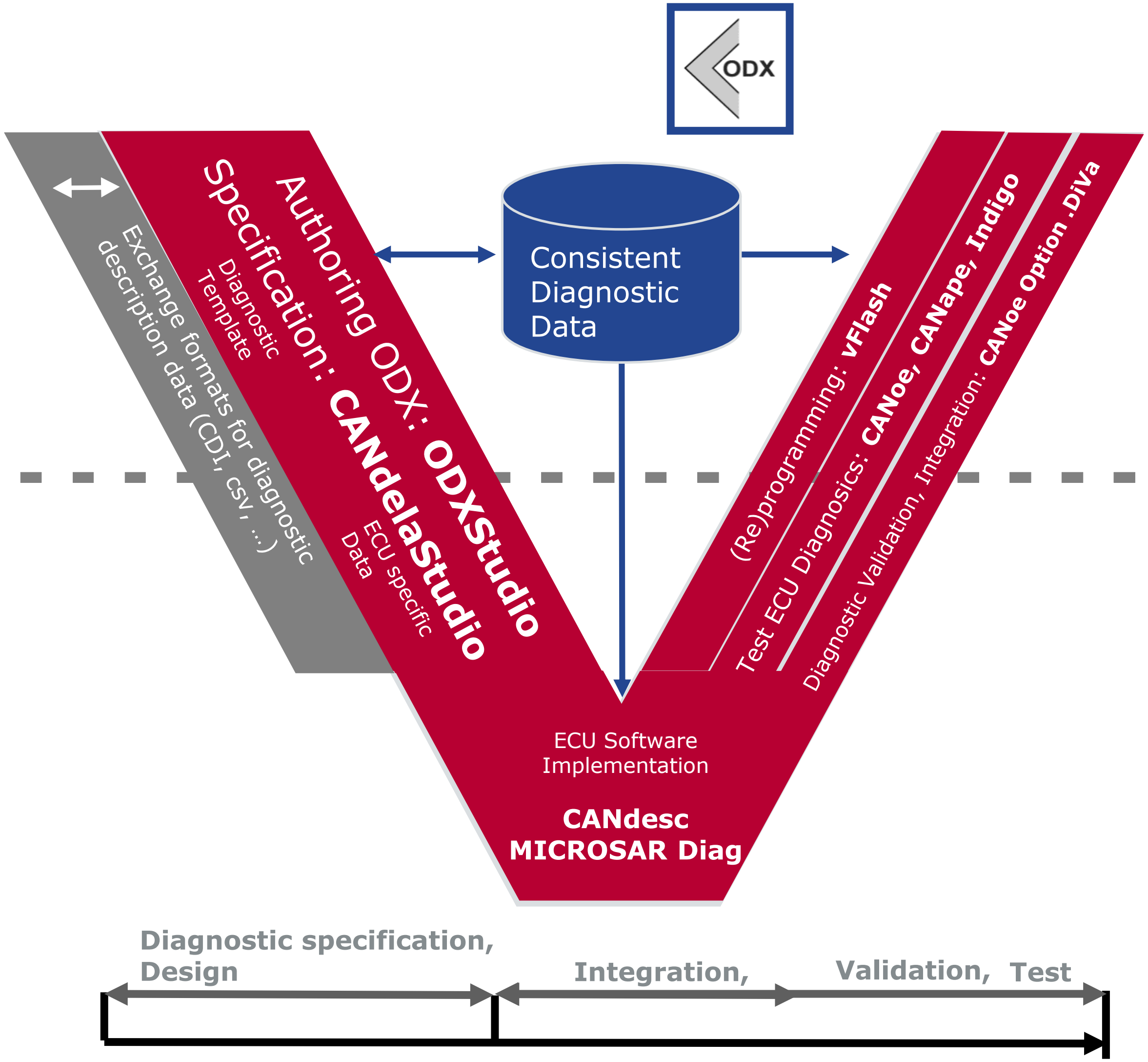
Diagnostic Protocol UDS

- Services Overview
- Diagnostic Session Control
- Security Access
- Read/Write Data By Identifier
- Fault Memory
- Communication Method in Diagnostics
- Review/Summary

Vector Diagnostic Solution

- Solution
- Tool-chain
- **Summary**

Specification/Process/Tool-chain



VECTOR

Selection of Customers

AAPL
A.D.C.
AW
ALFA ROMEO
ALPS
ALPS
ALPS NORDIC AB
Autoliv
AL AUTOMOTIVE LIGHTING

BEHR
BERU
BITRON
BorgWarner Automotive
BOSCH
catem
CHERRY

CLAAS
Continental TEMIC
Continental TEVES
DAEWOO
DAEWOO PRECISION INDUSTRIES CO. LTD
DAIMLER

DELPHI
DENSO
FIAT
Ford
fuba
GM

ISUZU
HARMAN/BECKER AUTOMOTIVE SYSTEMS
HELBAKO Kompetenz in Elektronik
HELLA BEHR THE MODULE COMPANY
HELLA
JOHNSON CONTROLS

kiekert
KOSTAL
LANCIA
LUK
MAGNA STEYR
MAGNETI MARELLI

MARQUARDT
mazda
Mecel
MITSUBISHI
PORSCHE
RHEINMETALL
SAAB

smart
SIEMENS VDO Automotive
SCHRADER ELECTRONICS
TRW
Visteon
Valeo
ZF

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Author:
Lisa,Liu
Vector China